

MACROSCOPIC LIFE

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Book One

*The Search for Life Beyond the Scale of Human
Perception*

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The limits of human observation are not the limits of reality.

PART I

THE OBSERVER

Before we can ask what reality contains, we have to understand the limits of the instrument doing the asking.

CHAPTER 1

The Window

What if we are living inside something alive and simply lack the scale of perception required to recognize it?

Imagine shrinking until you are smaller than a human cell. Not merely microscopic, but small enough that a single cell becomes a landscape. Its membrane stretches beyond your horizon. Proteins become structures. Chemical gradients become geography. Electrical changes across the membrane become events in your environment. If you could somehow survive there and build instruments, you might become an excellent scientist of that microscopic world without ever seeing the organism whose body contains it.

From where you stood, blood flow might resemble weather. A hormone arriving from somewhere beyond your observable region might appear as a mysterious environmental change. A pressure wave generated by a heartbeat could seem like a recurring law of nature. You might measure temperature, chemistry, electrical activity, fluid movement, neighboring cells, nutrients, waste, and countless other variables with extraordinary precision. Your measurements could be correct while your picture of the larger organization remained incomplete.

The thought experiment proves nothing about our own position in the universe. A human being is not literally a cell in a planetary body simply because the comparison is imaginable. But it exposes a problem that science has encountered again and again: an observer can accurately measure the part of reality available to it while remaining unable to perceive important organization outside that observational window. The

limits of an observer are not necessarily the limits of the system being observed.

That distinction is where this book begins.

The Instrument We Were Born Inside

FIGURE 02 - APPROVED POSITION

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We usually think of the human body as the thing doing the observing, but scientifically it is also an instrument. Our eyes respond to a limited range of electromagnetic radiation. Our ears respond to a limited range of pressure oscillations. Our skin detects particular forms of pressure and temperature. Our nervous system combines those signals into the world we experience, and because that experience is continuous and convincing, it is easy to mistake our biological interface with reality for reality itself.

Other organisms immediately expose that mistake. Many bats navigate and hunt using echolocation, extracting information from returning sound that human beings cannot naturally use in the same way. Pit vipers possess infrared-sensitive pit organs that provide access to thermal information outside ordinary human vision. Electric fish can generate or detect electrical fields and use them in behavior. Other

animals make use of ultraviolet light, polarization, chemical gradients, vibration, or magnetic information in ways radically different from ordinary human perception.

None of these animals occupies a more real universe than we do. They occupy the same physical world while sampling different portions of it. What exists and what an organism can naturally detect are therefore two different questions. Once that distinction becomes obvious, human perception stops looking like the reference standard for reality and starts looking like what it actually is: one evolved observational window among many.

Science has spent centuries extending that window. Microscopes revealed structures too small for unaided vision. Telescopes expanded the spatial horizon in the opposite direction. Radio astronomy opened a sky that human eyes cannot see. Infrared instruments, particle detectors, seismometers, magnetometers, spectrometers, electrocardiograms, and thousands of other devices convert inaccessible physical events into records we can examine.

The instruments did not create those phenomena. They changed what the observer could detect.

The limits of human observation are not the limits of reality.

That sentence is not permission to believe whatever we want. It means almost the opposite. If something lies outside ordinary perception, imagination cannot tell us what is there. We need another detector.

The Clock Is Part of the Window

Scale is not only about size. It is also about time.

Stand beside a tree for ten seconds and almost nothing seems to happen. Watch it across seasons and a different process appears. Follow a forest across decades and succession, disturbance, competition, regeneration, migration, disease, fire, and climate become visible in ways that no afternoon walk could reveal. Extend the observation across centuries or geological intervals and the apparent object changes again.

The same problem works in the opposite direction. Some events happen too quickly for ordinary human perception. High-speed cameras reveal processes that our eyes blend together. Slow-motion changes the observer's access to a fast event just as time-lapse changes our access to a slow one. Neither technique changes the underlying process. It changes the temporal window through which we observe it.

This matters because a process can appear static when our observation is too short, chaotic when our sampling is too coarse, or instantaneous when our temporal resolution is too slow. A geological process does not become alive because it unfolds slowly, just as a rapid chemical reaction does not become intelligent because it exceeds our visual resolution. Temporal mismatch can hide organization from an observer. It cannot tell us what kind of organization we have found.

If we are going to ask whether meaningful organization can exist at scales larger than familiar organisms, then our clock cannot automatically be the clock of the system. A process that matters over a century may look like background noise across a week. A process that matters over ten thousand years may look almost motionless across a human lifetime. Time-lapse changes the observer, not the process.

Where Does One Thing End?

There is another assumption hidden inside ordinary perception: we are accustomed to seeing individuals.

A dog looks like one thing. A tree looks like one thing. A person looks like one thing. The boundaries feel obvious enough that we rarely ask why those boundaries deserve to matter. Yet even the human body complicates that intuition. Matter continually enters and leaves us. Oxygen crosses the lungs. Food becomes tissue and energy. Water moves through us. Heat leaves us. Microorganisms live on and within us. Cells die and are replaced, while the person persists.

So an individual cannot simply mean a perfectly sealed object made from the same material forever. What persists is organization. The boundary of the organism matters not because nothing crosses it, but because crossing it has consequences and because processes on one side

are integrated differently from processes on the other.

That observation becomes more difficult when we look upward. A forest has no skin. An ecosystem exchanges organisms, water, carbon, nutrients, heat, and energy with its surroundings. A city has roads instead of membranes. A civilization has no obvious outer wall at all. Earth appears to offer a wonderfully convenient boundary because it is a planet floating in space, but convenience is not evidence of individuality.

We are not allowed to draw a circle around Earth and then announce that we have discovered the thing inside the circle. Earth's visual enclosure is only the beginning of the question.

If Earth, a forest, a city, a civilization, or any other large system is ever treated as a meaningful individual in this investigation, its boundary will have to do scientific work. It will have to help us explain organization, predict behavior, understand disturbances, or distinguish internal relationships from external ones. The boundary must emerge from what the system does, not merely from the shape we find attractive.

The Seduction of Resemblance

Nature is full of patterns that invite comparison. Rivers branch like blood vessels. Lightning branches like roots. Fungal networks can resemble other distributed networks. Spiral forms appear in organisms, weather systems, and galaxies. Similar mathematical structures can arise in systems separated by enormous differences in size and mechanism.

Human beings are exceptionally good at noticing those similarities. That ability is useful, because analogy can suggest questions that would otherwise never occur to us. It is also dangerous, because resemblance is cheap.

The old intuition *as above, so below* captures something emotionally powerful about recurring forms in nature. Used carefully, it can inspire a search for mathematical relationships, scaling laws, common constraints, or mechanisms that produce similar geometry. Used carelessly, it becomes a shortcut from “these things look alike” to “these

things are the same kind of thing.” Science cannot take that shortcut.

A river network is not a circulatory system merely because both branch. A galaxy is not a cell because both can be drawn as bounded structures. The cosmic web is not a nervous system because photographs of the two can be visually suggestive. Fractal or self-similar geometry can be scientifically important without implying functional self-similarity.

Analogy gives us somewhere to look. Evidence decides whether there is anything there.

Frequency, Resonance, and Electricity

The same discipline becomes especially important around words such as *frequency*, *resonance*, *energy*, and *electricity*. These words belong to real science, but their scientific meanings are precise enough that they can be damaged when they are used as universal explanations.

Frequency describes how often a repeating event or oscillation occurs over time. A heart has rhythmic activity. Neural populations can exhibit oscillatory electrical behavior. Chemical and mechanical systems can oscillate. Electromagnetic radiation has frequency. These are measurable physical quantities associated with specified processes.

Resonance is also real. Under appropriate conditions, a system can respond strongly to forcing at particular frequencies. Coupled systems can synchronize. Oscillators can influence one another through physical pathways. Resonance can be crucial in mechanical, electrical, acoustic, chemical, and biological systems.

But resonance is a physical relationship before it is an informational relationship. Two systems oscillating at related frequencies are not automatically communicating. Synchronization does not establish shared identity, intelligence, intention, or consciousness. Before frequency can explain anything, we need to know what is oscillating, at what rate, through what mechanism, how the systems are coupled, and what measurable consequence follows.

Electricity deserves the same care. Human bodies are profoundly electrical in the ordinary biophysical sense. Membrane potentials, ion gradients, action potentials, cardiac conduction, muscle activation, and many other processes depend on electrical phenomena. Those facts are extraordinary enough without turning electricity into a mystical substance.

Electrical activity does not automatically imply information processing, information processing does not automatically imply cognition, and cognition does not automatically imply consciousness. The interesting scientific question is not whether electricity exists in living systems. It plainly does. The question is what particular electrical organization does within a particular system.

What About the Aura?

This is where a subject like the human aura becomes useful—not as an established explanation, but as an example of how this investigation will treat experiences that sit near the boundary between ordinary perception and extraordinary interpretation.

People across cultures have described sensations, impressions, colors, fields, presences, or other experiences associated with bodies. Those reports are experiences, and experiences can be studied. But the report and its explanation are not the same thing. A person can sincerely experience something without yet knowing its physical cause.

Human bodies already produce measurable phenomena that ordinary vision does not directly reveal. We radiate infrared energy. Our hearts, nerves, and muscles generate electrical activity. Electrical currents have electromagnetic consequences. Bodies produce chemical emissions, mechanical vibrations, sound, and changes in surrounding air and temperature. Some of those phenomena can be measured with instruments.

None of that, by itself, establishes a colored spiritual field, a personality-bearing energy layer, or any other stronger interpretation. If someone proposes an additional physical phenomenon, the scientific response is neither ridicule nor automatic belief. It is measurement.

What quantity is being proposed? What produces it? How does it vary with distance and time? What detector should respond? Can blinded observers distinguish its presence from its absence? Does the effect replicate?

An experience can begin a question. It cannot decide the answer.

That rule protects wonder rather than destroying it. If an unfamiliar phenomenon is real, disciplined measurement gives us a chance to discover what it actually is instead of forcing it into the first explanation that feels satisfying. If the proposed phenomenon disappears under controlled testing, that matters too. Either way, reality gets the final vote.

We Are Not the Reference Frame

The deeper issue beneath all of these examples is surprisingly simple. Human beings experience the universe from one range of sizes, one range of lifetimes, one sensory architecture, and one position inside the systems we are trying to understand. There is no established scientific reason to assume that our particular observational scale is the privileged scale at which every meaningful form of organization must become obvious.

That does not mean meaningful organization exists at every scale. It certainly does not mean larger systems are secretly organisms. It means only that absence from ordinary human perception is not enough to settle the question.

We can call this **observer-relative detectability**. A phenomenon may be physically real while its detectability depends on the observer's sensory channels, spatial resolution, temporal resolution, duration of observation, instruments, and chosen boundary. Changing those conditions can reveal relationships that were previously inaccessible.

The idea is familiar in science even if the phrase is not. A bacterium became observable when microscopy extended spatial resolution. Radio galaxies became observable when instruments opened another part of the electromagnetic spectrum. Rapid physical events became analyzable through high-speed imaging, while slow processes became visible

through long records and time-lapse. Reality did not appear because humanity became clever enough to create it. Our observational window became larger.

That leaves us with a possibility worth investigating and a warning worth carrying with us. The possibility is that important organization may exist outside the scales at which human intuition naturally recognizes individuals. The warning is that perceptual limitation provides no positive evidence that such organization exists. It tells us only that looking with unaided human senses is not enough.

So we will not begin by declaring that Earth is alive, that the universe is conscious, that resonance is communication, or that an unfamiliar sensation proves the existence of an unseen field. Those are conclusions, and conclusions belong at the end of evidence rather than the beginning of curiosity. We will begin with the harder task: building ways to distinguish organization from resemblance, interaction from integration, and a convenient boundary from an actual individual.

If organization existed at a scale our senses and intuitions were never built to recognize, what would we have to measure before we could honestly say we had found it?

That is where the real investigation begins.

CHAPTER 2

The Search

Accepting that human observation is limited creates a strange freedom. Once we admit that important relationships might be hidden by the wrong scale, the wrong clock, the wrong variable, or the wrong boundary, almost anything can be imagined somewhere outside the window. That freedom is intellectually exciting. It is also where an investigation like this can become unfalsifiable.

The wider the search becomes, the stricter the discipline has to become with it. Perceptual limits justify looking farther. They do not protect whatever we hope to find. The problem of Chapter 1 was that we may fail to see what is there. The problem now is almost the opposite: once we search enough places in enough ways, we may start seeing whatever we want to see.

That is the central problem of the search: how do we expand observation without turning flexibility into a machine for manufacturing patterns?

A Forest Full of Possible Answers

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Imagine we instrument a forest. We can measure air temperature, humidity, soil moisture, fungal activity, sap flow, leaf chemistry, canopy

temperature, animal movement, microbial composition, electrical potentials, acoustic activity, water tables, rainfall, wind, fire history, nutrient flow, and hundreds of other variables. We can measure them by the minute or by the month, at one tree or across a watershed, and we can choose where the candidate forest begins and ends.

Each choice changes the question. A thermometer gives one forest. A soil probe gives another. A fungal assay gives another. Satellite imagery gives another. A century of records can reveal relationships that disappear in a week of observations, while very fast sampling can expose dynamics hidden by daily averages.

It helps to think of this as a search through **Scale-Space**. Let

$$\mathbf{S} = (\mathbf{x}, t, \mathbf{v}, b)$$

where x represents spatial scale, t temporal scale, v the variables or channels measured, and b the boundary used to define the candidate system. We do not need the notation to make the idea true, but it helps keep the dimensions separate. Moving through Scale-Space can reveal a relationship that a single observational setting would miss.

It can also manufacture a relationship that disappears as soon as the search freedom is removed.

The Beautiful Pattern

Suppose sensors reveal that trees separated by kilometers show a striking change in activity several hours before a measurable local drought response. The pattern repeats enough times to be interesting. It is tempting to say that the forest is signaling across distance or coordinating a system-level response.

But a long-distance correlation has many possible explanations. The two regions may share weather. They may be responding to the same hydrological change. A third variable may influence both. The apparent delay may result from sampling. The boundary may have been chosen after the pattern was found. The relationship may be real while the story attached to it is wrong.

Distance makes the pattern worth investigating. It does not make it a message.

That is where the search becomes adversarial. We ask for the strongest common driver we can build. We include known hydrology, topography, soil, weather, physiology, ecological interactions, and measurement artifacts where they are relevant. If an ordinary explanation absorbs the pattern, that is progress. The goal is not to keep the mystery alive. The goal is to learn what survives when the mystery is attacked.

The Machine That Finds Too Much

Artificial intelligence makes this problem more important, not less. A human researcher might examine dozens or hundreds of relationships. A sufficiently automated search system can examine millions. It can test different time lags, spatial groupings, variable combinations, nonlinear transformations, and candidate boundaries, and it can find relationships no person would ever have thought to ask about.

That capability could be transformative. It also multiplies the number of coincidences available for discovery. Worse, a generative system can often produce a persuasive narrative explaining the pattern after it has already been found. The machine can therefore automate both sides of one of science's oldest temptations: discover an unlikely relationship and immediately tell a beautiful story about why it had to happen.

The cure is not to stop searching. It is to separate discovery from confirmation.

During discovery, explore. Search many scales. Let algorithms propose relationships. Let boundaries move. Let surprising patterns surface. But once a candidate relationship has been selected, freeze the relevant variables, lag, spatial region, boundary, expected direction, error measure, and competing explanation. Then ask whether the relationship survives withheld data, a new time period, another site, or a prospectively specified disturbance.

The more freedom used to discover the pattern, the less evidentiary weight that pattern deserves until it survives an independent test.

When Many Pieces Become Fewer

Sometimes a higher-scale variable becomes useful because it compresses many lower-level measurements into something more stable. Weather science does not stop being physical because meteorologists use pressure systems, fronts, or storm tracks rather than listing every molecule in the atmosphere. Neuroscience uses population activity because some questions are more tractable at that level than at the level of every ion channel. Economics uses unemployment or inflation because certain large-scale regularities can be studied without tracking every transaction independently.

That kind of compression matters for Macroscopic Life, but it does not settle the central question. A higher-level variable can be scientifically useful without being alive, agentic, or even a genuine individual. The first question is much smaller: does a proposed macro-state capture organization that reliably improves our understanding of the system under a specified comparison?

Suppose a forest-level drought index predicts future regional response better than an otherwise strong collection of local measurements. That is interesting. It does not mean the forest has a mind. It means we may have found a scale at which some relationships become easier to model, and perhaps more causally stable, than they are when every component is treated independently.

The next task is to determine whether the apparent whole was discovered or drawn.

Things That Can Fool the Detector

A hurricane has a boundary, internal flow, persistence, energy throughput, and coordinated structure. Traffic can form waves that move through a road network while the vehicles composing the wave continually change. An adaptive engineered network can sense variables, respond to disturbances, and reconfigure itself. None of these examples should be dismissed as trivial, because they are exactly what a detector for unfamiliar organization must survive.

If our criteria declare every persistent dynamic structure to be an individual, the criteria are too weak. If every feedback controller becomes an agent, agency has lost its meaning. If every information-bearing network becomes a mind, the framework has become a metaphor generator.

Calibration must therefore begin with systems whose mechanisms are comparatively well understood. A detector that cannot say no to a hurricane or thermostat cannot be trusted when it says yes to a forest or planet.

The Forest Changes Shape

Now imagine that our original forest boundary begins to fail. Perhaps vegetation inside the mapped forest responds more strongly to groundwater and soil systems extending beyond the visible tree line than to neighboring vegetation inside it. Perhaps disturbances propagate along a watershed boundary that cuts across the map. Perhaps repeated analyses recover the same hydro-ecological region even when researchers begin with different human-defined borders.

That does not prove we have found an individual. It does show something important: the organizational boundary may not respect the category we started with.

The candidate system is now allowed to push back against the observer.

This matters because unfamiliar organization will often be missed if we insist that political borders, ecosystem labels, coastlines, property

lines, or visual edges must also be causal boundaries. The map is a proposal. The system gets to disagree.

The Empty Search

A serious search must also be allowed to end with nothing. Suppose we define a reasonable region of Scale-Space, specify which variables and timescales will be examined, search for candidate higher-level organization, and find no pattern that survives independent confirmation. We do not get to rescue the hypothesis by moving endlessly outward: perhaps the forest failed because the real system is the watershed, then the continent, then Earth, then the solar system, then the galaxy.

A new scale can always motivate a new hypothesis. It cannot retroactively save the one that failed.

That is why the search needs boundaries of its own. We specify where we will look, how much flexibility discovery is allowed, and what outcome would count as failure. A useful theory must be capable of dying.

What the Search Is Really Looking For

The goal is not to find the largest thing in the data. It is not even to find the most complicated pattern. We are looking for cases in which ordinary descriptions begin to perform badly and a particular larger organization starts doing scientific work.

The signals might include unusually stable integration across components, persistent regulation, boundary effects, memory, compensatory dynamics, or higher-level variables that improve prediction under strong comparison. Those signals are invitations to further testing, not confirmations of life.

Eventually, if enough of them survive, we are allowed to draw a circle around the candidate.

But now the circle itself becomes the hypothesis.

The search has brought us to the edge of a possible whole. The next question is whether there is actually an individual there.

PART II — THE WHOLE

PART II

THE WHOLE

Before a system can possess a property, we must know
what system we are talking about.

CHAPTER 3

The Boundary

You are one thing, or at least that is how it feels. You wake as one person, move as one body, and persist through changes that would make a strict inventory of your material nearly useless as a definition of identity. Molecules enter and leave. Cells die and are replaced. Microorganisms live in and on you. Water, oxygen, food, heat, waste, signals, and pathogens cross the edges of the body continuously.

Yet the boundary of the organism is not meaningless. The fact that matter crosses it does not erase it. What makes the boundary scientifically important is that events on one side tend to participate in a different network of regulation, persistence, repair, and mutual dependence than events on the other.

If we want to investigate individuality above familiar organisms, this is where the problem becomes serious. A line on a map is not an individual. A category in a dataset is not an individual. A candidate whole has to be discovered from organization rather than manufactured by the researcher's decision to draw a circle.

The Circle We Draw

FIGURE 06 - APPROVED POSITION

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Think about skin. It is porous, metabolically active, chemically complex, and constantly crossed by matter and energy. Still, cutting across the skin of a person is not equivalent to cutting through the surrounding air. The boundary predicts consequences. It separates regions with very different patterns of regulation and integration.

Now consider a forest. We can draw its edge where trees become sparse, where a property boundary ends, where a satellite classification changes, or where a watershed divides. Those lines may or may not correspond to the organization we are trying to study. If the candidate system is real in the relevant sense, different ways of drawing the boundary should not all work equally well.

That gives us a basic principle: a meaningful boundary should earn predictive or explanatory advantage over competing boundaries. It should help distinguish stronger internal relationships from weaker or differently structured external relationships. It should matter to the way perturbations spread, resources move, information becomes functionally available, or the system maintains itself.

This is where the most visually seductive candidate has to lose its privilege.

Earth must earn Earth.

The planet is physically bounded in an obvious astronomical sense, but physical enclosure does not by itself establish the kind of organizational individuality we are investigating. If Earth is ever treated as an individual in this framework, planetary scale must outperform serious competing descriptions for the relevant property.

What Happens When We Disturb It

Boundaries become more informative when something goes wrong. Cut a finger and a coordinated response unfolds across tissues, blood, immune processes, nerves, and behavior. Break a granite boulder and no corresponding organism-level restoration follows. Disturb a forest and the response may propagate through roots, water, soil, microbes, herbivores, fire regimes, seed banks, and weather interactions.

The key question is not whether a disturbance causes effects. Everything affects something. The question is whether the proposed boundary predicts a distinctive pattern of effects that competing boundaries do not capture as well.

Suppose drought in one region of a forest produces compensatory changes elsewhere. That may indicate integration, but only if common environmental drivers and ordinary local responses fail to explain the coordination. If the same response appears when the candidate forest is divided differently, the original boundary weakens. If removing a region has no consequence for system-level organization, its membership becomes harder to defend.

A boundary should therefore be allowed to fail. The candidate individual is not whatever survives our prose. It is whatever survives perturbation and comparison.

Who Belongs to the Whole?

Membership is harder than influence. A passenger influences a train but is not part of the train in the same way the braking system is. Fuel is necessary for an engine but is not organized as a durable engine component. Oxygen is essential to a human but the entire atmosphere is not one of your organs.

For a candidate macroscopic individual, membership should have consequences. Removing a component, replacing it, moving it, or disconnecting it should alter the proposed whole in ways that differ from manipulating a merely external influence. Some members may be replaceable; biological organisms replace components constantly. Replaceability is not the same as irrelevance.

The deeper question is whether the component participates in the organization that makes the whole the same kind of thing across time. A tree in a forest can die and be replaced without destroying the forest. But if the hydrological, microbial, reproductive, or trophic relationships that sustain the larger organization collapse, the identity of the system may change even while many of the same trees remain.

Membership therefore cannot be reduced to material inclusion. It is a functional and historical relationship.

The Hospital Problem

Imagine a hospital. Inside one building are surgeons, nurses, laboratories, pharmacies, computers, maintenance staff, ambulances, oxygen systems, power supplies, patients, cafeterias, accounting departments, and hundreds of other functions. Many are necessary for the hospital to operate. Yet the fact that a function occurs within the hospital does not mean the hospital itself owns that function in the stronger sense relevant to agency or individuality.

A surgeon makes a decision. The hospital contains the decision-making person. That does not automatically mean the hospital as a whole decided. A computer stores information. The hospital contains memory. That does not automatically mean the hospital remembers as one integrated agent. A backup generator restores power. The hospital contains recovery capacity. That does not automatically mean the hospital repairs itself as one organism.

This distinction is easy to lose when we move to larger systems because ordinary language compresses levels of organization. We say a company decided, a nation remembers, a city wants, or civilization invented. Sometimes those phrases are useful shorthand. Sometimes they conceal the very ownership question we are trying to investigate.

A function can exist inside a system before the system owns the function.

Functional ownership must therefore be tested, not inferred from containment. We ask whether the function depends on integrated states of the candidate whole, whether it survives turnover of particular components, whether removing the larger organization destroys the function in a way that cannot be explained merely by removing individual agents, and whether the system-level description improves intervention or prediction.

What Persists When the Pieces Do Not?

The Ship of Theseus is often treated as a philosophical puzzle about identity, but biology turns the puzzle into ordinary life. Organisms persist while components change. Development can transform body form dramatically. Cells turn over at different rates. Molecules are continuously exchanged with the environment. The individual is not a frozen inventory of matter.

What persists is a history of organization. The organism carries forward constraints, relationships, developmental continuity, regulatory architecture, and patterns of mutual dependence. When those relationships collapse irreversibly, we say the organism has died even if much of its matter remains physically present.

That gives us another test for candidate macroscopic individuality. If the components change while an organized higher-level identity persists, the case becomes stronger. If the supposed whole disappears whenever particular components are replaced, perhaps the boundary merely captured a temporary collection.

Conversely, if the components remain but the integrated organization dissolves, the candidate individual may have ended. A dead body and a living body can contain almost the same kinds of matter at adjacent moments. The difference is organization, not substance alone.

Individuals Within Individuals

Biology already teaches us that individuality can be nested. Cells can be individuals in some contexts and components of multicellular organisms in others. Organisms host symbionts. Social insects can form colonies with striking degrees of integration while individual insects remain biologically significant. Ecological and evolutionary boundaries can overlap without being identical.

This matters because Macroscopic Life does not require one universal scale to win. A forest-level organization could be scientifically meaningful while trees remain individuals. A planetary organization, if one ever earned support, would not erase organisms, ecosystems, or societies beneath it. The framework allows multiple levels to matter

simultaneously.

Nested individuality is therefore not evidence for a largest individual. It is simply a warning against assuming that one successful level invalidates all others.

When the Circle Begins to Push Back

A candidate whole becomes scientifically interesting when the boundary stops behaving like a researcher's convenience and starts constraining what we can successfully predict. Disturbances propagate differently across it. Membership matters. Competing boundaries perform worse. Organization persists despite component turnover. Removing the proposed whole from the model costs something.

That still does not earn agency. A hurricane has a coherent dynamic boundary. A flame maintains organization while material flows through it. A traffic jam can persist while every car changes. Individuality and agency are separate questions.

By the end of this chapter, the most we may have earned is a candidate whole: an organization whose boundary, persistence, membership, and integration survive serious comparison.

Earning the whole does not earn the mind.

Now we have to ask what the whole does when more than one future is available.

CHAPTER 4

The Verb

It is cold in a house. The thermostat detects that the temperature has fallen below a set point, the furnace turns on, and the room warms. In ordinary conversation we say the thermostat sensed the cold, knew the room was too cold, decided to activate the furnace, and tried to restore the target temperature.

Nothing about the machine changed while we were talking. Only the verbs did.

That is the danger. Human language is built for agents, and it can make agency appear long before the evidence has earned it. If we want to ask whether a forest, ecosystem, city, or planet possesses agency, we need a definition that survives the thermostat.

The thermostat did not change. Our verbs did.

The Minimum Adversary

FIGURE 07 - APPROVED POSITION

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A simple thermostat detects one variable, compares it with a set point, and changes an output. That is regulation. It is already more interesting than passive persistence, because the system participates in a feedback loop that tends to restore a condition.

Now make the controller harder to dismiss. Give it heating and cooling. Let it use occupancy, energy price, weather forecasts, and measured heat loss. Let it choose among multiple available actions. It can now produce different responses to the same temperature depending on context.

Give it retained history. It learns that one room warms slowly, that afternoon sunlight changes heat load, or that a particular building empties at predictable times. Past states now alter future responses.

Finally, allow the controller to change its own weights or strategies through learning. It no longer follows one fixed lookup rule supplied in advance. Its policy adapts based on performance.

We have created something far more flexible than the thermostat we began with. We still have not automatically created a conscious agent. Adaptive control, machine learning, and reinforcement learning show why complexity cannot be the threshold.

Information Is Not One Thing

The word *information* becomes dangerous at this point because it can mean several different things. Two variables can carry statistical information about one another simply because they covary. A physical state can persist and later affect another process. A signal can be functionally useful to a system because acting differently on different values improves performance. Semantic meaning adds another layer involving representation and interpretation.

Those senses can overlap, but they are not interchangeable. A temperature difference contains statistically usable information for a thermostat without implying that the thermostat understands winter. A hormone can alter cellular behavior without becoming a sentence. A forest may contain correlations among water, roots, fungi, and growth without those correlations constituting a language.

For agency, the relevant question is functional: does information available to the candidate system alter which trajectory is selected among physically available alternatives? If not, information language may be doing no more work than saying that variables are related.

Give the Thermostat a Past

History dependence alone is also cheap. Materials exhibit hysteresis. Soil moisture depends on prior conditions. A bent spring carries a physical history. The fact that past state affects present behavior does not automatically establish memory in the functional sense we care about.

A stronger form of memory appears when retained physical state becomes available to alter future response under relevant conditions. An immune system can respond differently after prior exposure. An adaptive controller can change future action because stored history altered its parameters. A forest can respond differently to the same drought because prior disturbance changed species composition, roots, soil structure, fungal networks, nutrient stores, or canopy architecture.

The important point is not the word *remembers*. The important point is whether a trace of the past remains physically present and causally changes what the system does later.

The Marble in the Bowl

Place a marble in a curved bowl and push it away from the center. It rolls back. The system has a stable attractor, but we do not say the marble wants the bottom of the bowl.

Living systems often do something much more sophisticated. Body temperature, blood chemistry, water balance, and other variables can be maintained within viable ranges through multiple mechanisms. When one pathway is unavailable, another may compensate. Different disturbances can converge toward similar outcomes.

That flexibility matters, but even **equifinality**—different routes to a similar end state—does not by itself establish agency. Feedback networks can produce it. Evolution can build robust control architectures. Distributed systems can maintain target ranges without one centralized chooser.

A stable outcome can therefore look goal-directed long before we know whether the candidate system owns anything resembling a goal.

So Where Is the Choice?

For this project, operational agency begins to become a serious hypothesis when a candidate system repeatedly selects among available trajectories in an information-dependent, system-level, adaptively flexible manner. That definition is intentionally weaker than conscious intention and intentionally stronger than simple feedback.

Several burdens follow. There must be meaningful alternatives. The state of the system must affect which alternative is selected. The information used in selection must be functionally available to the proposed whole, not merely to one component we happen to have placed inside the boundary. The selection architecture should persist despite turnover of some components. And competing lower-level explanations must be allowed to explain the same behavior.

The strongest question is ownership. Does the proposed whole merely contain agents making local choices, or does the organization of the whole contribute something that disappears when we remove the higher level from the explanation?

The System That Owns the Selection

Consider an ant colony. Individual ants act. Local interactions generate colony-level patterns of foraging, defense, nest construction, and labor allocation. In some cases, colony-level variables may help predict collective behavior. That does not mean every action belongs to a single colony mind.

The same problem becomes sharper in a forest. Trees regulate water and carbon. Fungi respond to chemical conditions. Animals move. Microbes alter nutrient cycles. Natural selection has shaped countless local adaptations. A forest can therefore display extraordinarily coordinated-looking behavior even if every apparently purposeful feature is fully explained by interacting organisms and ordinary ecology.

To attribute agency to the forest, the higher-level organization has to earn ownership of the selection. We would need evidence that forest-level state constrains available trajectories, that information distributed across the system alters which trajectory occurs, and that

interventions at the higher level predictably change those selection relations beyond what the strongest local models already explain.

A Forest Under Pressure

Imagine a drought. Some trees close stomata. Roots alter water uptake. Fungal relationships shift. Species differ in tolerance. Herbivores move. Mortality changes canopy structure. Fire risk rises. Seedlings fail or survive. The forest may later recover along several possible trajectories.

It is easy to tell the story as though the forest chose a strategy. But before using that language, we have to show that there really were alternatives at the system level, that the larger state of the forest influenced which alternative occurred, and that the effect cannot be reduced without loss to ordinary physiology, hydrology, ecology, evolution, and local interaction.

If those ordinary mechanisms explain everything equally well, the agency language adds decoration rather than explanation.

Agency Without Free Will

This framework does not require metaphysical free will. Agency here is operational. A system can qualify for an agency hypothesis without possessing language, self-reflection, centralized control, or subjective experience. The point is to identify information-dependent selection that belongs scientifically to the candidate system.

That also means life, agency, intelligence, and consciousness must remain separate. A bacterium can be alive without anything resembling human intelligence. An adaptive system can display agency-like selection without evidence of consciousness. A system could solve problems flexibly without telling us whether there is anything it is like to be that system.

No property higher in that chain can be inferred solely from one below it.

Remove the Agency Language

There is a brutally simple test for anthropomorphic inflation. Remove the words *chooses*, *wants*, *decides*, *knows*, *tries*, and *protects*. Rewrite the explanation using feedback, optimization, adaptive control, network dynamics, ecological interaction, learning, and known physical mechanisms.

If nothing explanatory, predictive, or interventionally useful disappears, the agency interpretation has not earned its keep.

The stronger agency claim becomes meaningful only when removing it costs us something that the mechanistic replacement cannot recover without effectively rebuilding the same system-level selection architecture under another name.

The Thermostat Wins Again

The thermostat matters because it tells us where agency does not begin cheaply. Regulation is not enough. Retained history is not enough. Multiple inputs are not enough. Flexible control is not enough. Even learning is not enough by itself.

The candidate whole must own the relevant selection architecture, and that ownership must survive comparison with stronger ordinary explanations.

At that point a different objection appears. Suppose the forest passes every agency test we can devise. Every tree, fungus, microbe, animal, and molecule still obeys lower-level physics. Perhaps the higher-level forest is only a convenient summary of what the parts are doing.

If we remove the higher level from the science, what do we lose?

That is where we go next.

PART III

THE TEST

A larger-scale explanation matters only if it can do scientific work - and survive the alternatives.

CHAPTER 5

The Level

A driver taps the brakes on a crowded highway. The car behind slows. Then the next. A compression wave grows and begins moving backward through traffic even while every vehicle in the wave is traveling forward. Drivers enter the jam and leave it. The particular cars change. The jam can persist.

No new traffic-jam force has appeared. Every event is implemented through vehicles, roads, drivers, reaction times, and ordinary mechanics. Yet traffic engineers can still measure, model, and sometimes intervene on jam-level patterns.

That is the higher-level problem in its cleanest form. If every event in a larger system is physically implemented by smaller parts, can the larger organization ever become scientifically important for more than convenience?

A World Made Entirely of Parts

FIGURE 08 - APPROVED POSITION

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Imagine an impossible intelligence that knows the position, state, history, and relevant interaction of every tree, root, fungal filament, soil particle, water molecule, microbe, animal, and atmospheric variable in a forest. Would that intelligence need the word *forest*?

Perhaps not. If complete microscopic knowledge were possible, maybe every higher-level regularity could in principle be reconstructed from lower-level physics and initial conditions. That possibility is worth granting because the framework does not depend on metaphysical irreducibility.

But science is not conducted by omniscient observers. We compare models that operate at different levels of description and ask which variables provide robust prediction, intervention, and explanatory compression across changing lower-level implementations.

The fact that temperature is implemented by molecular motion does not make temperature scientifically useless. The fact that a traffic wave is implemented by cars does not make congestion an illusion. Reducibility of implementation does not automatically imply redundancy of organization.

The Pattern and the Pieces

The traffic jam is not independent of the cars. Remove every car and the jam disappears. Yet the jam is not tied to one particular car either. Different vehicles can realize similar congestion dynamics.

That distinction matters. A higher-level variable becomes interesting when relationships survive changes in the particular components realizing them. Body temperature can remain meaningful while individual molecules exchange. An organism can persist while cells turn over. A traffic wave can recur with different drivers.

The higher level may therefore capture patterns that are stable across many lower-level configurations. Stability across implementation is not proof of causation, but it is one reason higher-level models can generalize.

When the Higher Level Predicts

Return to the droughting forest. Build the strongest lower-level model we reasonably can from local soil, hydrology, weather, physiology, species composition, fungal relationships, topography, and other known mechanisms. Then build a model that also includes a candidate forest-level state.

The higher-level model does not win because it sounds holistic. It wins only if it predicts prospectively better under a fair comparison. That improvement is what we can call **Prediction Advantage**: the difference in predictive performance between the candidate higher-level model and the strongest relevant baseline.

Prediction Advantage needs a predictor, a target, an error measure, and a baseline defined in advance. Retrospective fit is not enough. Adding more variables until the model wins is not enough. The comparison has to survive withheld data or new observations.

Even then, prediction is not causation. Clouds can predict rain without causing it. A rooster can predict sunrise without controlling the Sun. A higher-level forest state might merely summarize lower-level conditions we have not modeled well enough.

So the skeptic gets better tools.

Give the Skeptic Better Tools

The local model should improve. Add better hydrology. Better soil measurements. Better fungal data. Better physiological state. Better topography. Better disturbance history. If the supposed higher-level advantage disappears, that is exactly what should happen.

A fair test does not freeze the skeptic at a weak baseline so the favored model can keep winning. The strongest ordinary explanation gets resources too.

If the higher-level variable continues to add predictive value after the rival improves, the case strengthens. The question then becomes whether the higher level merely forecasts or whether interventions defined at that level reveal stable causal structure.

Touching the Whole Without Inventing a Force

A higher-level intervention does not mean a mysterious force descends from above. Consider room temperature. It is a macro-variable realized by enormous numbers of molecular interactions. Yet a thermostat can measure it and participate in a control mechanism that changes furnace activity, fuel use, airflow, and the future molecular state of the room.

Nothing supernatural happened. The intervention relation is physically implemented all the way down.

For a forest, we might compare different lower-level manipulations that produce a similar higher-level state. If interventions implemented through different mechanisms lead to reproducible outcomes that are better organized by the same forest-level variable than by each implementation separately, the higher-level description gains causal credibility.

This is a **cross-realization** test. The higher-level relation matters because it survives differences in the microscopic route used to realize it.

Constraints and the Shape of Possibility

Higher levels often matter by constraining what lower-level components can do. A membrane shapes molecular exchange. Railway tracks constrain the paths available to trains. A river channel constrains water flow. Computer architecture constrains which operations are easy or possible for lower-level components.

Living systems add another feature: some constraints are actively maintained. Membranes are repaired. Regulatory structures are rebuilt. Development restores organization after disturbance. But the existence of a constraint does not automatically mean the system owns its maintenance, and ownership does not automatically mean agency.

The causal question is whether the higher-level organization captures stable constraint relations that disappear when we treat components as independent or when we replace the higher-level variable with a weaker

description.

Remove the Whole

Now perform the conceptual removal test. Explain the drought using tree physiology, soil, fungi, hydrology, weather, evolutionary history, and local ecological interactions. If those descriptions reproduce the same prediction, generalization, and intervention relations without needing a forest-level variable, then the higher level has not demonstrated nonredundant work for that property.

But if removing the forest-level state degrades prediction, destroys generalization across different component configurations, or obscures intervention relationships that recur across implementations, the higher level has earned scientific standing.

That still does not mean the forest is an agent. Higher-level causation and agency are separate claims.

Three Possible Endings

The investigation now has several legitimate outcomes.

In **Model A**, lower-level mechanisms account for the evidence without a meaningful higher-level advantage. That is not a failed experiment. It tells us the candidate whole was unnecessary for the property tested.

In **Model B**, a higher organizational level repeatedly improves prediction, generalization, constraint description, or intervention while stronger agency tests fail. The whole matters scientifically without becoming a mind or organism in the stronger sense.

Model B is a discovery, not a consolation prize.

In **Model C**, a particular stronger property survives its dedicated tests. That classification must be property-specific. A system might earn individuality without agency, agency without intelligence, or intelligence without any justified consciousness claim.

There may be a great deal of scientific territory between a pile and a person.

The Forest May Be Enough

The project does not need to race upward. If forest-level organization proves scientifically nonredundant, that result matters even if Earth never earns individuality and no planet ever chooses anything.

The hierarchy may stop, branch, overlap, or change properties at different scales. Larger does not automatically mean more integrated. Integration does not automatically mean agency. There may be no single ladder at all.

Earth Returns

Planetary science already studies real global coupling among atmosphere, ocean, rock, ice, chemistry, life, and energy flow. The Gaia tradition helped focus scientific attention on feedback and planetary regulation. Feedback, however, is not by itself life or agency.

The opposite mistake would be to assume that because all planetary processes are physically implemented through ordinary mechanisms, planetary organization can never become scientifically meaningful. That claim also has to be tested rather than assumed.

We can make Earth compete. Does a planetary boundary outperform alternatives for the property under study? Do planetary variables improve prospective prediction beyond strong subsystem models? Do natural quasi-experiments reveal stable global constraint relations? Do different physical mechanisms realize similar higher-level consequences?

If not, Earth loses that test. If yes, it earns another question, not a title.

No Throne at the Top

Macroscopic Life is not searching for the rung above humanity. Nature need not be arranged as a ladder ending in a largest organism. Properties can emerge, disappear, overlap, or become scientifically useful at different scales.

A forest may possess stronger integration than a continent for one property. A city may display sophisticated adaptive coordination while lacking biological individuality. A planet may show robust regulation while failing agency. The landscape of organization may be uneven.

That is a richer possibility than a hierarchy.

The Check Comes Due

We have now moved from observer limitation to disciplined search, from search to boundary, from boundary to agency, and from agency to higher-level causal standing. Every step has made the original question harder to answer casually.

The method, however, is still not evidence. A beautiful framework can organize ignorance just as easily as it can organize discovery.

That is where Macroscopic Life either becomes science or remains an idea.

Now we build the experiment.

CHAPTER 6

The Experiment

There is a forest somewhere waiting to disappoint us, and that is the forest I want to study.

Not the forest that produces the most dramatic pattern. Not the forest that looks most like a body from the air. Not the forest that gives us the best story. I want the one that makes the hypothesis fight for every inch of interpretation.

Science becomes serious when reality is allowed to make us lose.

Before Anything Happens

FIGURE 09 - APPROVED POSITION

Structural proof placeholder. Insert only the frozen publication-authority visual at final composition; do not invent or substitute artwork.

We begin before the drought. The candidate property is not consciousness, intelligence, or even agency. Those claims are too strong. We begin with something narrower: integrated regulation at the forest level.

We define the candidate boundary in advance. We record the variables that count as forest-level state. We specify the strongest ordinary alternatives we can build from weather, soil, hydrology, physiology, species composition, fungi, topography, and known ecological interaction. We decide what predictions would distinguish the higher-level model from those rivals.

Most importantly, we freeze the boundary before seeing the result.

The First Drought

The drought begins ordinarily. Rainfall declines. Soil dries. Tree water stress rises. Stomata close. Growth slows. Species respond differently. Some regions lose canopy while others remain comparatively stable.

Then a larger-scale pattern appears. A forest-level state measured before the strongest local stress predicts which regions later reorganize and which recover. The result survives initial controls and improves prediction beyond the first version of the local model.

The tempting sentence writes itself: *the forest redistributed resources to protect itself*.

We do not write that sentence yet.

The data justify a smaller claim: a candidate higher-level state improved prospective prediction. That earns another test.

The Ordinary Explanations Return

Now the skeptics improve the rival. Better hydrology explains part of the effect. Soil structure explains another part. Fungal relationships absorb some of the apparent coordination. Weather history eliminates a relationship that originally looked extraordinary.

The mystery shrinks.

Good.

An interpretation becoming less dramatic while the evidence becomes more precise is not a failure of science. It is what science is for.

After the stronger ordinary models are included, one smaller forest-level contribution remains. It is not enough for agency. It is enough to justify replication.

The Pattern Leaves Its Discoverers

A second research group receives the preregistered model and prediction. They use different analysts and independent data. Some of the original relationships weaken. One disappears. The central higher-level effect survives, though smaller than first reported.

That matters because the pattern is no longer protected by the judgment of the people who found it.

It survived its discoverers.

Another Forest

The next test occurs in a forest with different species, soils, hydrology, disturbance history, and climate. We should not expect identical component behavior. The question is whether some organizational relationship survives across the different implementation.

Again the higher-level state predicts part of the response. The particular trees and pathways differ. The organizational role is similar.

Now the evidence begins to support something more interesting than one local correlation. A relationship may persist across different lower-level realizations.

We Drew the Wrong Forest

Then the boundary fails.

A major disturbance propagates in a way the original forest boundary predicts badly. Relationships across one edge are stronger than relationships within another. When researchers compare alternative boundaries, a vegetation-water-soil region extending beyond the original forest map performs better.

The temptation is obvious: redraw the forest and claim success.

We do not.

The original prediction failed. That failure stays in the record. The newly discovered boundary becomes a new candidate and must face a prospective test of its own.

In the next season, some predictions based on the new boundary succeed and some fail. Confidence falls from where the first exciting result might have pushed it. The candidate becomes narrower.

That is probably what a real science of unfamiliar organization would look like: not a procession of astonishing confirmations, but a narrowing process in which the most dramatic interpretation repeatedly loses territory while some smaller and harder result refuses to disappear.

When the Past Remains Present

After several disturbances, the forest responds differently to droughts of similar physical severity. It is tempting to say the forest remembers.

So we ask what physically remains. Species composition changed. Root architecture shifted. Soil structure was altered. Fungal communities reorganized. Nutrient stores moved. Canopy gaps changed light and moisture. Those retained states alter later response.

The word *memory* is useful only if it points to this causal persistence. The mechanism is more interesting than the metaphor because it gives us something to measure, perturb, and compare.

Then We Ask About Choice

By now the candidate has survived several lower-level attacks. It has a more defensible boundary, organizational persistence across component turnover, a higher-level contribution to prediction, and functional history that alters future response.

Now we ask the question Chapter 4 prepared us for: does the forest select among available trajectories in an information-dependent manner that belongs to the forest as a whole?

The answer is not strong enough.

Local physiology, ecological interaction, hydrology, evolutionary adaptation, and distributed organism-level behavior can still account for the apparent selection architecture. We cannot demonstrate that the forest-level whole owns the choice in the stronger sense required by the

agency criterion.

The agency claim fails.

We leave it failed.

The Discovery We Were Not Looking For

For a moment that can feel disappointing. The original imagination wanted a forest that chose, remembered, perhaps even possessed some primitive mind-like unity.

But look at what remains after the stronger claim dies. A defensible higher-level boundary outperformed competing descriptions for some questions. Organizational state contributed to prospective prediction. Some relationships persisted across different lower-level realizations. Retained system history altered later response. Interventions and disturbances revealed constraints that were easier to describe at the higher level.

The forest is neither a mere pile nor a giant creature.

It may occupy scientific territory our ordinary categories handle poorly.

That is Model B.

If so, Macroscopic Life could become scientifically important even if no forest ever thinks and no planet ever chooses.

And Sometimes Nothing Survives

Now imagine another candidate forest. The original pattern disappears when hydrology improves. The boundary proves arbitrary. Independent replication fails. No higher-level variable beats the strengthened lower-level model. Every apparent system-level effect dissolves into known component interactions.

Stop.

Do not move automatically to a larger forest, hidden channel, slower timescale, continent, or planet and pretend the original hypothesis remains alive. Archive the negative result. It tells us where the detector

found nothing beyond ordinary organization.

A map of failures is part of the science.

A Detector That Can Say No

Macroscopic Life should therefore produce profiles, not one magic LIFE score. A system may show strong boundary evidence and weak regulation. Another may show functional memory but no agency. Another may exhibit higher-level prediction without individuality. The properties can separate.

That is desirable. A single score would encourage us to add weak signals together until an exciting label appears. The detector should instead ask which specific property has earned support, at what scale, under what boundary, against which rival, and with what evidence level.

Calibrate Before Going Exotic

Before trusting the detector on Earth or the cosmos, test it on cases we understand better. A thermostat should fail strong agency. A hurricane should show organization and persistence without automatically becoming an individual. A known animal agent should satisfy agency criteria that the thermostat fails.

Calibration is not decorative. If the detector cannot recover obvious differences among known cases, its verdict on unfamiliar cases is meaningless.

AI can help search Scale-Space and generate candidate relationships, but it cannot confirm its own discoveries. And one experiment that bears on boundary, memory, and regulation remains one evidence unit, not three independent confirmations.

The Hardest Laboratory

Only after calibration do we return to Earth. Planetary science presents a special problem because we have one Earth. We cannot create a population of planets with controlled experimental treatments.

Unique systems can still be studied, but the burden shifts toward triangulation. Independent measurements, different natural experiments, multiple model classes, cross-mechanism convergence, temporal replication, and successful prospective predictions become especially important.

Planetary agency would require extraordinary convergence: a defensible boundary, integrated persistence, functional information use, genuine alternative selection, system-level ownership, predictive advantage, stable intervention relations, independent confirmation, and repeated failure of strong ordinary rivals.

Nothing in Chapters 1–6 has established that result.

And Beyond Earth

The same restraint applies as scale expands. A solar system is not an individual merely because its parts orbit together. A galaxy is not alive because it has persistent structure. The cosmic web is not a nervous system because images of the two can resemble one another. The universe is not conscious merely because information exists within it.

Bigger is not deeper. Larger is not more alive.

If large-scale organization exists at those levels, it has to earn the relevant property under tests appropriate to the system. Visual wonder can start the question. It cannot finish it.

The Framework Must Be Vulnerable Too

The candidate systems are not the only things on trial. The framework is also a hypothesis about how to detect unfamiliar organization.

If it repeatedly classifies known non-agents as agents, it is badly calibrated. If results depend on arbitrary representation choices, confidence falls. If stronger rival models continually erase higher-level effects, the empirical domain shrinks. If criteria move whenever a favored candidate fails, the framework has stopped being scientific.

Macroscopic Life has to be capable of losing not only individual experiments, but its own research program.

What Would Change Our Minds?

The cleanest version of the investigation may require two researchers who want different outcomes. One thinks the higher-level interpretation is plausible. The other thinks ordinary mechanisms will explain everything.

Before the next experiment, each writes down what would change their mind. The advocate names the failure that would weaken the higher-level claim. The skeptic names the result that would force the higher level into the explanation.

Then both stop moving the target.

Nature referees.

The Original Question

We began with a question I still find difficult to shake:

What if we are living inside something alive and simply lack the scale of perception required to recognize it?

After everything we have done, I still love the question. But I no longer trust it in the way I did at the beginning.

At first the question felt powerful because it was enormous. Now it is powerful only if it can be broken into smaller obligations: What boundary? What variables? What timescale? What alternatives? What information? What prediction? What intervention? What rival explanation? What failure would make us stop?

That transformation—from awe to obligation—is the beginning of a science.

We do not know whether the strongest forms of Macroscopic Life exist. We do know more precisely what discovering them would mean, and we know that many scientifically interesting outcomes lie below the most dramatic one.

Ignorance does not require belief, and it does not require disbelief. It requires a detector capable of surprising both.

Wonder opens the door. Measurement decides what walks through it.

The next step is not belief.

It is experiment.

PART III — THE PROPERTIES

PART IV

THE FUNCTIONS

The familiar verbs of life become dangerous when
resemblance is mistaken for evidence.

CHAPTER 7

The Signal

Before dawn, a forest can look almost motionless. The air is cool. Leaves hang still. Soil temperatures change slowly. The trunks seem fixed in place, and if you were standing among them with nothing but your eyes and ears, you might reasonably conclude that very little is happening.

Imagine an experiment we have not yet run. Wire that forest for observation. Soil probes measure moisture at different depths. Sensors track sap flow, leaf temperature, humidity, light, wind, pressure, electrical potential, volatile compounds, and minute changes in chemistry. Cameras watch the canopy. Other instruments record changes too subtle, too slow, too fast, or too distributed for unaided human attention. Data arrive every second, every minute, every hour, across hundreds of locations.

At 4:17 in the morning, something changes. A cluster of sensors on one side of the forest shifts first. Minutes later, another region changes, then another. Electrical measurements alter. Sap flow changes. Chemical concentrations move. A pattern travels across the map. If we saw something similar inside a nervous system, we might call it signaling. If we saw it in an endocrine system, we might ask what information was being carried. In a forest, the temptation is enormous: maybe one part has detected something and the rest is responding. Maybe the pattern is a message. Maybe the forest is talking.

Or maybe nothing has spoken to anything at all.

A pattern can cross a system without the system sending a message.

The Easiest Explanation

Suppose the changes line up beautifully. The western trees alter first, then the center, then the eastern edge. A machine-learning system predicts the order. The correlations survive repeated measurements. There is structure in the data. Then we place a weather station beyond the forest and discover a pressure front moving across the region.

The observed sequence no longer requires tree-to-tree signaling. A common physical driver is sufficient to generate the pattern we saw. Additional signaling might still occur somewhere in the system, but this observation no longer demonstrates it. Nothing about the measurements was false. The correlation was real. The sequence was real. The prediction was real. Only the explanation changed.

Imagine hundreds of automated lights along a highway, each switching state when the same external sensor network reports that darkness has crossed a threshold. A map could show a wave of activation. A model could predict which lights change next. Yet the lights need not be sending messages to one another. They participate in a shared pattern because they share a driver.

The same caution applies everywhere. Birds may begin calling because dawn arrives. Plants may change physiology because humidity falls. Animals may move because a storm approaches. Markets may shift because participants respond to the same announcement. Similar behavior can emerge from common causes without direct communication among the responding parts. Correlation tells us where to look; it does not tell us what moved.

Even propagation is not enough. Drop a stone into a pond and a wave travels outward. Energy and a physical disturbance move through the water and affect distant points. That alone does not mean the pond is transmitting a message to itself. Propagation is not communication. The distinction becomes harder when the event is electrical, chemical, biological, or exquisitely patterned, because interesting patterns invite

interesting verbs.

When Signaling Is Not a Metaphor

Biology gives us cases in which signaling is experimentally tractable rather than merely poetic. Plants are useful because their internal coordination does not resemble the animal nervous system most people carry as their intuitive model. Local stimuli can trigger long-distance electrical and calcium dynamics associated with systemic responses elsewhere in an already established organism.[1] The point is not that plants contain tiny hidden brains. A physical event occurs in one region, measurable biological processes propagate, distant tissues change state, and those changes can participate in regulation.

For this book, we can use a practical scaffold: Source → Signal → Channel → Receiver → State Change → Functional Consequence. That is an operational tool for asking questions, not a claim that every scientific discipline uses one universal definition of communication.

Bacteria provide another positive control. In quorum sensing, cells produce and respond to signaling molecules whose detection alters gene regulation and behavior. Mechanisms differ among organisms, but production, detection, regulatory circuitry, and downstream consequences can be manipulated experimentally.[2] The word communication is not doing the scientific work. The mechanism is. Alter signal production, reception, or downstream circuitry and ask what changes.

The danger begins when we carry the vocabulary upward without carrying the evidence with it. A forest contains organisms that signal. It contains fungi, microbes, insects, plants, animals, chemical gradients, water movement, electrical changes, volatile compounds, and innumerable interacting processes. Communication can therefore occur inside a forest without the forest itself being the communicator. A city provides the same warning: people, computers, institutions, and machines exchange information constantly, but their communication does not by itself establish the city as one communicating individual.

A function can exist inside a system before the system owns the function.

The Resonance Temptation

Some words invite larger interpretations almost automatically. Frequency and resonance are among the strongest because oscillation is everywhere. Hearts beat. Neural systems generate rhythms. Circadian systems cycle. Chemical reactions can oscillate. Mechanical structures resonate. Electrical circuits have characteristic frequencies. Populations fluctuate. Coupled systems can synchronize or become entrained.

These phenomena need to be separated before we ask what they mean. Resonance concerns the response of a system to forcing, often with enhanced response near characteristic dynamics. Synchronization and entrainment concern relationships among oscillators or between an oscillator and a driver. They can interact, but they are not synonyms, and neither automatically establishes communication.

Several mechanical oscillators coupled through a shared support can move toward coordinated behavior because each alters the physical conditions experienced by the others. A biological rhythm can become entrained by an environmental cycle. None of that requires symbols, meaning, intention, agency, or consciousness. Synchronization is a dynamic relation before it is a communication claim.

Frequency is a measurement, not an explanation.

If an oscillation is proposed as an information-bearing channel, the questions become concrete. What is oscillating? What is the coupling mechanism? Can the pattern be modulated? Do distinguishable signal states produce distinguishable downstream responses? Can a responsive subsystem be identified? Does a specific perturbation alter the effect while appropriate controls remain intact? Can a simpler physical mechanism explain the same observation?

That is where resonance becomes useful to Macroscopic Life. An oscillation does not become a message because it is beautiful. But if controlled changes in its pattern are selectively used by a regulatory architecture and reliably change what the system does, then the

dynamics may participate in information processing. The mechanism, not the metaphor, has to earn that interpretation.

[FIGURE 7 — Local mechanism and higher-level function. Frozen V3. Scientific brakes: A mechanism can be entirely local while the function it implements belongs to a larger organizational level. FREQUENCY IS A MEASUREMENT, NOT AN EXPLANATION. A SHARED WORD IS NOT A SHARED MECHANISM.]

The Network Beneath the Story

Few modern scientific stories show the danger of language inflation better than the idea popularly called the “wood-wide web.” The underlying biology is genuinely remarkable. Mycorrhizal fungi form associations with plant roots, and fungal structures can connect plants. Field experiments have measured carbon transfer between ectomycorrhizal tree species under particular conditions.[3] That is real evidence worth taking seriously. It is not evidence for every larger story told about the network.

Popular accounts have sometimes taken much larger steps. Trees are described as talking, warning one another, feeding offspring, or participating in a forest-scale network analogous to a brain or social system. Some narrower hypotheses beneath those descriptions are testable, but the stronger the verb, the more evidence it requires. A critical 2023 assessment found that evidence for several broad claims about common mycorrhizal networks in forests was too variable, limited, or confounded to support the generalizations often made about them, and reported no peer-reviewed published evidence for the specific claim that mature trees preferentially send resources and defense signals to offspring through CMNs.[4]

That does not mean the networks are unreal or the research program has failed. It means several questions must remain separate. A fungal connection can be real without carrying a biologically used message. Material can move without establishing encoded communication. An interaction can benefit a recipient without demonstrating altruism. Genuine communication between organisms can occur without creating

a communicating forest-level individual.

If two trees change after one is damaged, perhaps a signal moved between them. Perhaps both responded to an airborne compound, shared pathogen, soil disturbance, common environmental change, or another pathway. If labeled carbon later appears elsewhere, we still need to identify the route, magnitude, mechanism, functional consequence, and plausible alternatives. The attractive story is not the enemy; it is the hypothesis generator. Reality decides how much survives.

The same problem has now appeared twice. With resonance, detecting a physical relationship was easier than establishing that the relationship carried information. With a biological network, detecting a connection is easier than establishing what role the connection plays for the larger system. Measurement gives us the relation. Experiment has to earn the function.

Information for Whom?

The word information creates another trap because scientists use it in several legitimate but different ways. Suppose humidity at dawn strongly predicts a later biological state. In a statistical sense, one variable can reduce uncertainty about another. That formal relationship does not depend on a scientist being present. But a scientist may exploit the relationship for prediction long before we have shown that the biological system itself uses the variable in a regulatory process.

That is the intended meaning of a useful shorthand:

Information can exist for the scientist before it exists for the system.

Modern analysis can find predictive structure across enormous collections of variables. Those relationships can be scientifically valuable even when no subsystem uses them as signals. Prediction is therefore not enough to establish functional information.

Intervention helps, but causal influence is not yet functional information either. A falling branch can alter another branch without serving as a signal for it. To make the stronger claim, we should look for

state-dependent or selective use within an identifiable regulatory architecture. Manipulate the candidate signal while controlling alternatives. Ask whether distinguishable signal states produce reproducibly different downstream responses and whether those responses contribute to a measurable function.

Even then, meaning requires restraint. Scientists may label a signal “danger,” “density,” “injury,” or “hunger” because those words summarize conditions associated with it. The system does not need a human concept of danger for a physical signal to participate in regulation. Statistical information, functional information, and semantic interpretation should not be collapsed into one word.

There May Be No Control Room

Human beings often expect important organization to have a center. Governments have capitals. Companies have headquarters. Machines often have controllers. That familiarity can make a centralized command point feel like a requirement for large-scale organization. Biology complicates the intuition.

Ant colonies can produce organized foraging, construction, defense, task allocation, and network maintenance through interactions among individuals. Deborah Gordon's review of collective behavior across ant systems emphasizes feedback, local interactions, environmental dynamics, and variation in how information regulating behavior is distributed or spatially centralized.[5] No single worker needs to possess a complete plan of the colony for collective behavior to emerge.

Development offers another example. Cells contain and respond to molecular, positional, mechanical, and regulatory information, yet no individual cell contains a miniature finished organism directing construction from above. Reliable anatomy can arise through distributed interactions among changing cellular states. Engineered networks and decentralized control systems can likewise solve problems without one component representing every state of the whole.

A system-level function does not require a system-level control room.

The rule cuts both ways. The absence of a control room cannot dismiss higher-level organization, but neither can decentralization prove it. Distributed dynamics occur in systems we would never call organisms. What matters is whether the collective function belongs to an independently justified higher-level unit.

Who Owns the Message?

Imagine two neighboring organisms exchanging a chemical signal. The molecules are identifiable, responsive machinery can be found, and behavior changes. Who owns the function?

If each organism uses the interaction in service of its own independent regulation, communication can occur within a larger ecological network without belonging to that network as an integrated individual. The stronger possibility begins only after the larger unit has earned an independent boundary. Then we can ask whether selectively disrupting the exchange degrades a stable higher-level variable or function, whether the effect survives turnover or substitution of particular components, and whether a higher-level model improves prospective prediction or intervention against fair lower-level rivals given comparable information and modeling resources.

This is the Macroscopic Life Ownership Test, and it is project synthesis rather than a borrowed consensus definition: Does communication merely occur inside the proposed system, or does its function belong to the organization of the proposed system?

The test cannot bootstrap itself. We cannot draw a circle around a forest because signals occur within it, call the enclosed region a communicator, and then use the communication as proof that the circle was correct. The boundary must be earned independently. Ordinary mechanisms must remain free to win. The higher-level description must do scientific work rather than merely provide a more dramatic noun.

Return to the Forest

Return to our imagined forest before dawn. The mysterious wave still crosses the sensor network, but now the experiment has three jobs.

First, establish the phenomenon. Reproduce the pattern, identify what physically changes, measure plausible carriers, and specify in advance which signal states should distinguish which conditions. At the same time, give weather, light, soil moisture, temperature, hydrology, topography, seasonality, pathogens, diffusion, fungal structure, and measurement artifacts a fair chance to explain the observation. Discovery data do not get to confirm the pattern that discovered them.

Second, attack the mechanism. Perturb the proposed pathway with interventions designed to be as specific as possible, alongside sham controls and measurements of collateral damage. Where feasible, use more than one independent way of changing the pathway. A severed connection may alter water, tissue damage, chemistry, and stress at once; disappearance of an effect after such a manipulation is not automatically evidence for the proposed signal. Convergent perturbations, selective downstream responses, and recovery of function can strengthen the causal case without turning any single intervention into proof.

Third, test the level. Compare competing boundaries. Perhaps the meaningful organization follows hydrology rather than the visual edge of the forest. Perhaps a fungal topology matters. Perhaps no higher-level boundary survives. Then ask prospectively whether treating the candidate whole as an organized information-using unit improves prediction or intervention over a strong lower-level rival given fair access to relevant data, model capacity, and experimental resources. Finally, take the prediction somewhere else: another site, season, disturbance, realization, or research team.

The point is not to protect the signal. The point is to give it every reasonable opportunity to disappear.

What Could Survive

Several outcomes would count as scientific progress. In one, the apparent communication collapses under common-driver or local-mechanism controls. The pattern remains real and perhaps useful, but the stronger interpretation is unnecessary. That is consistent with the lower-level-sufficiency outcome already defined in this book.

In another, genuine communication among components survives. Pathways can be manipulated, responses are selective, and the interactions have measurable functions, but the higher-level ownership claim fails. The forest contains communication without becoming one communicator. That is still a discovery. Real organization can exist below the stronger claim we hoped to test.

A stronger result would require much more: communication survives common-driver controls; its disruption selectively degrades an independently justified higher-level organization; competing boundaries perform worse; the result survives component turnover or alternative realizations where relevant; the higher-level account provides comparative prediction or intervention advantage; and replication succeeds. Only then should the canonical Model A/B/C framework be applied according to the property actually demonstrated. Chapter 7 does not redefine those models.

Even an extraordinary communication result would leave the property ladder intact. Communication is not agency. Agency is not intelligence. Intelligence is not consciousness. Evidence for one property neither establishes nor automatically excludes another; each claim must face its own operational tests.

The Thing That Remains

Everything we have described can happen in the present: a signal appears, another part of a system responds, and the interaction ends. Living systems can do something more difficult. Their next response can depend on what happened before because the earlier event changed something physical that persisted.

A scar persists. A crater persists. Neither persistence nor physical history alone is memory. But memory cannot exist without some trace of the past remaining capable of re-entering future causal processing. A retained trace is necessary; whether it becomes functional, selective, or adaptive memory is a harder question.

Suppose our forest experiences a disturbance. The immediate signal disappears. Months later the disturbance returns, and the response is different. Did an earlier physical change alter what happened next? Does that retained influence belong to individual organisms, to local ecological history, or to some larger organization?

The signal has carried us as far as it can. Now we need to know whether yesterday can remain inside tomorrow.

Can the whole remember?

CHAPTER 8

The Memory

There is a scar on your hand that has outlived the moment that made it.

Maybe it came from a knife that slipped while you were cooking, a piece of broken glass, a fall against something sharp, or an accident you have almost forgotten. The event lasted seconds. The wound lasted days or weeks. Years later, the scar may still be there. Run a finger across it and the past is no longer abstract. Something that happened then is physically present now.

The tissue is different because of its history. Its structure changed. It may feel different or behave differently under later strain. The original event is gone, but a consequence remains embedded in matter.

Does the scar remember the knife?

Probably not in the ordinary sense of memory. Yet if memory means only a persistent physical change caused by the past, the scar qualifies. So does a crater. So does bent metal. Scientific fields can legitimately use *memory* for some retained material states, and nothing here requires policing that language. Our problem is narrower: persistence alone cannot establish the stronger functional and higher-level memory claims we will eventually ask an unfamiliar system to earn.

TRACE ≠ MEMORY.

The distinction seems obvious until we try to say exactly what more is required.

Everything Has a Past

FIGURE 10 - APPROVED POSITION

Structural proof placeholder. Insert only the frozen publication-authority visual at final composition; do not invent or substitute artwork.

Almost nothing in the physical world arrives in the present without history. A river channel carries the work of earlier floods. Soil reflects what grew, burned, decomposed, washed through, or was deposited there before. A magnet can occupy a state that depends on fields it experienced earlier. If all such history dependence counts equally as memory, the word begins to collapse into a broader fact: present states have causes, and some of those causes occurred in the past.

Start instead with a trace. An earlier event changes something that remains detectable. Retention is stronger: the changed state lasts long enough to matter to later dynamics. A changed soil layer may alter where water travels during the next storm; a deformed material may respond differently to the next load. The past is affecting what happens later through something physically retained in the present, but the scar has already warned us that this is not enough for the strongest claim.

The past leaving a mark is not yet the system using the past.

The Machine That Looks Like It Remembers

Consider a thermostat that does not switch on and off at exactly the same temperature. Suppose the heat turns on when the room cools below one threshold but does not turn off until the temperature rises above another. At a temperature between those thresholds, knowing the temperature alone may not tell you whether the heat is running. You also need to know which direction the system came from. Its current behavior depends on its path.

Physics and biology contain far more interesting versions of this general phenomenon. Magnetic materials can exhibit hysteresis. Regulatory networks can have stable alternative states. Biochemical feedback can preserve a state after the original trigger has disappeared, and biological literature legitimately describes some persistent bistable

or hysteretic states as forms of cellular memory.[1] It would therefore be a mistake to declare that hysteresis simply is not memory. The useful point is that history dependence, even when real and mechanistically important, does not by itself establish the stronger claim we will eventually ask a larger system to earn.

For Macroscopic Life, it helps to arrange the evidence as a ladder: **Trace → Retention → Functional Memory → Selective Memory → Adaptive Memory**. This is a project framework for comparing evidence, not a universal taxonomy or a claim that nature divides cleanly into five boxes. A retained difference becomes more interesting when it later participates causally in a specified response; stronger still when different histories remain discriminable; stronger again when the history-dependent response improves an independently specified outcome over a stated time horizon. In that last project category, *adaptive* means function-improving under the tested conditions. It does not by itself mean evolutionary adaptation, and it does not establish learning.

The point is not to win a vocabulary dispute. It is to prevent the strongest interpretation from arriving before the evidence.

Memory Without Remembering

The easiest way to escape the idea that memory requires conscious recollection is already inside your body.

When the adaptive immune system encounters an antigen, the response can alter populations and states of B cells, T cells, and plasma cells, together with antibodies and other retained immune effects. After the immediate episode ends, some of those consequences persist. A later encounter can therefore produce a response different from that of an antigen-inexperienced system.[2] There is no little archive where the immune system keeps a picture of the first encounter. The later difference is distributed across physical populations, molecules, and states.

Some bacteria and archaea with CRISPR-Cas adaptive immune systems make the point stranger still. Sequence derived from invading

genetic material can be incorporated into CRISPR arrays. That retained sequence can later contribute to sequence-specific defense when matching genetic material appears again.[3] The organism does not need an autobiographical recollection of infection. A prior encounter has altered a physical system so that a later matching challenge can be handled differently.

That is worth lingering over. Memory, in a scientifically useful biological sense, can exist without a brain and without conscious remembering. The first event can disappear from the environment while a physical difference remains available to change what happens during the second.

But the expansion of memory beyond neurons cannot become permission to call every persistent biological change memory, much less to conclude that an ecosystem remembers. The lesson is more disciplined: the past can remain causally available in many substrates. We now have to find out what remains, how it matters later, and who—if anyone above the component level—owns the function.

Something Must Remain

Past events influence later events through consequences that persist or propagate. The carrier need not resemble a written record. It might be a changed cell population, regulatory feedback state, receptor distribution, molecular modification, structural difference, network configuration, chemical concentration, bioelectric state, community composition, or a pattern continually reconstructed while particular components turn over. Different systems can retain history in very different physical forms.

What we cannot accept as an explanation is *the information is stored somewhere* when neither the somewhere nor the mechanism can be defended. If a past condition matters later, some physical difference must survive or be reconstructed. For the stronger causal claim, we should be able to identify a later condition under which changing that retained state changes a specified downstream response while relevant rivals are controlled. That is what *re-entry* will mean here: not a

mysterious return of the past, but an intervention-sensitive role for a physically retained difference.

FIGURE 10 — THE PAST MATTERS LATER ONLY IF SOMETHING PHYSICAL PERSISTS AND CAN RE-ENTER CAUSAL PROCESSING.

This creates a subtler problem. Memory is not an alternative to present physical state. Whatever memory exists must be physically instantiated now. If every causally relevant present variable could be described perfectly, the physical effects of history would already be somewhere in that description.

The scientific question is therefore whether organizing part of the present as a retained consequence of history gives us useful causal structure. Does that description compress what would otherwise require a sprawling inventory of states? Does it identify an intervention target? Does it improve prospective prediction against a fair rival? The function being evaluated cannot be invented after the result appears; the relevant response and time horizon have to be specified independently, prospectively where feasible.

Memory is physically present. The question is whether history provides a useful causal organization of that present state.

Selectivity can make the case stronger. Suppose prior exposure A and prior exposure B leave distinguishable retained states, and a later probe produces reproducibly different responses. Now the system carries more than a generic consequence that *something happened*. Some differences among past conditions remain causally discriminable. Immune specificity and CRISPR sequence specificity give us clean intuitions for this, but selectivity is not being declared a universal requirement for every legitimate scientific use of memory. It is an especially useful discriminator when we are trying to separate organized history dependence from damage, depletion, aging, drift, or wear.

The Better Second Response

Plants complicate the story. Prior environmental stress can alter later plant responses, and researchers use terms such as stress priming and stress memory for some of these effects. Yet the phenomenon and the proposed storage mechanism have to be separated. Evidence that prior stress changes a later response can be strong while evidence that a particular chromatin state, methylation pattern, or other epigenetic mechanism carries the relevant memory remains case-dependent.[4] Within-generation priming also does not establish transgenerational memory. Inheritance across generations requires its own evidence and controls.

This is exactly the kind of uncertainty the framework must tolerate. We can observe a history-dependent response without pretending we already know what stores it.

It is also where the language of learning begins trying to enter the room. Experiments with the slime mould *Physarum polycephalum* have been interpreted as evidence that some hallmarks of habituation can occur without neurons.[5] The possibility is fascinating, but a declining response under repeated exposure can also reflect adaptation, fatigue, depletion, damage, or changing conditions. A changed second response is evidence to investigate; it is not, by itself, a verdict that learning occurred.

A better second response is not automatically learning. Memory and learning are not synonyms. For now we need only establish that physical history can persist and later matter functionally. Learning asks more.

The Population That Changed

Imagine a population of microbes exposed to severe stress. Most susceptible individuals die. A smaller resistant fraction survives, the population grows again, and the next exposure produces a dramatically better population-level response. The temptation is immediate: it learned.

But several histories can produce that same appearance. Pre-existing resistant types may simply have been selected. Exposure may have

induced persistent physiological states within surviving lineages. New heritable changes may have arisen. Or the main difference may be demographic: the population now contains different proportions of the types it already had. The second response can change profoundly without any individual having learned in the behavioral sense.

Yet composition is not fake history. The changed population is a physical legacy of the first event, and at the population level that distribution can help determine what happens next. The question is not whether the effect is real.

Who owns the effect?

A compact population-level variable may predict the second response better than a cumbersome list of individual histories, but compression alone does not establish ownership. If independent component states plus the shared environment reconstruct the effect completely, a stronger population-memory interpretation may add little. If interactions among components produce stable predictive and intervention structure at an independently justified population boundary, the higher-level description becomes more useful. Prediction can earn scientific utility. It cannot, by itself, earn individuality.

This is where memory stops being only a question about time and becomes a question about level.

The Material Can Change

Memory also cannot be tied to preservation of the exact material that first experienced an event. Biological systems replace molecules, cells, and members while some regulatory differences persist because later dynamics recreate or maintain them. The continuity that matters need not be identity of every original part.

The deeper invariant is the causally relevant difference carried forward through the changing material.

MEMORY NEED NOT PRESERVE THE MATERIAL. IT MUST PRESERVE THE CAUSAL DIFFERENCE.

But even that sentence needs a brake. The difference must be carried or reconstructed by a defensible substrate or process attributable to the system being tested, not continuously imposed by an unchanged external driver. A forest can replace trees while a persistent soil condition continues to force the same response. A population can replace members while one external exposure remains. Turnover can coexist with memory, but turnover cannot prove it.

And if memory can be distributed and reconstructed through turnover, the absence of a brain will not settle the forest question for us. We have to return to the larger scale.

The Forest That Has Been Here Before

Years ago, our hypothetical forest experienced a severe drought. During the drought, signals moved through individual plants. Roots changed. Growth slowed. Some organisms died while others survived. Soil moisture collapsed, microbial communities shifted, seed production changed, fungal relationships changed, and canopy gaps appeared. Eventually the immediate drought ended.

Years pass.

Then another severe drought arrives, and the forest responds differently.

There is a phrase waiting for us: *the forest remembers the drought*. Ecology already uses **ecological memory** for ways antecedent conditions and biological or material legacies influence later ecological responses.[6] That idea does not belong to this book. The Macroscopic Life question is narrower: when, if ever, does an ecological legacy become evidence for memory owned by a higher-level organized unit?

The difference between the first drought and the second could be carried by surviving organisms, changed community composition, soil and hydrological legacies, altered microbial or fungal states, structural changes, pathogens, or differences in the disturbances themselves. Some retained differences may lie inside one proposed forest boundary and outside another. That distinction cannot be chosen after seeing which boundary produces the most impressive result.

All of these histories can matter without requiring the forest to be one remembering individual. Ecological memory is established terminology; higher-level individuality remains a separate claim.

Who Remembers?

A function can exist inside a system before the system owns the function.

Now the function is memory.

The ownership problem cannot be solved by finding memory somewhere inside the forest. The proposed whole must first earn a boundary independently of the memory result. Then a retained or reconstructed state must matter to a response specified at that level, and changing the retained state must change that response. If component memories plus shared environment explain the result just as well, the stronger whole-memory claim has not earned its work. If the higher-level organization adds stable prediction or intervention structure across changing components, the claim becomes more interesting.

Even then, the language must remain modest. Saying that a retained state is *used* by the forest is shorthand for a causal relationship, not evidence of awareness or intention. The experiment has to make that relationship measurable and capable of failing.

Return to the Experiment

Imagine that we could run the forest experiment properly. First, we create or identify the past. Matched regions experience a well-characterized disturbance, or a natural disturbance provides a design strong enough to compare exposed and control systems. We measure immediate responses and candidate retained states across the relevant scales.

Then we wait.

The obvious signals decay and acute stress responses subside. But we do not declare that the systems have returned to identical conditions. Total equivalence is neither realistic nor necessary. We measure the

variables relevant to the competing explanations—composition, soil and hydrological state, structural damage, pathogens, microbial state, resources, development, seasonality—and model important residual differences rather than pretending they vanished.

Then comes the probe.

A second challenge is applied. If previously exposed systems respond differently, that is where the experiment begins. We ask what retained the difference and whether the candidate state can be selectively disrupted or changed. Where technically and ethically feasible, erasure, transplantation, reconstruction, or independent perturbation can make the causal test stronger; in complex ecosystems, partial perturbations, natural experiments, mechanistic submodels, and convergent interventions may have to do the work instead.

The rival models receive the same generosity we demand for our own. A State-Only or lower-level model gets comparable data quality, relevant causal variables, model capacity, tuning, and validation opportunity. A label such as *previously drought-exposed* may improve prediction simply because it bundles uncontrolled differences. That is useful discovery evidence, not yet a memory mechanism. The stronger case moves from historical label to retained state to intervention-sensitive causal variable.

Suppose we write the comparison as:

$$\text{PA_memory} = \text{Performance}(\text{Memory Model}) - \text{Performance}(\text{State-Only Rival})$$

A positive value means the memory-level description has demonstrated predictive utility under that fair comparison. It does not establish ontological independence, individuality, agency, intelligence, or consciousness. Prediction Advantage is evidence that a level of description is doing scientific work; ownership requires the additional boundary and intervention evidence already demanded.

Then we attack the boundary. Perhaps the useful unit follows hydrology rather than the forest edge. Perhaps it follows a microbial community. Perhaps individual trees and their shared environment

explain everything. Boundary exploration can help during discovery, but the selected boundary and memory metric must then face prospective or held-out confirmation at another disturbance, period, site, or realization where feasible. Otherwise we can keep moving the circle until something looks as if it remembers.

A serious experiment must be able to end badly for the hypothesis. Imagine that, after individual-tree physiology, species composition, soil chemistry, microbial state, hydrology, pathogens, and other relevant present variables are included, the historical forest label adds no reliable held-out prediction. Perturbing candidate distributed states produces no intervention leverage beyond their component mechanisms. Alternative boundaries perform equally well or better.

The first drought left real ecological legacies. Those legacies mattered. Forest-level memory ownership was not demonstrated.

The goal is not to make the forest remember. The goal is to build an experiment in which it is possible to discover that it does not.

Other outcomes remain possible. We might find genuine functional memory in organisms or subsystems while higher ownership fails: the forest contains memory without becoming one remembering individual. Or a distributed retained state might reproducibly regulate an independently justified higher-level unit, survive appropriate perturbation tests and fair rivals, and continue to work prospectively across a new realization. Only then should the canonical Model A/B/C framework be applied according to the property actually demonstrated.

Even the strongest result in this chapter establishes memory only. Memory is not learning; learning is not prediction; prediction is not anticipation; anticipation is not foresight; foresight is not consciousness. The words become more seductive as we climb, which is exactly why the distinctions must survive.

Before Tomorrow Arrives

Look again at the scar.

It carries the past because something physical remains. But it does not prepare itself for the knife to return.

Some living systems change before a recurring condition arrives. A circadian system can become phase-locked to a periodic environment and prepare physiology around that cycle. But entrainment does not by itself require an internal predictive model, and a present cue can trigger preparation for a condition that usually follows without representing the future at all.

Memory lets the past alter what happens later. The next question is whether a system can use what came before to change what it does **before** the next event arrives—and what evidence would distinguish genuine anticipation from a clock, a cue, or a repeated environmental rhythm.

Can the whole anticipate?

CHAPTER 9

The Anticipation

At 4:59 in the morning, the valve is closed.

At five, it opens.

Water moves through a pipe, divides through smaller lines, and enters the soil while the sky is still cool. Nothing has happened yet. The air has not become dangerous. The plants have not begun losing water to the day's worst heat. Yet the system is already responding to an afternoon that has not arrived.

If you knew nothing about the mechanism, the sequence might seem almost impossible. The action comes first. The event that appears to explain the action comes later.

Then we move the heat.

Tomorrow the hottest part of the day does not arrive at the usual hour. The valve still opens at five. The next day the heat never comes at all. The valve opens at five again.

There was no message from noon traveling backward into the morning. There was a clock.

The distinction sounds obvious once we know the mechanism, but it is one of the easiest mistakes to make when we study systems whose histories, signals, and internal states are partly hidden from us. A response can occur before an event and still be generated entirely by causes already present. A schedule can be inherited. A cue can arrive early. An oscillator can carry phase. Evolution can build a response around regularities that existed long before the current organism was

born.

THE FUTURE CAN BE THE TARGET OF CONTROL WITHOUT BEING THE SOURCE OF THE SIGNAL.

That does not make anticipation unreal. In biology, anticipation is already a legitimate scientific word. Circadian clocks can entrain to environmental cycles and use their phase relationship to prepare for predictable daily changes. Physiologists also use predictive language for regulation that prepares for needs before they arise.[1][2] We should not define those uses away merely because this project needs to test a stronger higher-level claim.

The problem is not whether preparation before an event exists. The problem is what kind of preparation we have found.

The Clock Inside the Body

Replace the irrigation timer with a living organism. Day and night repeat. Temperature rises and falls. Food arrives at recurring times. Biological clocks coordinate physiology and behavior with recurring environmental cycles, and circadian rhythms can persist under suitable constant conditions rather than disappearing the moment an external cycle is removed.[1]

That is more interesting than a timer wired to a valve. The system carries temporal organization inside itself, and that organization can alter physiology before a recurring environmental transition. Science can legitimately call this anticipation.

Now change the schedule. Move the event away from its usual phase. An oscillator may continue to produce the old preparation because its present dynamics still carry the old phase relationship. Acting early therefore does not tell us whether the system can revise what early means.

A clock can anticipate a recurring event without establishing a flexible forecast of that event. That distinction matters when we eventually investigate a forest or ecosystem, because larger systems contain components already equipped with clocks and anticipatory

mechanisms. A landscape can change before winter without the landscape itself owning a forecast.

A function can exist inside a system before the system owns the function.

Now the function is anticipation.

When the Cue Lies

Time is not the only thing that can point forward. Present conditions can carry information about what tends to happen next. Day length can precede seasonal conditions; sensory or physiological cues can precede later demands. Predictive-regulation frameworks explicitly describe preparation before a need fully arrives.[2]

The causal chain still moves forward. A cue exists now. The system detects it now. A response begins now. The later event determines whether the preparation was useful.

And that gives us an experiment.

Make the cue lie.

Preserve the cue and omit the event. Move the event while preserving part of the cue structure. Change the probability that the event follows the cue. Reverse a relationship that used to be reliable. If behavior remains locked to the old cue, a rigid cue rule remains a strong explanation. If behavior updates as reliability changes, that rigid rule has lost ground.

Change the relationship between the cue and what comes next.

But updating is not a certificate for a model of the future. Extinction, habituation, altered sensitivity, state estimation, associative updating, and other mechanisms can also change behavior when cue reliability changes. The experiment eliminates some rivals only by forcing those rivals to make different predictions.

This matters because cue reliability is central to adaptive plasticity: organisms must assess noisy environments, and unreliable cues can

produce maladaptive responses or favor different cue use.[3] A system that prepares correctly when an old relationship holds is interesting. A system that revises preparation when the relationship changes gives us a stronger experimental problem—but still not a conclusion.

A Bacterium and Tomorrow

The temptation is to think that prediction must begin with a brain. It does not.

In 2008, Ilias Tagkopoulos, Yir-Chung Liu, and Saeed Tavazoie reported predictive behavior within microbial genetic networks. They described correlations in *Escherichia coli* transcriptional responses associated with environmental transitions and reported rapid decoupling of those internal correlations after exposure to novel environments.[4]

The phrase is startling only if we hear it through ordinary human language. A bacterium predicting sounds like a bacterium thinking ahead. The evidence requires no such thing. A regulatory network can be organized so that a condition encountered now activates machinery useful under a condition that historically tends to follow. The mechanism can remain physical, local, and nonconscious.

Predictive behavior can therefore be real without planning, and anticipation can be real without imagination. The bacterial example removes one prejudice and introduces another question: where did the useful relationship come from?

Evolution Got There First

Suppose a cue has reliably preceded a challenge across many generations. Organisms whose inherited regulatory architecture used that relationship effectively may have survived or reproduced differently from organisms whose architecture did not. The relationship may therefore have been built by selection across generations rather than learned by the organism alive today.

Other routes operate on other timescales. Developmental or physiological plasticity can alter an individual's phenotype in response to environmental assessment. Learned associations can update within an individual's history. A candidate higher-level organization could, in principle, update on yet another timescale. These possibilities can overlap, and they are not an exhaustive taxonomy.

The distinction matters because a future-appropriate response does not tell us where the relevant updating occurred. Selection can shape inherited architecture; plasticity can use current environmental information; learning can alter later responses after experience. The mechanism and timescale must be identified rather than inferred from the fact that preparation was useful.

The same problem becomes unavoidable when we move upward in scale. If trees, fungi, microbes, insects, and other organisms contain mechanisms that prepare them for recurring conditions, their combined activity can generate a striking landscape-scale pattern before drought or winter. That does not yet tell us whether anticipation belongs to the forest.

What We Can Predict

Imagine that we place sensors throughout the forest and collect years of data: soil moisture, canopy conditions, atmospheric measurements, and system fluxes. A statistical model finds a combination that predicts later drought stress unusually well.

We have discovered something important.

But who made the prediction?

The model did.

Predictive information is a statistical relationship defined over specified variables and temporal distributions. It can be measured in biological systems, including sensory populations.[5][6] But the ability of a scientist to decode a later state from a present one does not establish that the system uses that statistical structure to regulate itself. Nor does statistical predictive information automatically become functional or

semantic information for the system.

PREDICTIVE INFORMATION CAN EXIST FOR THE OBSERVER BEFORE PREDICTIVE CONTROL EXISTS FOR THE SYSTEM.

The difference is causal use.

Suppose a candidate state X predicts drought. A useful intervention test must do more than show that changing X changes behavior, because X may also be an ordinary part of the system's present physiological state. Where technically possible, the forecast hypothesis should make a more specific prediction: alter the candidate future-related state, or alter the distribution of later conditions while matching relevant present conditions, and test whether current action changes in the direction uniquely expected from the proposed future-sensitive mechanism. In large biological systems, clean interventions may be impossible; partial perturbations, natural experiments, and convergent mechanistic tests may be the best available evidence.

Engineered control gives us a positive example of the distinction. Model predictive control uses a model of system dynamics within an optimization procedure to choose current control actions while accounting for predicted evolution and constraints.[7] Biology need not implement explicit trajectory simulation to qualify as future-sensitive control; a different physical mechanism could satisfy the same causal test. Engineering shows that the distinction is coherent, not that nature uses the same implementation.

The Error After Expectation

One way to probe a candidate expectation is to violate it. An event normally follows a cue, then arrives late or not at all. Something inside the system changes, and later behavior changes with it.

That is not yet a prediction error. Novelty, homeostatic error, damage, adaptation, and ordinary mismatch responses can all follow an unexpected event. Prediction-error concepts are used across computational neuroscience, perception, learning, cognition, and action,

but the mechanisms and operational definitions vary.[8][9]

An error signal becomes relevant to this chapter only after a candidate prediction has been specified independently and a plausible comparison or updating mechanism has been identified. The experiment can then state what violation should do, create the violation, measure the resulting change, and ask whether that update alters later preparation in the predicted way.

For Macroscopic Life, it is useful to keep several evidentiary cases separate:

Reactive Regulation → Cue-Anticipatory
Regulation → Entrained Anticipation →
Forecast-Sensitive Control → Flexible
Model-Based Anticipation

These are project categories for testing progressively stronger evidence about the particular ownership claim, not a universal scientific taxonomy, developmental sequence, or ladder of biological superiority. A successful clock-driven anticipatory mechanism need never “progress” to model-based control. The last category simply demands the most evidence for the stronger claim this book is trying to test.

Now return to the forest we have been testing and give its history a new burden.

The Forest Before the Drought

After years of measurement, several variables begin changing before a period that historically preceded severe drought. Seen across space and time, the same candidate forest appears to reorganize before the most damaging conditions arrive. The experiment remains composite and hypothetical; the continuity belongs to our investigation, not to a claim that one real forest dataset has already demonstrated this sequence.

The sentence almost writes itself:

The forest knows drought is coming.

Do not write that sentence into the science.

Individual organisms may be following local cues or seasonal clocks. Soil and hydrological changes may already carry much of the signal. Interactions among organisms and environment may generate the apparent whole-system pattern without requiring forest-level forecast ownership.

The forest can look early without the forest owning the forecast. But dismissing the whole in advance would be equally premature. If system-scale variables capture causal organization that strong component-level models miss, the higher level should be allowed to compete.

The answer is not to believe the forest. The answer is to make the season lie.

Make the Season Lie

In a real forest we cannot drag summer across a calendar, but controlled subsystems, altered watering regimes, unusual climatic years, cross-site comparisons, and natural experiments can change pieces of the relationship between present cues and later conditions.

The experimental logic is simple: establish what normally predicts what, break or shift that relationship, then ask what the system follows and whether it updates. A clock may follow phase; a cue-response system may follow the immediate cue; a rich lower-level model may reproduce the pattern through interactions, history, spatial coupling, and environmental forcing. Those rivals receive the same fair treatment established earlier. The candidate higher-level state and boundary must also be fixed before confirmation and survive genuinely new times, sites, or altered temporal regimes; a variable invented after the outcome remains discovery, not confirmation.

Then ask two separate questions. Does the higher-level variable improve held-out prediction? And does it organize interventions that change outcomes in ways the best fair rival does not capture equally well? Prediction Advantage and Intervention Advantage are related but not interchangeable.

Where intervention is feasible, the forecast hypothesis should predict a specific change in present preparation when the candidate future-sensitive organization is altered. Where direct manipulation is not feasible, convergent natural experiments and mechanistic submodels must carry more of the burden.

This experiment can fail at every step. It should.

Who Owns Tomorrow?

Suppose the forest-scale pattern survives prospective testing. We still have the ownership problem.

The target property must be defined before we decide that the forest has passed: present organization at the proposed system level must causally modulate current action with respect to distributions of later states, and the organization must update in a discriminating way when predictive relationships change. Passing a test is evidence for that property; it is not the definition of the property.

Can strong lower-level models—complete with biologically plausible interactions and shared environmental forcing—reconstruct the result? If they can, anticipation may be abundant inside the forest without belonging to the forest as a distinct causal unit. If an independently justified forest-scale variable adds prospective prediction and intervention leverage that those rivals do not recover, the higher-level description begins to earn scientific work.

We can write the predictive comparison compactly:

$$PA_anticipation = \text{Performance}(\text{Future-Sensitive Model}) - \text{Performance}(\text{Best Fair Rival})$$

This is project synthesis, not a standard field equation. $PA_anticipation > 0$ earns predictive utility for this comparison, not individuality, agency, consciousness, or foresight. Lower-level sufficiency remains a legitimate result; higher-level predictive utility can matter without establishing anticipation ownership; and the stronger property-specific result must survive the changed cue relations, causal tests, boundary competition, prospective confirmation, and replication

already demanded by the experiment. Model B remains a discovery rather than a consolation prize.

The forest can also fail cleanly. Move the timing and preparation may remain locked to phase: the clock wins. A strong interaction model may reconstruct the apparent forest-level effect: the lower level wins. Or our decoder may forecast drought while perturbation and natural experiments fail to show that the proposed system uses the decoded structure: prediction existed for the observer, but system-level predictive control was not demonstrated. A boundary can fail, an advantage can disappear at a new site, or a stronger rival can absorb the effect.

The preparation can remain real. The components can remain genuinely anticipatory. The stronger forest-level claim can still be false.

That is scientific progress.

[FIGURE 11 ABOUT HERE]

PREPARATION $\neq$ PREDICTION $\neq$ FORESIGHT $\neq$ PROPHECY

Figure 11. Preparation can precede an event for many reasons. The visual sequence is a restraint on inference, not a claim that these terms exhaust the scientific taxonomy of anticipatory processes. “Prophecy” is a rhetorical endpoint in the figure, not a scientific category used by Macroscopic Life.

A Preferred Future

We began with a valve opening before the heat. The clock dissolved the appearance of backward causation without dissolving anticipation itself. The harder problem became whether a present system state merely tracks what tends to happen next or causally changes current control because later conditions differ.

If a higher-level future-sensitive description ever survives strong rivals, intervention, altered cue relations, boundary competition, prospective testing, and replication, the next question still does not become consciousness.

It becomes goal.

A system can act with respect to what is likely to happen, yet “preferred” can mean several very different things. A controller can have an operational target supplied by its design. A biological regulatory system can maintain variables within viable ranges. Dynamical systems can return toward attractors. None of those facts alone establishes subjective preference, intention, or purpose at the level being studied.

We encountered the distinction earlier:

Function is not goal. Goal is not purpose.

Anticipation does not erase it.

Tomorrow is one problem; a preferred tomorrow is another.

Can the whole have a goal?

FIGURE 11 - APPROVED POSITION

Structural proof placeholder. Insert only the frozen publication-authority visual at final composition; do not invent or substitute artwork.

CHAPTER 10

The Goal

A marble rests at the bottom of a bowl.

Push it halfway up one side and let go. It rolls down, crosses the bottom, climbs the opposite wall, returns, and eventually settles close to where it began. Push it from another direction and the route changes, but the destination does not. Start it higher. Start it lower. Give it a harder shove. Within limits, the same landscape keeps drawing the motion toward the same place.

It is difficult to watch this happen without importing a verb.

The marble returns. The marble seeks the bottom. The marble prefers the bottom.

Did the marble want the bottom?

Of course not. Gravity, geometry, friction, and the marble's initial motion were enough. The bottom mattered to the trajectory, but nothing in that fact requires the marble to detect a deviation or act with respect to a target. The shape of the system can produce convergence all by itself.

RETURN IS NOT YET A GOAL.

The word *yet* matters. In dynamical systems, trajectories can converge toward states or regimes we describe as attractors. In biology and philosophy, there are legitimate traditions that use goal-directed or teleonomic language for feedback, regulation, self-maintenance, and related forms of organization.[1] We do not need to outlaw that language. We need to ask a harder question: when does a proposed goal

belong to the organization of the system we are studying?

A Real Target

We have met the thermostat before. In Chapter 4 it taught us how easily our verbs can outrun a mechanism. Now it returns for a different reason.

Set a thermostat to seventy-two degrees. The control architecture is physically sensitive to a reference condition or rule; alter that reference and the controller's behavior changes. Depending on the design, this can be implemented through mechanical switching, hysteretic control, digital logic, or more elaborate feedback. Disconnect the relevant sensing or control pathway and regulation fails in predictable ways.

This is more interesting than the marble. It also prevents us from making the opposite mistake. Goal-directed control does not require consciousness. A controller can act with respect to a target without feeling cold, wanting warmth, imagining tomorrow, or knowing that it exists.

But the thermostat gives us another problem. Who chose seventy-two?

A person may have entered the setting. An engineer designed the architecture that can use it. The controller implements a reference condition, but the reference was not necessarily authored by the controller. A target can therefore be real, physically consequential, and nonconscious while still raising a question about ownership.

Living systems make that question much harder. Bodies regulate temperature, hydration, glucose, ionic conditions, and many other variables important to continued organization. Walter Cannon's homeostasis framework made physiological regulation explicit nearly a century ago. In influential allostasis frameworks, successful regulation need not mean returning every variable to one fixed value: operating conditions can change with context, and present action can prepare for expected demand.[2] The terminology and scope of allostasis remain debated, so this is not a claim that physiology has converged on one

universal regulatory theory. The narrower point is enough: one variable can move while another constraint remains within a viability-relevant range.

That changes the experiment.

If a system repeatedly returns toward one state, we can ask whether the return is passive or actively regulated. But if several candidate constraints cannot all be preserved at once, the system faces a more revealing test.

Which one gives way?

What Will It Sacrifice?

Imagine a living system pushed into a condition where two constraints conflict. Preserving one variable perfectly would drive another outside its pre-specified range. There is not enough energy, water, time, material, or control capacity to do everything.

Now recovery alone tells us less. The revealing question is what fails first.

A proposed hierarchy should predict the answer before the outcome is known. Under one disturbance, the system may tolerate a large deviation in A while B remains constrained. Under another, it may recruit a different mechanism while B again remains within range. Under a third, B may fail as a different constraint is preserved.

This does not establish consciousness or agency. Fixed network architecture, energetic scarcity, differential fragility, evolutionary history, local feedback, and hard-wired priority structure can all generate prospective tradeoffs. Predicting sacrifice is evidence for organized priority structure. It becomes evidence for higher-level goal ownership only if a higher-level target model adds prediction or intervention power beyond comparably rich mechanistic priority models.

A GOAL IDENTIFIED ONLY AFTER THE OUTCOME IS A STORY UNTIL IT PREDICTS THE NEXT TRADEOFF.

That gives us a way to force competing descriptions apart without assuming that the system has already chosen anything.

The Body That Rebuilds

Some of the most provocative examples come from development and regeneration.

Cut living tissue and repair begins. Alter developmental conditions and cells may compensate. In some regenerative systems, large-scale anatomical outcomes can be restored despite substantial perturbation to components and routes by which the outcome is produced. Research on morphogenesis and bioelectric signaling has examined how membrane-potential patterns, gap-junction communication, biochemical signals, mechanical interactions, and gene-regulatory processes participate in distributed anatomical patterning.[3]

The temptation is immediate. Somewhere, we want to say, the body must know what shape it is supposed to be.

That sentence is stronger than the experiment.

Terms such as *target morphology* and *anatomical homeostasis* are especially prominent in a particular research program associated with work on regenerative pattern control. They should not be mistaken for settled field-wide ontology. They can nevertheless be productive experimental ideas: they direct attention toward large-scale outcomes and variables that can be missed if we inspect only one cell at a time. Restoration of form does not automatically establish a stored blueprint, symbolic representation, intelligence, or system-owned goal.

Target morphology should therefore enter this book as an experimental hypothesis before it enters as an ontology.

That restraint does not make regeneration less astonishing. No individual cell need contain a literal, complete spatial description of the finished anatomy for distributed processes to build or restore coherent form. Different local mechanisms can contribute to a large-scale outcome. The causal organization can be real while the strongest interpretation remains unsettled.

[FIGURE 12 — DISTRIBUTED PROBLEM-SOLVING /
WHOLE-SYSTEM SOLUTION]

***THE WHOLE CAN SOLVE A PROBLEM THAT NO
SINGLE COMPONENT SOLVES ALONE.***

Here *solve* is functional language: distributed interactions achieve a constrained outcome under perturbation. It does not imply cognition, awareness, or an internal problem statement.

Biological robustness makes the same warning necessary. Feedback, redundancy, degeneracy, modular organization, plasticity, and response diversity can allow different mechanisms to preserve similar outcomes.[4] Multiple roads reaching the same destination are evidence of robust organization. They are not, by themselves, evidence that the destination is a target owned by the whole.

The Objective in Our Notebook

There is another way to manufacture a goal, and scientists are especially good at it.

Take a complicated system and record enough variables. We may discover that its behavior can be described as minimizing variance, preserving biomass, maintaining diversity, maximizing throughput, remaining inside a viability region, or minimizing some mathematical quantity. With enough skill, the resulting objective function may predict the system beautifully.

But who chose the objective?

Perhaps the proposed objective corresponds to a causally privileged variable or constraint in the physical organization. Or perhaps we chose it because it compresses the observations.

Those possibilities cannot be separated by fit alone. If many objective functions generate the same observed behavior, behavior alone cannot identify which, if any, is causally used by the system.

**AN OBJECTIVE FUNCTION CAN EXIST IN OUR MODEL
BEFORE A GOAL EXISTS IN THE SYSTEM.**

This problem becomes especially important in theories that use optimization language. A physical process can sometimes be represented as minimizing or maximizing a quantity. Evolutionary models use fitness-related quantities. Engineers specify loss functions. Active-inference and free-energy approaches use optimization-like formalisms to model self-organizing and adaptive systems. Those frameworks can be scientifically useful, and debates about their realist interpretation remain active.[5] Formal optimization by itself does not establish that the proposed objective is a causally privileged variable or constraint of the physical system.

The test is causal. If perturbing the physical organization associated with a proposed target changes action in the specific direction the model predicts, that is stronger evidence than discovering afterward that a trajectory can be fit by an objective function. If several objectives fit equally well, ownership becomes weaker still.

This is the same discipline we used with prediction. A forecast could exist in our model before predictive control existed in the system. Now an objective can exist in our notebook before goal-directed control belongs to the whole.

What Selection Preserved

Evolution adds another source of apparent purpose.

Eyes are for seeing. Hearts pump blood. Wings contribute to flight. Biological language is full of function, but *function* itself has several scientific and philosophical meanings. Selected-effect, causal-role, and organizational approaches ask different questions about why a trait exists and what it does in a present system.[6]

Historical selection and present control are not the same thing.

A trait can exist because ancestors possessing it left more descendants without the present organism carrying an internal scalar called *fitness* and trying to maximize it. Organisms can, of course, sense and regulate variables that are fitness-relevant proxies. That is different from treating evolutionary fitness itself as an online objective used by

the organism.

The same warning applies when we move upward in scale. Trees, fungi, microbes, insects, and other organisms can carry regulatory capacities shaped by selection. Their interactions can stabilize larger ecological variables. That stabilization may be scientifically important. It does not automatically establish that the containing forest maintains that variable as its own target.

What selection preserved is not automatically what the present system pursues.

Force the Tradeoff

Return to the forest, now carrying the history of the drought experiment forward. A severe disturbance pushes it away from a familiar regime. Perhaps drought changes water availability. Perhaps a storm removes canopy. Perhaps fire reorganizes the landscape. Over years, some variables move toward earlier ranges while others settle somewhere new.

We may be tempted to write: *the forest is healing itself*.

Treat that sentence as a hypothesis-generating metaphor, not a scientific conclusion.

This forest is a composite hypothetical experiment, not a report of one empirical site. Its apparent recovery could arise from succession, environmental forcing, interacting organism-level regulation, species replacement, redundancy, soil and hydrological dynamics, or an analyst's decision about which baseline counts as recovery. A serious lower-level rival gets those interactions, nonlinearities, histories, and comparable model capacity. The stronger model does not get weaker opponents.

So we make the experiment harder.

First, specify a candidate target before the disturbance. Call it *G*. *G* cannot mean something vague such as *forest health*. It must be measurable and bounded: for example, a hypothetical relation among

canopy function, soil-moisture buffering, and water or carbon flux that is expected to remain within a defined tolerance over a stated time horizon. The example is illustrative, not an empirical claim. The boundary, variables, tolerance, horizon, and failure criterion must be fixed before confirmation.

Second, create or exploit a condition in which two candidate constraints cannot both remain within their pre-specified ranges. The experiment forces a tradeoff, not agency. If the goal hypothesis is useful, it should predict which deviation will be tolerated and which compensatory routes will be recruited.

Third, predict the tradeoff before the outcome is known.

That is the heart of the test.

Different perturbations may recruit different mechanisms while preserving the same prospectively specified constraint. Redundancy, degeneracy, attractor dynamics, distributed feedback, and fixed priority networks remain serious rivals. The question is whether a target-structured model predicts the route, the tradeoff, and the consequences of intervention better than those rivals do when resources and flexibility are matched fairly.

Then we separate discovery from confirmation. Candidate target and boundary can be developed in one dataset or regime, but they must be frozen before prospective tests in held-out disturbances, times, sites, or implementations. Serious alternative targets considered during discovery are recorded rather than disappearing when they lose. We keep the losing goals, too. Otherwise a high-dimensional forest gives us nearly unlimited opportunities to discover a goal after the fact.

When the Goal Cannot Be Reached

Now make the experiment harder again.

Suppose G cannot be restored.

A simple controller may continue pushing because its actuator is saturated or because a fixed feedback rule has no alternative. Persistent

corrective action therefore proves little by itself. A stronger target-structured model should prospectively distinguish among continued correction, context-sensitive reallocation, transition to another viable regime, graceful degradation, collapse, or ordinary dynamics predicted by a lower-level rival.

None of those outcomes is frustration, disappointment, or surrender unless psychological capacities are independently established. They are physical responses under an unreachable constraint.

This matters because a return curve can no longer carry the argument. When the familiar destination is unavailable, competing models must predict behavior away from the easy case.

If A and B cannot both be maintained, we specify the expected tradeoff before the conflict. A goal identified only after the outcome remains a story until it predicts the next tradeoff.

Who Owns the Goal?

By now the ownership test can be stated without mystery.

First, specify the candidate target or constraint independently of the result. Second, identify physical organization through which deviation from that target changes action. Third, show that different disturbances produce discriminating compensation and tradeoffs. Fourth, compare the target-structured account with strong component-level, dynamical, evolutionary, fixed-priority, and externally imposed alternatives. Fifth, require prospective prediction and, where possible, intervention advantages that survive new conditions.

The target does not need to be a symbolic number stored in one location. Organizational approaches to biological function and teleology examine how self-maintaining networks and closure of constraints can ground goal-directed or normative descriptions without a little internal representation labeled *goal*.^[7] Macroscopic Life need not defeat that framework. It asks whether organization of comparable causal relevance exists at the proposed macroscopic level and whether the chosen boundary is justified.

A function can exist inside a system before the system owns the function.

Goal-directed regulation can exist inside a system before the containing system owns the goal.

A forest full of organisms regulating themselves does not automatically establish a self-regulating forest. A tissue containing cells that respond to local signals does not automatically establish that every larger boundary we draw around them owns the resulting anatomical outcome. Boundary, causal organization, and intervention evidence must travel together.

We can formalize one part of the comparison:

$PA_{goal} = \text{Performance}(\text{Goal-Structured Model}) - \text{Performance}(\text{Best Fair Rival})$

The inherited Prediction Advantage standard still applies: the prediction target is specified prospectively; rivals receive comparable data, history, capacity, and tuning opportunity; evaluation occurs on held-out or genuinely new conditions; and the performance difference is assessed with uncertainty and calibration appropriate to the task. A flexible goal model does not earn anything merely by having more parameters.

If $PA_{goal} > 0$, the goal-structured model has earned predictive utility under that comparison. Ownership requires independent boundary and causal or intervention evidence. Prediction Advantage remains different from Intervention Advantage. Neither establishes consciousness, desire, agency, purpose, or ontological independence.

Remove the Goal

There is a simpler way to ask the same question.

Remove the higher-level goal variable from the model.

This is not a vocabulary trick. Keep the strongest mechanistic dynamical account available, with the same data, histories, interactions, model capacity, and opportunity to succeed. Ablate the proposed

higher-level target or constraint and ask what scientific capability disappears.

Perhaps nothing does. If the mechanistic account predicts novel disturbances, tradeoffs, compensatory routes, and intervention effects just as well, stronger goal ownership has not earned necessity in the science.

But perhaps something is lost. A higher-level target variable might compress changing implementations into a stable predictive structure. It might predict sacrifice under conflict. It might identify interventions difficult to see from component descriptions alone. It might generalize across turnover or new disturbance types.

If so, the higher level has done scientific work.

That still leaves more than one outcome. Lower-level sufficiency remains legitimate. Model B remains available when a higher-level target-structured description improves explanation, prediction, or intervention without earning stronger goal ownership. A property-specific Model C would require the independently justified whole to regulate with respect to a prospectively specified constraint through discriminating organization that survives serious rivals and generalizes beyond the cases used to discover it.

That is a discovery even if the stronger property remains unearned.

Let the Forest Fail

The forest should be allowed to disappoint us.

Perhaps an attractor or succession model predicts recovery just as well. Perhaps a capacity-matched component-interaction model predicts every tradeoff. Perhaps the apparent target changes whenever we change the analyst's metric. Perhaps the target model works on known disturbances and collapses on the first prospective test. Perhaps its advantage vanishes when excess flexibility is removed. Perhaps intervention on the proposed target-maintaining organization adds no leverage. Perhaps another boundary works equally well or better.

Then the stronger claim loses.

That would not mean nothing happened. The forest may still be robust. It may still contain extraordinary organism-level regulation, distributed feedback, ecological memory, anticipatory responses, and compensatory dynamics. A higher-level description may still earn Model B utility. What failed was the stronger claim that the forest itself owns the goal we proposed.

This is exactly what a useful theory must permit.

Purpose

There is one more word waiting behind goal, and it is even more dangerous.

Purpose.

We established the distinction earlier and should keep it now: function is not goal. Goal is not purpose. Even strong evidence for a system-owned regulatory target would not establish subjective desire, conscious intention, moral purpose, cosmic purpose, or a reason the system exists.

If one day a larger-scale system earns a goal claim, that would be scientifically remarkable. It would not mean the forest wants to survive, Earth has a mission, or the universe has a plan.

The stronger the word, the more it must earn.

A goal tells us what counts as success, but it does not yet tell us who selects among possible paths.

Can the whole choose?

FIGURE 12 - APPROVED POSITION

Structural proof placeholder. Insert only the frozen publication-authority visual at final composition; do not invent or substitute artwork.

PART V

THE INDIVIDUAL

A process can happen inside a system before the system
itself owns the process.

CHAPTER 11

The Choice

A switch can select.

Give it two routes. Under one condition, current goes left. Under another, it goes right. The alternatives are physically real, the difference in input is real, and the resulting pathway is real. Calling that selection does not require a mind. Something in the mechanism discriminated between available routes and produced one of them.

But almost immediately we want to add another word. We want to say it chose.

SELECTION IS REAL. HIGHER-LEVEL CHOICE HAS NOT YET BEEN EARNED.

That sentence needs care. Scientists and philosophers do not agree on one universal definition of agency, and some serious accounts place basic agency much lower than ordinary language does. One influential framework, for example, emphasizes individuality, interactional asymmetry, and regulation in relation to norms rather than requiring deliberation or a represented menu of options.[1] Macroscopic Life does not get to redefine those traditions out of existence simply because we want a harder test.

So this chapter asks something narrower. When several physically realizable courses of action are available, what evidence would justify saying that the selection among them belongs causally to the proposed whole?

The Selector

We can make the switch much more impressive: give it memory, many inputs and outputs, changing thresholds, probabilistic behavior, and enough internal state that a human observer cannot predict the next output by inspection. None of that creates a shortcut. Unpredictability is not freedom, randomness is not a chooser, and determinism does not automatically rule agency out under legitimate scientific frameworks. The question is how physical organization produced the action and whether a proposed higher level adds something testable to that explanation.

Fixed controllers can look flexible; adaptive controllers can alter their mappings; learning systems can update action probabilities after experience. Those are real differences in capability, but the level question remains: what selected—a component, distributed interactions, environmental forcing, an already identified controller, or a larger system whose organization adds stable causal and predictive work?

We need a case in which *decision* is already legitimate scientific language.

A Swarm Between Homes

Imagine a mass of honeybees hanging from a branch.

They are between homes.

A swarm has left its old colony and must establish a new nest. Scouts search the surrounding landscape and inspect possible cavities. Different scouts encounter different sites. They return and advertise sites through waggle dances, with dance persistence related to site quality. Other scouts can be recruited to inspect advertised sites, so support for candidate locations changes through repeated local interactions. Eventually the swarm reaches a decision and later departs for a selected site.[2]

Researchers explicitly study this process as group decision-making in social insects.[2] That matters because we are not beginning with a

macroscopic metaphor and asking science to tolerate it. We are beginning with a distributed biological decision process already studied on its own terms.

There is no little bee at the center holding a complete map of every cavity and issuing the final order. Individual scouts carry local information. Recruitment, decay of signaling, cross-inhibitory processes described in the broader literature, and quorum-related mechanisms contribute to the collective process. In the classic quorum experiment, the specific question was narrower than the whole decision: how scouts trigger preparation for the swarm's move after a site has accumulated enough visitors. Delaying quorum formation was predicted to delay worker piping, a preparation-for-takeoff signal, and swarm takeoff.[3]

The physical implementation is distributed. That does not make the collective decision unreal; it tells us how the decision is produced.

THE QUESTION IS NOT WHETHER A DECISION OCCURRED. THE QUESTION IS WHERE THE DECISION BELONGS.

Here *belongs* is an experimental claim, not a metaphysical possession. We ask which level supplies a stable causal variable that survives serious lower-level and alternative-boundary accounts and improves prospective prediction, intervention, or compression. A mechanism at the component level does not automatically erase a useful higher-level description; nor does a useful higher-level description automatically establish a new individual.

The quorum work is instructive precisely because it links a proposed part of the decision process to a risky intervention-sensitive prediction. Alter the formation of the relevant quorum while leaving as much of the preceding decision process intact as possible, and the timing of preparation for departure should shift.[3] That is much stronger than pointing at coordinated movement and naming it choice.

Where the Decision Belongs

Suppose a swarm-level variable predicts commitment better than a list of isolated scout observations. That would be interesting. Suppose it remains useful across different swarms or changing individual participants. More interesting still. Suppose intervention on the mechanism represented by that variable changes the predicted collective transition. Now we are doing causal science.

None of those results, alone, establishes consciousness, free will, purpose, or biological individuality.

COLLECTIVE DECISION-MAKING CAN BE REAL BEFORE COLLECTIVE INDIVIDUALITY IS ESTABLISHED.

There is useful middle territory here. A group-level decision process can be scientifically legitimate while stronger claims about the group as one biological individual or one agent under a particular theory remain unresolved. We do not have to choose between “nothing but bees” and “a giant mind.”

Work on animal groups supplies an important adversary. In a well-known 2005 model, Couzin and colleagues showed that a small proportion of informed individuals could guide a group with high accuracy even when other members did not know who possessed the information. The model could also generate consensus when informed individuals favored different directions.[4] Complex-looking group performance can therefore arise from relatively simple local interaction rules.

That is a reason to take collective organization seriously and a reason to resist being impressed too quickly. Accuracy, coordination, and consensus do not by themselves identify the level at which choice should be owned.

Did We Invent the Options?

There is another possible source of the apparent choice: the scientist's description.

We decide where one behavior ends and another begins. We choose the time window. We label trajectories A and B. We define a state space. In formal models we may assign costs, rewards, values, or utilities and then fit a model that predicts which action will occur.

The model may be useful and still leave an uncomfortable question: did the system physically distinguish the alternatives in a way relevant to its dynamics, or did we merely draw useful categories across a continuous process?

A stronger test requires more than labels. Candidate alternatives should correspond to physically discriminable response regimes or trajectories. Their definitions should be fixed before the outcome is known. And perturbing a prospectively specified system state should change their occurrence in a predicted way while serious hidden-state and component-level alternatives remain in the comparison.

Vocabulary can outrun evidence here. Reinforcement learning makes the problem visible because *policy*, *value*, and *reward* have formal computational meanings. A numerical reward signal does not by itself establish subjective pleasure or desire. In neuroscience, many midbrain dopamine neurons exhibit reward-prediction-error responses important to learning, but that does not make “dopamine is desire” an adequate scientific translation.[5]

Formal variables can organize a model without acquiring every ordinary-language meaning of their names.

When the Parts Disagree

Chapter 10 forced candidate goals to reveal themselves by making desirable outcomes compete. We can use disagreement among components as a related stress test for action selection.

If every local response tendency points the same way, a collective result may follow directly from aligned component dynamics. That can still be important organization. But disagreement can expose more of the mechanism by which a collective outcome is resolved. Components may receive different information, face different local costs, or favor

different response trajectories.

This is not a universal requirement for agency or choice. It is an experimental stress test. If a proposed higher-level variable predicts how conflicting component tendencies are resolved, especially in new conditions, it earns more than it would from a case in which every part already points toward the same result.

Honeybee nest-site selection supplies an intuition rather than a universal template. Scouts can advertise different candidate sites, and the swarm's eventual coordinated movement need not begin with every scout supporting the same destination. The question is how the distributed process reaches a state from which coordinated departure can occur.

We also need to be careful with the phrase *could have done otherwise*. This chapter does not require metaphysical freedom. It requires physically realizable alternatives: under controlled changes in state or context, more than one response trajectory is available to the system. A deterministic hidden-state controller can satisfy that condition. Good. It remains a serious rival rather than being excluded by definition.

Return to the Forest

Now we can make the forest experiment harder.

Return to the same hypothetical forest, now carrying forward the candidate constraint G from Chapter 10. Suppose several physically realizable response trajectories could preserve it after disturbance—changes in growth, water use, phenology, species contribution, or resource exchange. The new question is not whether G is maintained, but whether organization of the proposed whole contributes causally to which viable trajectory occurs.

First, justify the candidate forest boundary before observing the response. Second, define multiple physically realizable response trajectories before the disturbance. Third, specify the candidate higher-level state and the conditions under which the model predicts

divergence among those trajectories. Fourth, perturb the proposed organizing state where a scientifically interpretable and ethically appropriate intervention is possible, then test whether trajectory probabilities change in the predicted direction.

Now bring in the rivals at full strength. Give the component model nonlinear interactions, thresholds, history, heterogeneity, local adaptation, and feedback. Test environmental forcing. Fit hidden-state and fixed-rule accounts with comparable capacity. Include stochastic alternatives. Move the proposed boundary and test whether another candidate whole performs as well or better.

If the forest-level model wins only because every rival was made stupid, it did not win.

The Same Whole in Different Parts

Any forest-level state is physically implemented through trees, fungi, microbes, water, atmosphere, soils, chemistry, and flows. The higher-level question is not whether we found a new force, but whether a prospectively defined variable remains scientifically productive across changes in its lower-level implementation. Species composition may change. Individual organisms may turn over. Local pathways may differ. If the same prospectively defined higher-level variable continues to predict response organization across those changes, it may provide causal compression that a brittle component-specific description does not.

That still does not establish choice ownership. Prediction and causal ownership must remain separate.

We can formalize the predictive test:

```
PA_choice = Performance(Higher-Level  
Action-Selection Model) - Performance(Best Fair  
Rival)
```

The Chapter 5 Prediction Advantage rules still apply: prospective target, fair rival resources, genuinely new evaluation, and visible uncertainty. Positive `PA_choice` earns predictive utility for the

action-selection model; stronger ownership still requires independent boundary and causal evidence.

Remove the Chooser

Now perform the replacement at the level of the model.

Do not merely delete the word *choice*. Remove the proposed higher-level action-selection variable. Give the lower-level rival the strongest justified nonlinear interactions, histories, hidden states, local controllers, adaptive processes, environmental drivers, and comparable modeling resources.

Then ask what is lost. Does prospective prediction degrade in novel contexts? Does the lower-level account become brittle across component turnover or implementation change? Does it lose reliable prediction of conflict resolution? Do interventions organized around the higher-level variable generalize in ways the rival fails to anticipate? Or does nothing scientifically important disappear?

If nothing important disappears, the stronger whole-level choice claim has not been demonstrated.

That is a legitimate result. Model A may win: lower-level sufficiency. Model B may win: a collective or whole-level action-selection description provides useful prediction, compression, or intervention while stronger ownership remains unresolved. Or evidence may support the project's property-specific Model C for higher-level choice: an independently justified whole-level variable contributes causally to selection among physically realizable response trajectories, survives fair component, environmental, stochastic, fixed-rule, and alternative-boundary rivals, and retains prospective prediction and intervention value under replication.

Even Model C establishes only the property tested here. It does not establish consciousness, libertarian free will, moral responsibility, intelligence generally, personhood, or purpose.

Let the Decision Survive

A theory that interprets every outcome as success cannot teach us much.

Routing or threshold dynamics may be sufficient. A hidden-state controller may replace the whole-level variable. Local adaptation may explain apparent flexibility. Stochastic dynamics may explain variation. External forcing may explain divergence. A capacity-matched component model may match every relevant prediction. Another boundary may outperform ours. The alternatives we defined may fail to correspond to physically meaningful response regimes. A retrospective fit may collapse prospectively. An apparent higher-level advantage may disappear when the rival receives equal resources.

Any of those outcomes must be allowed.

And some discoveries can survive them. If the strongest whole-level ownership claim fails in a honeybee case, the empirically studied collective decision process does not evaporate. We have learned more precisely how it is implemented and what stronger interpretation the evidence does or does not support. If a forest-level action-selection variable turns out to be unnecessary, the ecological dynamics that motivated the hypothesis do not become unreal. We have learned where not to place the stronger property.

That is the territory Model B protects: real higher-level organization without automatic promotion to every stronger property.

A function can scale before an individual does.

The Individual

There is one complication we should not hide. Some influential theories make individuality part of agency itself.[1] From that perspective, testing choice-like action selection before individuality can look backward.

We separate them deliberately because the macroscopic problem is vulnerable to circular reasoning. If we define the agent by assuming the

individual and then use apparent agency to prove the individual, the conclusion has entered through the definition. So the project temporarily separates properties that some theories legitimately combine and asks what evidence each can earn independently.

The swarm shows why this can be useful. A distributed collective can gather information, support competing alternatives, cross decision-related thresholds, and produce coordinated movement. Those functions can be investigated without first settling whether the swarm, colony, or another boundary should count as one biological individual.

[FIGURE 13 — frozen figure placement]

FUNCTIONAL ANALOGY $\neq$ BIOLOGICAL IDENTITY

>

***WE HAVE SHOWN THAT THE FUNCTIONS CAN SCALE.
WE HAVE NOT YET SHOWN THAT THE INDIVIDUAL
DOES.***

That is the next boundary we have to cross.

A decision process can be real before we know whether the system implementing it is one individual.

Can the whole become an individual?

FIGURE 13 - APPROVED POSITION

Structural proof placeholder. Insert only the frozen publication-authority visual at final composition; do not invent or substitute artwork.

CHAPTER 12

The Individual

We are used to beginning biology with the individual already given.

There is one person standing in front of us. One dog crossing the yard. One oak beside the road. One bee landing on a flower. We can point, count, and draw a line around each body. The world appears to arrive predivided into things.

Your own body seems like the easiest case of all. Skin provides a visible boundary. Wounds are repaired. Temperature is regulated. Parts depend on other parts. Extensive turnover occurs at multiple material levels while the larger organization persists. Your heart, liver, immune system, nervous system, blood, bones, and muscles participate in an organization that is recognizably yours.

None of this is an argument that you are secretly many people. It is the opposite. You are such a convincing individual that we rarely ask how much organization is required to make that fact possible.

Evolution did not have the luxury of beginning with the individual already supplied.

The Hurricane Problem

Imagine looking down from orbit at a hurricane.

There is a recognizable structure. Pressure and temperature gradients help define it. Air and water move through it continuously. Energy flows through the system. Feedbacks help maintain its organization for a time. Its material constituents are replaced while the storm persists. It

can strengthen, weaken, reorganize, respond to its surroundings, cross an ocean, make landfall, and disappear.

If we tried to establish individuality using only a boundary, persistence, energy flow, feedback, organized structure, and material turnover, the hurricane would perform uncomfortably well.

That does not make a hurricane a biological individual. It tells us that several properties we associate with organisms are not sufficient by themselves.

A BOUNDARY CAN BE REAL BEFORE AN INDIVIDUAL EXISTS.

This takes us back to Chapter 3. There we forced a proposed whole to earn its boundary. Here the requirement becomes harder. Even if a boundary is scientifically useful, we still have to ask what kind of biological unit, if any, lies inside it.

A watershed can have a useful boundary. So can a population, a fire, a tumor, a nation, an ecosystem, and a weather system. A meaningful edge does not settle the individuality question.

One Genome, Many Bodies

Biology makes the problem harder without leaving life at all.

Consider a clonal plant. What appears to us as several plants can sometimes belong to one genetic individual, a *genet*, produced through clonal growth. The visible shoots or modules are *ramets*: potentially autonomous modules whose degree of physiological integration can vary with species, development, and circumstance. Ramets may remain connected or become physically separated; neither connection nor separation alone tells us exactly how physiologically independent they are.[1]

Now count the individuals.

One clonal lineage? Several physiologically functioning modules? One connected network at one stage and several more autonomous units later? Even somatic mutation can introduce genetic variation within a

long-lived clone. The answer therefore depends partly on which biological property we are trying to individuate.

The point is not that individuality is imaginary. The point is that one clonal lineage, one visible body, one continuous skin, and one physiological unit need not always identify exactly the same entity.

Our everyday counting rule has broken.

A Colony That Looks Like One

We have already watched honeybees make a collective decision. Now change the question. Do not ask where the decision belongs. Ask what kind of thing the colony is.

A highly integrated social-insect colony can divide labor among members. Reproductive output and reproductive roles may be concentrated in a subset of colony members while others perform different functions. Colonies distribute resources, defend themselves, regulate aspects of their internal environment, rear offspring, and coordinate collective activity.

Under influential cooperation-and-conflict approaches to organismality, highly cooperative groups with sufficiently low internal conflict can move toward the organismal end of a continuum.[2] That makes some social-insect colonies serious higher-level individuality cases. It does not mean the word *superorganism* ends the debate.

The worker is still biologically real. The queen is real. Individual insects have bodies, nervous systems, behaviors, and evolutionary histories. Yet some causal and evolutionary questions may become clearer when the colony is also treated as a unit.

This is our first warning that individuality can be nested. The existence of lower-level individuals does not automatically prohibit a higher-level individual. But neither does coordination automatically create one.

More Than One Individual

The word *individual* can hide several empirical questions that do not always have the same boundary.

An evolutionary biologist may ask which entities exhibit the reproduction, heredity, variation, fitness differences, and response to selection relevant to the problem. A physiologist may care about integration, regulation, development, repair, or functional dependence. Other approaches can emphasize autonomy, immunity, ecology, development, or organizational closure.[3]

These are not merely different opinions about vocabulary. They can be different scientific questions asked of the same material system. A colony might be strongly integrated physiologically while its evolutionary individuality remains more complicated. A clonal lineage may be one genet while its ramets behave as partly autonomous physiological units. A host and its microbial partners may be deeply interdependent in one domain while failing a stronger criterion in another.

Evidence for one kind of individuality cannot silently substitute for evidence for another.

Macroscopic Life therefore cannot ask whether a candidate system is simply “an individual” and leave the term undefined. We must state which individuality claim is being tested. At larger scales that discipline becomes essential, because ambiguity otherwise favors whatever conclusion we already wanted.

Many Becoming One

Now the problem turns inside out.

The literature on major evolutionary transitions examines how cooperative groups of lower-level entities can become more integrated higher-level entities. Different frameworks formalize that transition differently. For evolutionary individuality, however, higher-level reproduction, heritable fitness differences, selection, division of labor,

interdependence, coordination, and the reorganization of lower-level conflict are among the central issues.[4]

Multicellularity gives us the easiest intuition. Multicellular lineages descend from unicellular ancestors in which cells existed as complete reproductive individuals. In later multicellular organization, cells can participate in development, specialization, signaling, regulation, reproduction, repair, and constraints on lower-level proliferation. Across evolutionary transitions these relationships can become progressively reorganized rather than crossing one universal instant at which “many” suddenly becomes “one.”[4]

Cooperation can deepen. Reproductive roles can specialize. Dependencies can accumulate. Lower-level conflict can be altered, limited, or redirected. The balance of evolutionary interests can change. Under particular evolutionary-individuality frameworks, the higher-level entity can thereby acquire increasing significance as a unit of selection and reproduction.[2][4]

INDIVIDUALITY MAY BE AN EVOLUTIONARY ACHIEVEMENT, NOT A STARTING CONDITION.

The claim is strongest here: in evolutionary transitions in individuality. It is not a declaration that every physiological, developmental, ecological, or other kind of individuality must be historically achieved in exactly the same way.

Still, the scale reversal matters. “Can many become one?” is not a question invented because we wanted to imagine forests or planets as organisms. Evolutionary biology already contains cases in which many-becoming-one is part of the problem scientists are trying to explain.

The warning matters equally. Evolutionary transitions do not mean every network, ecosystem, society, or cooperating group is beginning such a transition. Possibility is not evidence.

When the Parts Can Win

Cooperation is not enough because cooperating entities can retain partly divergent interests.

A useful stress test appears when a lower-level trajectory can conflict with persistence of the proposed whole. In multicellular organisms, cell proliferation provides a simple illustration: proliferation is necessary for development and repair, while dysregulated proliferation can contribute to disease and threaten organism-level persistence. That example is not a theory of cancer. It simply makes lower-level versus higher-level consequences easy to see.

Evolutionary-transition research pays close attention to cooperation and conflict. In relevant systems, relatedness, reproductive specialization, policing, bottlenecks, partner fidelity, or other mechanisms can alter opportunities for disruptive lower-level competition. But active conflict suppression should not be treated as a universal prerequisite that must appear in the same way at the origin of every transition; recent work continues to refine how conflict and conflict resolution matter across different transitions and stages.[5]

The question is narrower: does the proposed individuality framework predict a pattern of lower-level conflict, alignment, or constraint that the system actually exhibits?

What happens when success for the part and persistence of the whole stop pointing in the same direction?

A Temporary Body

Slime molds make our counting problem stranger.

In the social amoeba *Dictyostelium discoideum*, amoebae feed as individual cells when resources are available. After food depletion, cells aggregate, form a multicellular motile slug, and later produce a fruiting body. A fraction of the cells contribute to the stalk while others become reproductive spores.[6]

The collective phase is real. So is the earlier cellular phase. And because aggregation can bring together genetically different cells, the system can expose conflict and cheating: genotypes can differ in their representation among reproductive spores.[6]

Individuality here cannot be understood simply by finding one permanent skin and following it from birth to death. Organization changes across the life cycle. Cells become incorporated into a temporary multicellular organization, differentiate, and return reproductive propagules to the next cycle.

The boundary moves because the biology moves.

You and Your Microbes

Now return to your own body, but widen the frame.

Animals live in intimate association with microorganisms. Microbial communities can influence metabolism, development, immune function, nutrition, and resistance to pathogens. In many systems, host biology cannot be understood well without studying those associations.

It is tempting to take the next rhetorical step: therefore the host and its microbiome are one evolutionary individual.

That step is not free.

The holobiont and hologenome literature contains a genuine dispute about what follows from host-microbe association. Douglas and Werren argue against treating host and microbiome automatically as a single cooperative unit of selection. Theis and colleagues defend a broader eco-evolutionary holobiont/hologenome framework and emphasize multilevel processes rather than requiring every association to constitute one primary unit of selection.[7][8]

So separate the questions. Does the microbial community contribute physiologically? Does it influence development? Is the association ecologically important? Are partners reliably transmitted? Is community structure heritable at the relevant level? Are fitness interests aligned in the way the hypothesis predicts? Is there selection at the host-microbe

collective level in the sense required by the particular evolutionary-individual claim?

Those questions may produce different answers. Dependence, association, and evolutionary individuality cannot be silently collapsed into one another.

This is why “everything is connected” is scientifically inadequate. Connection is the beginning of the measurement problem, not the solution.

Before the Verdict

We need a way to keep different dimensions visible without pretending biology has agreed on one master score.

For Macroscopic Life, the safest approach is a notebook scaffold: ask separately about boundary consequence, integration, persistence, conflict or alignment where relevant, recovery or reconstitution, stability across component turnover, and evolutionary individuality when that is the claim. These are prompts for investigation, not validated axes with common units. Another scientific framework might choose different dimensions entirely.

We will not assign arbitrary weights, add them together, and announce that a forest is 73 percent individual. The point of keeping the dimensions separate is to prevent exactly that kind of false precision.

A system can be strongly integrated yet weak as an evolutionary individual. Another can have a clear reproductive lineage while remaining physiologically modular. A multidimensional description can therefore precede a framework-specific verdict.

And every claimed dimension still needs a way to lose.

Membership and Continuity

Return to the boundary. Suppose we identify components whose relation to a proposed whole can change. What follows if they become members of that organization rather than remaining outside it?

Membership need not mean literally crossing a sharp line in space. It can be developmental, genealogical, physiological, functional, or organizational. Depending on the framework, membership might alter resource access, signaling relationships, developmental fate, reproductive opportunity, regulation, recovery, survival probability, or exposure to higher-level constraints.

The stronger the claim that a boundary marks an individual, the more useful it becomes to ask whether membership predicts systematic causal consequences. Where the biology permits, transplantation, grafting, migration, fusion, succession, natural turnover, or other membership changes can provide tests.

Then let the parts change. Bodies persist despite extensive turnover at multiple material levels. Colonies replace workers. Plants grow and lose modules. Ecological communities change composition. If a proposed higher-level identity remains scientifically useful across the relevant turnover, that supports organized continuity rather than identity with a fixed inventory of parts.

But the hurricane returns as our adversary. Its material turns over too. So does an institution's membership. Turnover stability supports higher-level identity; it does not establish biological individuality by itself. The question is whether the relevant integration, constraints, functions, development, or evolutionary continuity remain predictively organized as components change.

Split It. Join It.

Where the biology permits interpretable division or fusion, intervention can make an individuality claim expensive.

Divide the candidate system. Perhaps one fragment retains the relevant organization while the other fails. Perhaps both reorganize into viable units. Perhaps both persist but lose the claimed higher-level function. Perhaps the pieces later fuse. Surviving division does not automatically disprove prior individuality; many legitimate biological systems reproduce or propagate through fission or fragmentation.

The value of the split test is prospective. A framework should tell us what division is expected to disrupt, preserve, duplicate, or reconstruct.

Then reverse the question. If two distinguishable systems become tightly coupled, when—if ever—does a new individual emerge under the framework being tested? Aggregative multicellularity and symbiosis give biology versions of that problem. We should not wait until the result looks organism-like and then declare victory. Freeze the criteria first.

Return to the Forest

Now risk the same hypothetical forest again, this time as an individuality candidate. Call the proposed boundary *H* and specify serious alternatives before confirmation—*H*1, *H*2, *H*3, or other empirically motivated partitions. Depending on the site, alternatives might follow a watershed, soil domain, disturbance regime, species-interaction network, mycorrhizal network, or another empirically motivated partition. The preferred forest outline does not compete against nothing.

Define the candidate membership relation prospectively rather than assuming that every tree, fungus, or patch inside a mapped outline is biologically “incorporated.” Specify what *H* predicts about component fate, coupling, recovery or reconstitution, persistence across turnover, and response to disturbances that cross the proposed boundary. Freeze the recovery variables, turnover horizon, disturbance comparisons, and results that would favor another partition before the confirmation test.

Then perturb organization within, near, and across the candidate boundaries where such interventions are scientifically interpretable and ethically acceptable. Ask whether *H* predicts recovery trajectories or component outcomes better than the alternatives, whether those predictions survive component turnover, and whether they generalize to another site or period.

The component and environmental rivals receive full resources: spatial gradients, species interactions, nonlinear feedbacks, network

structure, local histories, hydrology, soil variation, latent states, and disturbance history. If a simple forest-level variable wins only because the ecological rival was crippled, the forest did not win.

The predictive comparison keeps our familiar form:

$$PA_{\text{individual}} = \text{Performance}(\text{Individual-Level Model H}) - \text{Performance}(\text{Best Fair Partition/Rival})$$

The familiar Prediction Advantage discipline applies. Positive $PA_{\text{individual}}$ means H has earned predictive utility for that task; it does not mean we have discovered an organism in the woods.

Remove the Individual

Now remove H as a privileged individual-level partition. Give the rival model the components, interactions, network topology, environment, histories, latent states, and comparable statistical capacity.

Ask what disappears. Can the rival still predict recovery across turnover, outcomes after membership changes, responses to boundary-crossing disturbances, and interventions across different lower-level implementations? Can it compress the same range of outcomes without rebuilding a separate explanation for every local configuration?

If nothing important disappears, the stronger individuality claim has not earned explanatory necessity under that test. That result is allowed.

Model A may win: component, network, population, and environmental descriptions suffice. Model B may win: H is usefully modeled as an integrated higher-level system, but stronger individuality remains unsupported or framework dependent. Model C may be earned only for the explicitly named individuality framework. An evolutionary-individual claim must satisfy evolutionary criteria; a physiological-individual claim must satisfy physiological criteria. Success in one cannot be imported into the other. The result may also remain unresolved, or the test may be invalid because the boundary was post hoc, the rivals unfair, or the criteria circular.

What we cannot do is collect suggestive dimensions after the fact and use them to rescue a losing claim. The notebook scaffold is not an escape hatch.

What One Thing Means

Return to your body.

It still makes sense to call you an individual. Nothing in this chapter was designed to dissolve you into cells, microbes, organs, or molecules. The opposite has happened.

“One” has become more impressive.

A biological individual can require coordination across enormous numbers of components, persistence through turnover, mechanisms of development and repair, constraints on lower-level behavior, and an evolutionary history in which relationships among parts became organized so that a higher-level unit mattered.

We began by asking whether a large system could become one thing. Biology gives us a stranger answer: some of the things we call one were not always one.

That does not make the forest an individual. It does not make Earth an organism. It does not give us permission to enlarge the word *life* until everything fits inside it.

It gives us something better: scientific precedent for asking how many can become one, and an experimental obligation to determine when they have not.

If a whole can become an individual, another question follows. Can that higher-level organization generate another system of the same relevant kind?

Can the whole reproduce itself?

FIGURE 14 - APPROVED POSITION

Structural proof placeholder. Insert only the frozen publication-authority visual at final composition; do not invent or substitute artwork.

PART VI

THE LINEAGE

An individual becomes an evolutionary question only when descent, inheritance, and differential success can be tested at its level.

The Reproduction

Hold a seed in your hand.

It is not a miniature adult tree. Most of the matter that could someday compose a trunk, roots, branches, and leaves is not inside it. A growing tree will take carbon from the atmosphere, water and minerals from the ground, energy from sunlight, and countless influences from an environment the parent did not package into the seed.

Seed production is part of the tree's reproductive life cycle, yet the offspring need not begin by resembling the adult parent.

So reproduction cannot require an exact material copy. What, then, makes one system an offspring rather than merely another thing of the same kind?

More of the Pattern

Place the seed beside a crystal. Under the right conditions, new material joins an ordered structure and takes on a pattern related to what is already there. That kind of templating and pattern extension is scientifically important, but pattern extension alone does not establish the parent–offspring reproduction claim we are testing here.

Now strike a match. Fire introduces a different problem: propagation. One region of combustion can contribute causally to another beginning as a flame front spreads through new fuel. Hurricanes introduce another: recurrence. Similar organized processes can form repeatedly because similar physical conditions recur.

More of the pattern has appeared in every case. That is still not enough.

RECURRENCE IS NOT REPRODUCTION.

A pattern can appear again because a previous system generated a descendant. It can propagate through a medium. Or the world can repeatedly generate conditions from which similar structures form. Those possibilities cannot be collapsed into one verb.

When One Becomes Two

Repair makes the problem harder. A salamander can regenerate a lost limb without producing another salamander. In other biological systems, however, a separated fragment can regenerate into another individual and function in an asexual reproductive life cycle.

The initiating event cannot decide the category. Cutting something does not make reproduction happen, and surviving a cut does not prove that reproduction did not happen. Individuation, life history, developmental organization, and parent–offspring lineage all matter under the framework being used.

The question inherited from Chapter 12 therefore returns in a new form: when did one become two?

Parent and Offspring

Strip reproduction down to a causal scaffold:

Parent $P \rightarrow$ generative process $G \rightarrow$ offspring O

P must be a unit we had reason to identify without using the reproduction claim itself to rescue its individuality. O must become a unit under a defensible framework rather than merely because it resembles P . And the generative process must support a causal historical relation at the level where reproduction is being claimed.

SIMILARITY DOES NOT ESTABLISH LINEAGE.

Two systems can resemble one another because they share constraints or assembly conditions. Different lineages can converge on similar forms. Communities can reassemble into similar states. A strong contested lineage claim therefore requires evidence of causal history, even when that history must be inferred indirectly rather than read from one unique marker.[1]

Reproduction Inside Reproduction

Your body contains enormous numbers of cell divisions and lower-level lineage events that are not reproduction of you. A skin-cell lineage can continue without producing another human being.

Now enlarge the frame. Trees reproduce inside forests. Fungi, insects, birds, microbes, and other organisms generate descendants while populations turn over and forest structure persists.

It is tempting to add those events together and promote the verb to the next scale. We cannot.

REPRODUCTION OF THE PARTS IS NOT, BY ITSELF, REPRODUCTION OF THE WHOLE.

The words *by itself* matter. Nested reproduction is biologically possible. Lower-level implementation does not automatically invalidate a higher-level reproductive description, but neither does it establish one.

The earlier rule applies again:

A FUNCTION CAN EXIST INSIDE A SYSTEM BEFORE THE SYSTEM OWNS THE FUNCTION.

The Colony Leaves Home

Social insects show why the level question is legitimate rather than merely philosophical. Across different taxa and life histories, colonies may be founded independently by reproductive individuals or through dependent processes such as budding, fission, or swarming in which groups from an existing colony participate in founding another.[2][3]

Under relevant organismality or evolutionary frameworks, some of these processes can be analyzed as colony-level reproduction. That does not erase the individual insects. It creates nested descriptions whose usefulness has to be evaluated at the level being claimed.

A new colony may therefore descend from an old colony through a process implemented by organisms within both. The question is not whether components carried the process—they must—but whether the colony-level organization contributes causal and explanatory work not exhausted by the component description.

The Scaffolding Problem

A photocopier complicates the vocabulary more than intuition first suggests. In a broad parent–offspring framework, externally scaffolded copying can be analyzed as reproduction of tokens even though the paper does not contain the machinery that makes the copy. That obviously does not make a sheet of paper an organism or establish that it is alive.[1]

Biological reproduction is also never absolutely self-sufficient. Reproducers depend on matter, energy, environments, developmental contexts, partners, and sometimes extensive external machinery. Godfrey-Smith's framework explicitly distinguishes simple, collective, and scaffolded forms of reproduction.[1]

So external dependence cannot decide the question for us.

Imagine instead a factory whose organized activity contributes to constructing another functioning factory. The new factory can later participate in generating another. We would still need to state the reproduction framework before deciding what the word means, but the causal-level question would now be serious rather than rhetorical.

Parental causal responsibility can also come in degrees. P need not be the sole cause of O, and P need not be sufficient by itself. The question is whether P occupies the relevant parent role in a causal lineage and whether the proposed level earns explanatory work beyond its scaffolds and components.[1]

Reconstructed

Imagine a hurricane disappearing. Weeks later, another hurricane forms in the same basin. Similar oceanic and atmospheric conditions can repeatedly generate processes from which hurricanes form. Similarity and recurrence do not make the earlier storm a parent.

That gives us the chapter's strongest adversary: **reconstruction without lineage**.

A system can appear again because components remain available, environments repeatedly channel assembly, and historical conditions make some states more likely than others. Ecology supplies serious versions of this problem. Communities can change after disturbance and later regain aspects of earlier organization through survival, dispersal, recolonization, environmental filtering, succession, and historical effects.

A later system may therefore depend causally on an earlier state without being its offspring in the stronger sense tested here.

Causal influence is cheaper than reproduction.

What the Parent Leaves Behind

Inheritance can involve more than direct transmission of genetic material. Organisms can modify environments in ways that alter conditions experienced later. Niche-construction research describes persistent organism-caused environmental modification as ecological inheritance and examines how such legacies can feed back into ecological and evolutionary dynamics.[4]

A forest can leave extensive legacies: altered soils, surviving seed banks and roots, microbial and fungal communities, dead wood, hydrological changes, resource distributions, and disturbance-created opportunities for establishment. A later community may therefore carry causal history from an earlier one even when much of the visible system disappeared.

But a causal legacy is not automatically a parent–offspring relation. The experiment has to distinguish continuing organization, inherited environmental structure, component lineage, and recurring assembly conditions rather than labeling all historical influence reproduction.

The Burned Forest

Imagine a forest before a severe fire. We have already done the Chapter 12 work: the candidate boundary and relevant individuality framework were specified before the disturbance. We are not allowed to invent the forest-individual after seeing what happens.

Then the fire comes. Years pass. Forest stands there again.

What happened?

Perhaps the same higher-level individual persisted through disturbance. Perhaps organization regenerated from surviving roots, seed banks, organisms, and relationships. Perhaps component lineages recolonized the site. Perhaps outside propagules arrived. Perhaps physical conditions and ecological history reconstructed a similar community. Perhaps environmental legacies biased the later system toward the earlier one. Perhaps an earlier forest participated in a genuine higher-level generative lineage.

Or perhaps the word *forest* is doing too much work, letting one category name masquerade as ancestry.

Those possibilities do not all make the same reproduction claim, and several require no forest-level reproduction at all. Observation that “the forest came back” cannot decide among them.

The Second Forest

So make the test harder. Leave Forest P standing.

In a deliberately composite hypothetical landscape, biological material and candidate ecological effects associated with P reach suitable habitat elsewhere. Seeds or other propagules disperse. Animals move material. Microbes or fungi may travel through relevant pathways.

Water may transport biological material. Over time, an independently justified Forest O develops.

Now O cannot simply be P persisting in the same place. But the strongest rival remains.

Did Forest P produce Forest O? Or did organisms from P reproduce and disperse into suitable habitat where environmental filtering and ecological interaction assembled another forest?

Those explanations assign generative work to different levels. This is where Macroscopic Life either earns the larger verb or gives it back.

Lineage

Before seeing the outcome, define P, define what will count as O, specify the candidate generative pathways and expected lineage evidence, choose the time horizon, state the component and environmental rivals, and record what would count against parentage.

Component ancestry, propagule provenance, transmitted symbionts, environmental modifications, developmental sequence, or other historical signatures might contribute. The relevant markers must be specified prospectively; we cannot shop afterward for whichever channel happens to connect P and O.

Lineage must be inferred from causal history rather than appearance alone. Even strong component ancestry remains component ancestry: if many trees in O descend from trees in P, we have established ancestry among trees before we have established reproduction of forests.

The Parent-Contribution Test

Now attack the hypothesis.

Call the project-specific experiment the **Parent-Contribution Test**. It asks not whether P is an indispensable cause of O, but whether the hypothesized contribution of P changes O formation in prospectively predicted ways after fair environmental, component, and historical rivals are preserved.

A useful comparison might separate continuing P-level organization, P-created environmental legacy, component propagules, and background assembly conditions. Natural experiments or historical/provenance evidence may substitute where direct manipulation would be impossible or unethical.

The question is not merely whether P matters. It is how P matters, how much the evidence supports that contribution, and whether the contribution belongs scientifically to the proposed whole.

Scramble the Package

Suppose successful establishment depends on a structured combination of biological materials, environmental modifications, spatial arrangement, and timing generated through P. Keep the ingredients but scramble the proposed organization. Does O still form with the same probability and trajectory?

For Macroscopic Life, call the prospectively hypothesized transmission structure a **founding package**. This is a project scaffold, not an established biological category, and it need not be spatially compact. It cannot mean “whatever variables happen to predict O.” It must correspond to a specified causal transmission structure generated by P and must compete against matched material-only or unstructured alternatives.

Otherwise the package can explain everything after the fact and nothing before it.

Make the Whole Compete

Let H be the higher-level parent–offspring model and B the best fair component/environmental rival:

$$PA_{\text{reproduction}} = \text{Performance}(H) - \text{Performance}(B)$$

The established Prediction Advantage discipline applies. H must predict something the best reconstruction model does not, but positive

advantage alone cannot certify reproduction. The stronger claim still requires independently justified parent and offspring units and causal lineage, using intervention where feasible and provenance, natural experiments, historical evidence, or convergent triangulation where it is not.

Let It Fail

Suppose the forest loses. Component reproduction, dispersal, environmental filtering, ecological inheritance, and local history may explain the result without forest-level reproduction: Model A. Forest-level organization may improve prediction or intervention while parent–offspring status remains unearned: Model B. **MODEL B IS A DISCOVERY, NOT A CONSOLATION PRIZE.** Model C requires the stronger property-specific result: P and O qualify as relevant units under an explicit framework, a causal generative lineage connects them, and the higher-level account survives the best fair rivals through evidence appropriate to the system.

Even then, we have earned only reproduction. We have not automatically earned intelligence, consciousness, purpose, life under every definition, or Darwinian evolution at the higher level. Heredity, variation, and differential success remain separate tests.

Properties do not get to smuggle one another across the border.

And failure can still be interesting. If ecological systems are repeatedly reconstructed through component lineages, environmental inheritance, network constraints, and physical conditions rather than reproduced as higher-level parents and offspring, that would still reveal something important about how large systems persist through time.

We do not need to win the word *reproduction* to discover a new level of order.

Earth Does Not Get a Child for Free

Terrestrial organisms and biological material have been carried beyond Earth's surface by human spaceflight, and material can move naturally between planetary bodies. Neither fact makes Earth a reproducing planetary individual.

If a technological civilization someday establishes a living system elsewhere, several nested descriptions may be available: organisms reproduced, humans acted, a civilization expanded, or an engineered biosphere was constructed. None automatically entails that Earth had an offspring.

Which unit was the parent? What organization generated the proposed offspring? What was transmitted? What independently individuates O? What scientific work is lost if the planetary-parent description is removed?

Earth is especially difficult because we currently have no known lineage of Earth-like planetary individuals through which to replicate such a claim. We cannot repair that absence with poetry.

Earth must earn Earth.

What Survives the Journey

Return to the seed.

The tree that emerges will not be identical to its parent. That is not a failure of reproduction. It is part of what makes evolution possible.

A lineage requires continuity, while evolution requires that relevant continuity coexist with variation. Something must persist strongly enough for ancestry to matter, and something must be capable of differing strongly enough for descendants not to be interchangeable repetitions.

Reproduction therefore opens another door without answering what lies beyond it. One whole following another is not enough. If higher-level lineages are to evolve, differences among those wholes

must somehow survive the passage between generations.

If the whole can reproduce, can what differs between wholes survive the journey?

The Inheritance

Open an old family photograph and the experiment seems almost too easy.

A child has her father's eyes. A grandson carries the shape of his grandmother's face. Other similarities—a voice cadence, a gesture, a way of standing in a doorway—may travel through learning, culture, shared surroundings, or several pathways at once. Family resemblance makes inheritance feel obvious precisely because the photograph hides the mechanism.

Families share more than ancestry. They can share food, language, homes, neighborhoods, expectations, pollutants, microbes, customs, and histories. A descendant can resemble a parent because ancestry causally shaped the descendant, because both developed under similar conditions, or because the descendant rebuilt a similar state through processes correlated with lineage.

Resemblance gives us a question, not an answer.

The question left by the previous chapter is harder than it first appeared. If a whole can reproduce, can what differs between wholes survive the journey?

More precisely: what causal dependence on ancestry survives the generational transition?

The Difference Is the Detector

Return to the seed.

If every parent and offspring were identical in every measured respect, heredity could still occur, but it would be difficult to dissect. Variation gives us contrast.

Imagine two parent lineages differing in a prospectively specified state D:

$P1[D1] \rightarrow O1[?]$

$P2[D2] \rightarrow O2[?]$

If descendants of P1 tend toward D1 while descendants of P2 tend toward D2, we have found a pattern worth explaining. We have not yet found its mechanism.

D also cannot be merely a convenient human label. It must be operationally specified before the descendant result, measured consistently, and—if claimed as a higher-level property—contain independently justified relational or system-level information rather than simply renaming the composition of the parts.

Variation is therefore the detector, not the property under test. It lets us ask whether descendant differences follow ancestry, components, current environment, inherited environment, developmental reconstruction, filtering, or some combination of them.

Memory Is Not Inheritance

We have met persistence before.

A tree can retain a wound. A nervous system can remain changed by an earlier event. A forest can carry the effects of drought, fire, disease, or disturbance into a later state.

Chapter 8 called this a memory problem: does history leave a physical difference that later changes what the continuing system does?

Heredity adds a lineage relation. The descendant state must depend causally on ancestry across a generational boundary under the framework being tested. No material token has to cross that boundary intact.

A state can persist for centuries inside one continuing system without being inherited. A descendant can also resemble its parent without the relevant resemblance being lineage-dependent.

Memory asks whether an earlier event remains causally available within a continuing system. Heredity adds a generational boundary: ancestry must help generate or constrain descendant state. The mechanisms can overlap; the properties do not.

The Same House

Imagine two generations living in the same house.

They drink water from the same pipes, eat similar foods, breathe the same air, hear the same language, and absorb many of the same routines. If a measurable trait appears in both generations, ancestry and environment are tangled before the experiment begins.

This is why biologists use designs such as common gardens and reciprocal transplants: lineage and present environment are made to stop pointing in exactly the same direction. If a lineage-associated difference persists under matched conditions, a simple current-environment explanation weakens. If it follows the exchanged environment, environmental causation gains strength.

For the forest problem later in this chapter, we borrow the causal logic rather than requiring a literal classical common garden. Mesocosms, provenance studies, matched sites, natural experiments, historical contrasts, and targeted manipulations can serve analogous purposes.

And a common environment never means “genes isolated.” Descendants can arrive carrying parental effects, epigenetic states, cytoplasmic factors, symbionts, microbiomes, developmental histories, transmitted resources, or other lineage-associated differences. Matching the present does not erase the past.

The experiment is not “genes or environment?” It is: which causal channels still carry lineage-associated differences when particular rivals are separated?

What Counts as Inherited?

DNA sequence inheritance is central to biology. It is not the only inheritance pathway discussed in evolutionary research.

Inclusive-inheritance literature examines genetic, epigenetic, parental, ecological, and cultural pathways, among others. These are heterogeneous mechanisms and theoretical categories, not one unified inheritance system. Their stability, fidelity, duration, mechanisms, and evolutionary importance must be established case by case.[1]

Epigenetics makes the need for caution especially clear. Molecular mechanisms capable of supporting transgenerational epigenetic inheritance are well documented in several model systems, but their operation and persistence differ substantially among taxa.[2] In mammals, direct exposure effects can also be confused with true transgenerational inheritance. Evidence for comparable mechanisms in humans remains much more limited and contested than popular summaries often suggest.[3]

An inheritance channel therefore has to be identified rather than assumed.

That channel need not consist of a mature trait physically copied from parent to offspring. Under some biological mechanisms and frameworks, descendant states can be reconstructed through lineage-dependent processes.

Rebuilt Each Generation

We often imagine heredity as copying.

Something in the parent is duplicated and placed in the offspring. That metaphor works well for some molecular processes, but complex traits and developmental systems can depend on ancestry without being copied as completed objects.

Epigenetic research, for example, distinguishes direct replication of some states from mechanisms in which descendant states are reconstructed from inherited signals or lineage-dependent machinery.[2]

The mature trait does not have to exist as a miniature completed object inside the parent. What matters for the stronger claim tested here is whether ancestry contributes reproducibly to descendant state through a causal architecture that defeats fair non-lineage rivals.

HEREDITY DOES NOT REQUIRE PERFECT COPYING. IT REQUIRES LINEAGE-DEPENDENT CAUSAL CONTINUITY.

Here that sentence is an operational rule for Macroscopic Life, not a claim that every scientific framework defines heredity in exactly those words.

A forest-level property would therefore not need to be packaged like a chromosome. But a claim of forest-level heredity would require evidence that forest ancestry contributes to rebuilding or constraining the descendant property beyond what component inheritance and environmental reconstruction already explain.

The Selection Counterfeit

Now imagine noisy beginnings.

A parent lineage has state D. Descendants begin with several states. Developmental or ecological filtering removes some variants; in other cases, variants possessing particular properties may survive or reproduce disproportionately. By the time we measure mature descendants, D-like states may dominate.

Parent and descendant now look remarkably alike, even though the founding distribution may have been much less faithful.

WHAT SURVIVES SELECTION IS NOT NECESSARILY WHAT WAS FAITHFULLY TRANSMITTED.

This is why timing matters. Where possible, measure descendants at founding or birth, during development, and after strong filtering:

founding/birth → development →
post-filtering state

But absence of an expressed D at founding does not defeat heredity. A lineage-dependent developmental process may predictably reconstruct D later. The sequence tells us when the lineage-associated difference appears and whether filtering can explain mature resemblance.

Environmental filtering and evolutionary selection are also not synonyms. A condition can remove or suppress variants without establishing the full higher-level selection claim that the next chapter will test.

The warning is narrower: if we inspect only survivors, filtering or selection can make inheritance appear more faithful than the founding process actually was.

Heredity Is Not Heritability

There is another word close enough to cause trouble: heritability.

In quantitative genetics, heritability is a population- and environment-specific ratio of variance components under a specified model. Narrow-sense heritability concerns additive genetic variance; broad-sense heritability includes total genetic variance. Neither is a percentage of an individual's trait “caused by genes,” and neither is a universal constant attached to the trait.[4]

Change the population, environment, measurement, or genetic structure and the estimate can change. A high heritability estimate also does not identify the causal inheritance mechanism.[4]

Heredity and heritability therefore answer different questions. One concerns causal dependence across lineage. The other describes the statistical structure of variation in a specified population and context.

Two Forest Families

Return to the forest.

Suppose Chapter 13's reproduction problem has survived far enough that candidate parent and offspring forest units can be identified

prospectively. Before descendants form, two candidate parent lineages differ in D:

$$P1[D1] \rightarrow O1[?]$$
$$P2[D2] \rightarrow O2[?]$$

D might be a network-level recovery trajectory, a response threshold, spatial interaction structure, resource-flow organization, or another independently justified relational variable. It cannot simply be the allele frequency of one tree species relabeled as a forest property.

And we cannot inspect hundreds of differences after O1 and O2 appear, choose whichever one happens to resemble the parent, and call it inherited. D, its measurement scale, expected descendant relationship, and failure conditions must be specified before the result.

Suppose years later O1 resembles P1 and O2 resembles P2.

Now try to destroy the inheritance claim.

Ground, Parts, Legacy

First attack the current environment. If D1 appears only where soil, hydrology, climate, disturbance history, or other present conditions resemble P1's setting, ancestry may be unnecessary. Matched sites, exchanged conditions, natural experiments, and model comparisons can help separate lineage from present environment. But matched current conditions can still preserve environmental legacies inherited from the past, so current environment and inherited environment remain distinct rivals.

Then attack the components. O1 may resemble P1 because the same tree genotype, fungal lineage, microbial consortium, seed source, or other component carries the relevant difference. That could be excellent component-level inheritance without forest-level inheritance. Where manipulation is impossible, provenance tracking, statistical decomposition, historical contrasts, and convergent natural experiments can still ask whether D follows particular parts or the proposed whole-level lineage.

Finally attack the legacy. Forests can modify soils, hydrology, seed banks, deadwood, microbial communities, resource distributions, and disturbance conditions in ways that persist. Niche-construction theory describes such organism-caused environmental legacies as ecological inheritance.[5] That causal continuity can be fully real at organismal or population levels while ownership by a forest individual remains unestablished.

If O1 develops D1 inside an environment partly produced by P1, the question is not whether the legacy exists. It is whether an explicit forest-level inheritance model explains something beyond the strongest component and ecological-inheritance accounts.

One more rival is easy to miss: the observer. If researchers divide forests into Type A and Type B using human-selected bins, recurrence of those categories can manufacture apparent continuity. D must support prospective, sufficiently observer-independent measurement rather than merely repeat our classification scheme.

Who Owns the Inheritance?

Call the project-specific test the **Inheritance Ownership Test**.

After accounting for component ancestry, current environment, inherited environment, developmental reconstruction, filtering, and observer-defined classification, does ancestry of the proposed parent whole add prospective predictive, explanatory, or causal leverage over descendant higher-level state?

The parent and offspring units must already be independently justified, D specified before the outcome, and lineage established without using D itself as proof. Then make the higher level compete.

Let H be the higher-level inheritance model and B the strongest fair rival:

$$PA_inheritance = Performance(H) - Performance(B)$$

The established Prediction Advantage discipline applies. If H wins, the higher-level variable is doing scientific work; the equation does not define heredity. Lineage, level, causal architecture, and the strongest replication or triangulation the system permits remain independent burdens.

Let It Fail

Suppose the forest loses again. Component inheritance, present or inherited environment, reconstruction, filtering, or lower-level survival and reproduction may explain descendant differences without a forest-level inherited property: Model A. A higher-level lineage variable may still improve prediction or explanation without establishing inheritance owned by the whole: Model B, still a discovery. Model C requires independently justified parent and offspring units plus prospective lineage-dependent continuity in D that survives the strongest component, environmental, reconstructive, filtering, and classification rivals and appropriate replication or triangulation.

Even that earns only the property tested here. It does not establish agency, intelligence, consciousness, life under every definition, or evolution by natural selection at the higher level.

A forest can have history without higher-level heredity. If higher-level heredity were ever demonstrated, selection would still have to earn itself separately.

Earth Cannot Inherit From Itself

Earth has a past. Its present atmosphere depends on earlier atmospheres, and geological, chemical, and biological processes leave consequences that shape later conditions. Organisms within Earth participate in vast numbers of genuine hereditary lineages.

None of that establishes heredity of Earth.

Inheritance among components inside a proposed whole cannot silently become inheritance by the whole. For the property tested here,

planetary heredity would require an independently justified lineage relation among planetary parent and offspring units under an explicit framework.

Without that, saying Earth “passes on” its traits remains metaphor. The scale of the system cannot rescue a missing generation.

The Difference That Changes the Future

Return once more to the family photograph.

A difference can survive a generation without being perfectly copied. It can be reconstructed, environmentally mediated, or carried through several interacting pathways. What matters is discovering why descendant state depends on ancestry and at what level that dependence belongs.

But heredity alone gives no difference an evolutionary direction.

For that, inherited differences must become connected to differential persistence or production of descendants under relevant conditions. That is another property and another test.

A difference that survives a generation has a history. A difference that changes who leaves descendants can begin to have an evolutionary future.

If differences among wholes can survive generations, can those differences change which wholes leave descendants?

The Selection

A race is easy to score.

The runners begin at the same line, travel the same course, and one reaches the finish first. Whatever else we might argue about—the weather, training, shoes, luck—we know what winning means because the competition has given us an explicit currency.

Now replace the runners with relay teams.

The fastest individual runner in the stadium does not necessarily belong to the winning team. One runner can lose her leg of the race while her team wins the event. A slower runner can belong to the faster whole.

Nothing mysterious has happened. We changed the level at which success was measured. Athletic winning is only an analogy here, not biological fitness, but it exposes the problem we need: success can be defined at more than one level.

In evolution, lower-level components can differ in survival, reproduction, and representation while larger collectives also differ in outcomes relevant to their own lineages. Sometimes those differences point in the same direction. Sometimes they do not.

Before calling any of this selection at a higher level, we need to answer a more basic question.

What, exactly, won?

The Team

Imagine two relay teams. Team A defeats Team B repeatedly. We measure dozens of properties and discover that Team A wears red uniforms, stretches in a particular sequence, and has a distinctive running order. Then we notice something simpler: Team A happens to contain the fastest runner in every race.

As a thought experiment, move that runner to Team B and imagine the advantage following the runner. Real biological transfers can change interactions and context, so this is not a universal causal experiment. It is a way to expose the by-product problem.

The teams really differed in success. But success measured between teams did not establish that a team-level property caused the difference.

The same danger appears when lower-level units are sorted into groups. Groups containing fitter components can show greater group-associated output even when the causal advantage belongs to those components. In some multilevel frameworks that output is tracked through member reproduction rather than literal reproduction of the group itself.

A whole can win because its parts were already winners. The difficult case begins when organization of the whole changes the fate of the relevant lineage in a way the parts alone do not explain.

Counting Is Not Explaining

Evolutionary theory has powerful ways to partition change across levels. The Price equation can express evolutionary change through covariance between character and fitness plus a transmission term, and multilevel formulations can recursively partition covariance associated with differences among and within groups.[1]

That is valuable. It can show where statistical change appears under a specified grouping and representation.

But a partition is not automatically a causal verdict, and changing scientifically plausible groupings can change what the partition reveals.

A between-group term can appear because fitter lower-level types are unevenly distributed among groups rather than because a higher-level property independently causes success.[2]

A PARTITION CAN LOCATE CHANGE WITHOUT SETTLING WHAT CAUSED IT.

Contextual analysis asks a related but different question: roughly, whether a group or contextual property predicts individual fitness after relevant individual properties are included. It can separate some by-product cases that a covariance partition alone cannot. Yet its conclusions depend on the causal structure and model specification, and local or soft competition can make a contextual effect difficult to translate into a simple story of greater group reproductive success.[2]

Counterfactual comparisons offer another route. How would the relevant components perform outside the organized collective, or inside a defensible pseudo-group lacking the organization being tested? If their expected success already accounts for the result, the higher-level claim weakens. If their performance changes because of the organized higher-level context, higher-level causal structure becomes more credible. This is evidence, not a universal criterion.[3]

No one decomposition gets to choose the causal level by itself.

Two Levels at Once

Now imagine selection processes operating in a nested collective. Inside it, component variants differ in representation across generations. Among collectives, the collectives may also differ in whatever reproductive success has independently been justified at that level.

Those processes can align. They can also conflict.

A component strategy might rise in frequency inside a collective by taking more resources, reproducing faster, or avoiding a costly cooperative task while simultaneously reducing collective reproduction. Conversely, restraint or specialization by components can sometimes reduce a component's counterfactual or within-collective advantage while improving collective success.

Many frameworks for major evolutionary transitions emphasize some combination of cooperation, conflict mediation, specialization, dependence, and altered alignment between lower- and higher-level fitness, although the mechanisms differ across transitions.[3][4]

When the part can lose while the whole wins, the level question becomes harder to ignore. But conflict is evidence, not definition. Higher-level selection does not require dramatic sacrifice, and the absence of visible conflict does not prove that only one level matters.

What Counts as Winning?

The relay race had a finish line. Biological fitness is rarely so visually obvious, even for familiar organisms, and the problem becomes sharper for proposed higher-level systems.

A forest can contain more trees, accumulate more biomass, remain stable longer, recover faster after fire, produce more component seeds, support more species, export more component propagules, or persist across centuries. Each measurement may matter. None becomes forest fitness merely because a larger number sounds like greater success.

SELECTION AT A LEVEL REQUIRES A DEFENSIBLE SUCCESS CURRENCY AT THAT LEVEL.

A quantity can be biologically important without being the reproductive currency of the level under test.

For the stronger claim here, that currency must connect to the reproductive and lineage architecture earned in the previous chapters. If we have not established what counts as a higher-level descendant, counting component offspring cannot quietly substitute for counting offspring of the whole.

Variation, lineage-dependent inheritance, and differential higher-level reproductive success form the operational scaffold for the stronger selection claim in this chapter. They should not be mistaken for a claim that every formal framework defines selection in precisely those terms.

Without a defensible level and currency, “the whole was selected” is a sentence looking for an experiment.

The Stronger Claim

Multilevel-selection theory contains several related frameworks. In the MLS-1/MLS-2 terminology developed in the literature, MLS-1 tracks effects in structured groups through the fitness of particles or lower-level entities, whereas MLS-2 treats collectives themselves as entities with fitness-bearing birth/death or reproduction/extinction properties. Authors differ in how they formulate and deploy these categories, so the labels are guides rather than universal definitions.[5]

A group context can genuinely alter individual fitness without thereby establishing that the group is a new organism or reproducing evolutionary individual. That is not lesser science. It is a different claim.

Macroscopic Life is testing the stronger property question. Suppose candidate higher-level individuals, parent-offspring lineages, and heritable higher-level differences have already survived their own tests. Do those differences contribute to which higher-level lineages leave more descendants?

If that relationship survives the causal attacks that follow, we may have crossed a much more demanding threshold than showing that individuals behave differently in groups.

The Forest Race

Assume provisionally that earlier gates have survived. Candidate forest individuals and parent-offspring relations have been independently justified, and D is a prospectively specified, observer-independent higher-level difference rather than a convenient label for component composition.

$F1[D1] \rightarrow \text{descendants: } n1$

$F2[D2] \rightarrow \text{descendants: } n2$

Suppose D1 lineages leave more candidate descendant forests than D2 lineages. That is the pattern we need to investigate. It is not yet selection for D.

Perhaps the components won. Tree genotypes, fungi, microbes, dispersers, or other lower-level lineages may explain the difference. Perhaps the groups were built differently, with D1 forests containing more of an already successful component from the beginning. Assortment and composition can create an apparent higher-level advantage without higher-level ownership.

Perhaps the environment sorted the lineages. D1 may occur on wetter soils, richer sites, favorable disturbance regimes, or climates that independently increase descendant production. A D-success association can therefore arise without D causing the outcome. Local competition can further complicate the interpretation: a group-context effect on member fitness need not imply greater total reproductive success of the proposed whole.

Perhaps the boundaries leak. Seeds, animals, fungi, and microbes move. Candidate wholes can mix, fuse, fission, disappear, or be recolonized. These processes can change lineage frequencies, erase between-whole variation, or make apparent higher-level ancestry a consequence of lower-level movement.

Perhaps chance won. Founder effects, bottlenecks, unequal lineage sizes, extinction/recolonization dynamics, demographic stochasticity, and rare disturbances can all change lineage representation without selection for D.

Perhaps we drew the teams incorrectly. If the signal exists only under one convenient forest boundary and disappears under plausible alternatives, selection cannot rescue an individual we never independently found.

Perhaps D is merely traveling with E. A lineage carrying D can become common because a correlated higher-level property E actually affects reproductive success. Selection of lineages carrying D is not automatically selection for D.

And perhaps we chose D after the race. Searching hundreds of forest properties after observing the winners and then promoting the strongest correlation manufactures an evolutionary story. Discovery and confirmation must remain separate.

Only after those rivals are treated seriously does the higher-level claim become interesting.

Who Owns the Selection?

Call the next step the **Selection Ownership Test**. It is a Macroscopic Life synthesis, not a standard named test.

After fair component, composition, environmental, local-competition, migration, demographic, grouping, linked-trait, and post-hoc rivals are considered, does a prospectively specified heritable property of the proposed whole causally contribute, under specified conditions, to differential reproductive success of higher-level lineages?

The test inherits the earlier gates: individuals and lineages are already justified, D has survived heredity, the success currency is fixed before the outcome, and component, environmental, assortment, mixing, drift, boundary, and post-hoc rivals remain live. Let H be the higher-level selection model and B the strongest fair lower-level or contextual rival.

$$PA_selection = Performance(H) - Performance(B)$$

The established Prediction Advantage discipline applies. Positive PA is predictive evidence, not causal ownership; environment, composition, or linked property E can also predict. The stronger claim therefore needs converging causal evidence that changing D, or the architecture D represents, changes higher-level reproductive success under the specified conditions. The causal claim must become stronger than the covariance that first suggested it.

Let It Fail

If component selection, assortment, environment, local competition, migration, drift, demography, linked traits, or grouping explain the pattern, Model A wins. If higher-level organization adds prospective information or causal structure while higher-level selection remains unearned, Model B wins—and remains a discovery. Model C requires independently justified higher-level individuals whose heritable higher-level differences causally contribute to differential higher-level reproductive success under specified conditions, survive the strongest fair rivals, and generalize or replicate appropriately.

One Price term is not enough. One regression coefficient is not enough. One successful lineage is not enough.

The level has to keep doing work after we try to replace it.

Selection Is Not Adaptation

Even if D really contributes to differential reproductive success now, D is not automatically an adaptation.

Selection can act on correlated traits. Traits can hitchhike. Historical constraints and drift matter. Showing current selection on D does not establish that D originated, spread historically, or was maintained because of the effect we observe today. Establishing adaptation requires a historical argument appropriate to the trait and claim.[6]

Selection, adaptation, optimization, progress, and purpose are not synonyms. Natural selection has no obligation to produce perfection, increasing complexity, intelligence, harmony, or moral improvement. Nor does an outcome acquire purpose merely because it becomes common.

If higher-level selection were demonstrated, we still could not slide from “this organization affected descendant success” to “the whole evolved in order to achieve something.” Function remains separate from goal. Goal remains separate from purpose. Selection remains separate from agency.

Building the Individual

Selection can nevertheless change the architecture of levels.

Major evolutionary transitions include cases in which units that once reproduced more independently became increasingly integrated into collectives. Across different transitions, cooperation, conflict mediation, reproductive specialization, dependence, bottlenecks, policing, or other mechanisms can alter the relationship between component and collective fitness. No one mechanism is universal, and “fitness decoupling” is not a single binary switch.[3][4]

The broad lesson is powerful: biological individuality can itself become an evolutionary outcome.

Chapter 12 reached this from the problem of individuality:

**INDIVIDUALITY MAY BE AN EVOLUTIONARY
ACHIEVEMENT, NOT A STARTING CONDITION.**

Selection does not operate only on finished individuals. Under some conditions, selection across nested levels can participate in changing integration, conflict control, specialization, dependence, and higher-level reproduction.

That possibility is scientifically established in some evolutionary transitions. It is not evidence that a forest is undergoing one.

Earth Is Not a Population

Earth contains evolution and countless selected lineages, but Chapter 15 needs something more specific: a defensible population or lineage of planetary individuals. One observed Earth does not supply it. Panspermia, artificial colonization, or hypothetical cosmic populations may motivate questions, but they do not establish planetary parent–offspring lineages, heredity, or differential planetary reproductive success. Scale cannot manufacture the missing comparison.

The Builder

Return to the relay track.

One runner could lose while her team won. That showed us that success can be measured at more than one level. It did not tell us which level evolution was acting on.

The stronger claim requires the whole to earn its individuality, lineage, heredity, and success currency before a higher-level difference is allowed to explain differential higher-level reproduction. Then that explanation must survive lower-level and contextual rivals.

That standard is demanding because the claim is demanding.

If higher-level selection exists, we expose it by making the alternatives stronger.

Selection is not the sole source of evolutionary novelty. Mutation, recombination, developmental change, migration, symbiosis, and other processes can generate or introduce variation. Selection changes what happens to that variation: across generations it can differentially retain and amplify some variants, remove others, reshape lineage distributions, and participate in transitions that alter what counts as an individual.

The whole may therefore be more than a contestant. It may also be something evolution can help build.

If selection can act on the whole, can it build the whole?

FIGURE 15 - APPROVED POSITION

Structural proof placeholder. Insert only the frozen publication-authority visual at final composition; do not invent or substitute artwork.

The Transition

You are made of parts whose evolutionary fate has become deeply tied to a larger whole.

That sentence sounds less strange than it should. You have spent your entire life experiencing yourself as one thing. You wake as one person, move as one body, become hungry in one stomach, feel pain in one place and recognize it as happening to you. Your skin gives the arrangement an obvious edge. From the inside, individuality feels almost self-evident.

Look more closely and the certainty becomes complicated. Your body is built from an enormous population of cells performing radically different jobs. A neuron and a liver cell share the history of the same multicellular lineage but live astonishingly different lives. Cells divide, communicate, compete for resources, repair damage, die, and sometimes produce variants that proliferate at the expense of organismal function. Yet most somatic cells participate in a life cycle whose reproductive continuity is organized at the level of the organism rather than through independent somatic lineages.

Multicellular life did not begin that way. Multicellular lineages arose from unicellular ancestry, and in some of those histories cellular reproduction became reorganized within a larger repeating life cycle. That does not make multicellularity an evolutionary destination. Humans are not the point toward which life was moving, and a human body is not the template every biological individual must resemble. Evolution wanted nothing. The question is how units that once reproduced more independently could become components of an

organization whose reproduction increasingly operated through the whole.

When Two Become One

Imagine two lineages that repeatedly help one another. One obtains a resource the other cannot easily acquire; the second transforms something the first needs. Generation after generation, the partnership persists, and eventually each performs badly without the other. Have two become one? Not necessarily.

Nature contains ancient mutualisms and deeply interdependent symbioses. Cooperation can become essential without automatically producing a new evolutionary individual, and dependence can become profound without settling where individuality belongs. At the same time, distinct component lineages are not an automatic veto: some theories of evolutionary individuality take highly integrated host-symbiont associations seriously as candidate higher-level individuals. The point is that neither cooperation nor dependence decides the issue alone.[1]

The stronger question is whether evolutionary change has reorganized reproduction, heredity, conflict, dependence, and selection so deeply that collective-level variables become necessary for specified evolutionary explanations. Nature is full of cooperation. The difficult problem is discovering when cooperation participates in a change in the organization of individuality.

Conflict Inside the Whole

Your body helps expose the next mistake because it is not the peaceful republic it appears to be. Cells can mutate, and some variants gain a within-organism proliferative advantage while damaging organismal function. Cancer is far richer than a simple story of “selfish cells,” but it makes one fact impossible to ignore: lower-level selection and conflict can occur inside an organism whose higher-level individuality is not seriously in doubt.[2]

Cancer is an illustration of present multilevel conflict, not a reconstruction of the historical mechanism by which multicellularity evolved. Across different evolutionary transitions, mechanisms such as common descent, relatedness, reproductive bottlenecks, policing, developmental control, reproductive specialization, partner fidelity, and sanctions can alter conflicts among components. They are examples from different architectures, not a universal package.[1]

A new individual does not require perfect peace among its parts. What matters for the lineage-based transition studied here is whether organization can keep lower-level conflict compatible enough with collective continuity and reproduction for a higher-level evolutionary process to persist. Even that wording carries a prior requirement: a mechanism cannot stabilize higher-level reproduction until higher-level reproduction has first been defined independently. Chapter 13's reproduction gate cannot be manufactured retroactively by transition theory.

That moves us from cooperation and conflict to the structure that connects reproduction, heredity, and selection: the life cycle.

The Life Cycle

Cells sticking together are interesting. A collective that repeatedly participates in a collective life cycle is a stronger evolutionary candidate. Imagine a cluster that begins from a founding stage, develops, produces one or more propagules, and generates descendants that repeat the cycle. A propagule need not resemble a miniature adult; depending on the system, it may contain one component, many components, or another transmissible founding organization.

Now differences among collectives can persist through lineage. Some organizations can leave more descendant collectives than others, allowing selection to change the distribution of traits associated with the collective life cycle. A repeating collective life cycle is therefore central to the stronger lineage-based transition claim used in Macroscopic Life. It is not offered as a universal definition of every kind of biological individuality.[3]

This is where the previous chapters lock together. Reproduction gives the proposed whole descendants. Heredity allows differences to persist across those descendant lineages. Selection changes the distribution of inherited differences. Transition asks whether those processes have reorganized the relationship between components and collective enough that the higher-level unit gains additional causal and evolutionary leverage.

A flock can reform. A microbial community can recur. A forest can regenerate. A partnership can be reconstructed by an environment. Persistence and recurrence remain weaker than lineage. Once that distinction is clear, however, pieces of the transition problem become experimentally visible.

Watch It Begin

One famous experiment began with ordinary baker's yeast. Ratcliff and colleagues selected populations for rapid settling through liquid. Under that regime, daughter cells evolved to remain attached after division, producing branching multicellular clusters. The clusters reproduced through multicellular propagules produced by fracture, establishing a simple repeating multicellular life cycle.[4]

Later snowflake-yeast research has examined the evolution of larger collective size and additional emergent multicellular traits, turning the system into a long-running experimental model for how selection can alter collective organization.[5] Those later findings should not be silently folded into the first settling experiment, and early observations involving cell death should not be treated casually as proof that a germ-soma division of labor had already evolved.

The experimenters chose the selection regime, but that does not make the evolutionary response unreal. Selection always occurs under environmental conditions. Experimental imposition limits what we may infer about ecological generality and ancient historical routes; it does not erase heritable evolutionary change observed under the imposed conditions. Scientists did not recreate animal evolution. They made parts of a transition experimentally manipulable.

That matters because the transition is no longer only an event reconstructed from deep history. We can watch collective life cycles and traits change under controlled evolutionary pressure—and we can watch evolutionary trajectories fail to follow the staircase our intuition wants to draw.

No Ladder

Volvocine green algae provide a different window. Related lineages display multiple combinations of cell number, developmental organization, extracellular matrix, cellular differentiation, and reproductive specialization. Some are unicellular, some colonial, and some, including *Volvox*, show strong differentiation between reproductive and somatic functions.[6]

It is tempting to arrange them in a neat row from simple to advanced. That would turn living cousins into museum exhibits of one another's ancestors. Phylogenetic reconstruction instead shows a branching history in which traits associated with multicellular complexity have multiple origins and reversals. Comparative systems are valuable because related lineages preserve different combinations of traits, not because they form a surviving evolutionary staircase.[6]

EVOLUTION HAS NO UPWARD DIRECTION. LEVELS CAN CHANGE WITHOUT BECOMING A LADDER.

The claim is not that directional evolutionary trends never occur. Particular lineages can move directionally under particular conditions. What fails is the stronger idea of a universal intrinsic drive toward greater complexity, larger individuality, or us. Evolutionary history branches, reverses, simplifies, and ends as well as elaborates. A transition can be consequential without being progress.

An Individual Without a Body

The human body creates another bias: we expect an individual to be physically continuous. Eusocial insects break that expectation. A worker ant can leave the nest; a bee can fly across a landscape. Colony members remain visibly separate organisms, yet in many systems reproduction is specialized, labor is divided, conflicts are regulated, colony-level traits matter, and reproductive individuals or propagules participate in founding descendant colonies.

This has motivated serious biological use of *superorganism* and broader theories of organismality. Queller and Strassmann, for example, frame organismality in terms of high cooperation and low internal conflict and explicitly argue that physical contiguity, clonality, development from a single cell, germ–soma separation, and even same-species membership are not universally necessary under their framework.[7]

That framework does not settle every question about physiological, evolutionary, or ecological individuality, and colony-founding systems differ among eusocial taxa. But it defeats one easy objection: physical fusion is not a universal requirement for every scientifically serious account of higher-level individuality. Separate components can participate in strongly organized collective reproduction without becoming one continuous body.

The human body is one architecture of individuality. Biology gives us reason to investigate others.

Fitness Did Not Move Upward

It is easy to imagine a transfer taking place during a transition: first the components have fitness, then fitness somehow moves upward into the collective. The lower level loses it; the higher level gains it. That picture is too simple.

Theories of major evolutionary transitions have productively used ideas such as fitness alignment, fitness decoupling, and export of fitness. But recent conceptual work has emphasized limits to treating particle-

and collective-level fitness as a single transferable quantity. The measures may answer different questions and need not be directly commensurable.[8]

What changes can be stated more concretely. Independent component reproduction may become constrained. Component fate may become increasingly tied to collective fate. Collective traits may become heritable. Reproductive specialization may deepen. Collective organization may add predictive or causal leverage for particular lineage outcomes.

A TRANSITION IN INDIVIDUALITY IS NOT A MOMENT WHEN FITNESS MOVES UPWARD. IT IS A CHANGE IN HOW REPRODUCTION, HEREDITY, CONFLICT, DEPENDENCE, AND CAUSAL ORGANIZATION ARE STRUCTURED ACROSS LEVELS.

Cells remain cells. Ants remain ants. Physical mechanisms do not disappear when a higher-level description becomes useful. The scientific question is narrower: for a specified target, which variables at which levels add causal, predictive, or explanatory leverage that fair rival descriptions lack?

That makes present organization testable. It does not make evolutionary history easy to reconstruct.

The Present Is Not the Past

A mature organism can make its own history look inevitable. Watch an animal embryo develop and multicellular individuality can feel like the natural destination of cells. Present integration, however, does not reveal the sequence of evolutionary events that produced it.

Historical inference therefore requires converging evidence: comparative phylogenetics, experimental evolution, developmental and genomic mechanisms, repeated independent origins, and fossils where the relevant features can be inferred. Species in a phylogeny are not independent data points simply because we can count them separately; shared ancestry creates statistical non-independence that comparative

methods must address.[9]

We can test present-day dependence without pretending to reconstruct history. Call one project-specific approach the **Organizational Dependence Test**. Perturb a candidate higher-level organizational mechanism while preserving component identity, resources, and environment as far as possible, then ask whether collective reproduction, inheritance, or persistence changes beyond component-level predictions. Where feasible, include sham perturbations, matched component damage, environmental controls, graded disruption, and recovery or rescue. Otherwise collateral damage can counterfeit organizational dependence.

A positive result shows that organization matters now. It does not, by itself, tell us how that organization evolved. Repeated independent origins can provide especially useful historical contrasts, but even they require explicit phylogenetic and mechanistic analysis.

That distinction becomes essential when we return to our deliberately difficult forest.

The Forest Trial

Suppose, only for the experiment, that Chapters 12 through 15 had already justified candidate forest individuals, parent–offspring forest lineages, inherited forest-level differences, and a defensible higher-level success currency. Only then would it make sense to ask whether a forest lineage had undergone a transition in individuality.

Imagine that candidate lineages differ in a prospectively specified organizational property D. Across generations, D predicts changes in collective reproduction, lineage-stable specialization, component dependence, conflict structure relevant to reproduction, higher-level heredity, and perhaps differential lineage success. If a defensible perturbation of D changes those outcomes beyond component predictions, the higher-level architecture becomes more interesting.

Then the hostile work begins. Perhaps component species independently adapted. Perhaps mutualism strengthened without

producing a new individual. Perhaps ecological inheritance is sufficient to reconstruct the organization without higher-level individuality—even though ecological inheritance could also be part of a genuine inheritance architecture. Environmental filtering, species composition, disturbance history, colonization history, founder effects, demographic turnover, migration, spatial autocorrelation, and climate can all counterfeit lineage-stable organization. The proposed boundary itself may still be wrong, and D may be a proxy chosen after the outcome was known.

A forest could communicate, remember, regulate, cooperate, and even show useful higher-level selection effects without having crossed an evolutionary transition to a new individual. That is not a retreat. It is what prevents one successful property test from silently dragging every other property across the line.

If the best explanation is extraordinary association rather than a new evolutionary individual, extraordinary association is what the science should find.

Who Owns the Transition?

The project-specific **Transition Ownership Test** now has to do two jobs separately. First comes historical reconstruction: after fair lower-level and environmental rivals are modeled, does converging evidence support inherited changes in higher-level organization as contributors to a historical shift in specified dimensions of reproduction, heredity, conflict, dependence, and lineage structure? Second comes present causal status: do variables describing that derived higher-level organization add explanatory, predictive, or intervention leverage over fair association-level rivals?

A historical transition is relational. We need some defensible baseline or ancestral state S_0 , a derived state S_1 , inherited organizational changes linking them, changed reproductive or causal structure, and evidence connecting those changes to lineage dynamics. Without an $S_0 \rightarrow S_1$ contrast, “transition” can collapse into a description of current individuality.

Prediction can make the present-day models compete. Let H_transition use variables associated with the proposed higher-level organization, and let B_association be the strongest fair model in which lower-level individuals connected through cooperation, mutualism, ecological inheritance, composition, and environment remain sufficient.

$$PA_transition = Performance(H_transition) - Performance(B_association)$$

The established Prediction Advantage discipline applies. Positive PA_transition means higher-level variables earn scientific work on the specified present-day target; it does not prove the historical S0→S1 transition.

Where experiments are possible, interventions can strengthen causal inference. Ancient transitions may not permit them, leaving comparative, phylogenetic, mechanistic, and historical evidence to carry more of the burden.

Let It Fail

Association Sufficiency remains Model A: lower-level evolution, mutualism, ecology, composition, environmental filtering, and ordinary selection explain the organization without a transition at the proposed level. Model B is Higher-Level Organizational Utility: the higher level does scientific work now while the historical transition remains unearned—and that is still a discovery. Model C is the property-specific historical claim: converging evidence supports an S0→S1 shift in which inherited organizational changes altered specified dimensions of reproduction, heredity, conflict, dependence, selection, and lineage structure, while the derived higher-level unit gains relevant causal or evolutionary leverage. The models remain reversible, and none establishes progress.

No Staircase to Earth

We are now standing at one of the most seductive points in the argument. Cells formed multicellular organisms in some lineages. Organisms formed highly integrated societies in others. Ecological systems can become deeply coupled. Life has altered the atmosphere, oceans, soils, nutrient cycles, and climate, while technological civilization moves information and material around the planet at extraordinary speed.

The mind wants to draw a staircase: cell, organism, society, ecosystem, planet. Do not draw it. Every arrow must earn its own evidence; individuality at one level cannot be inherited by the level above. Earth contains major evolutionary transitions, but that is not evidence that Earth underwent one. Even empirically supported planetary regulation or Gaia-style life–environment feedbacks address regulation and coupling, not the lineage architecture required for evolutionary individuality.[10]

Nesting is not a ladder.

Return to the body with which we began. Your individuality once seemed like the easy case. It should now look historically stranger. Reproduction became organized in new ways. Components specialized. Conflicts were constrained without disappearing. Development repeatedly rebuilds the collective. Heredity and selection operate through a life cycle that cannot be understood by treating every somatic cell as an independent organism.

That history teaches something profound but narrower than the grandest version of Macroscopic Life. It does not show that every collection can become an individual, that larger systems are climbing toward life, or that Earth is becoming an organism. It shows that individuality itself can have an evolutionary history.

The organisms we know cannot therefore serve as timeless templates for every form an individual must take. Evolution did not discover a final size, shape, or architecture for individuality. Under some conditions, it changed the organization through which individuality

could emerge.

Scale alone cannot tell us where that process is possible.

**If individuality can be reorganized across levels, what
determines where a new individual can—and cannot—emerge?**

FIGURE 16 - APPROVED POSITION

Structural proof placeholder. Insert only the frozen publication-authority visual
at final composition; do not invent or substitute artwork.

NOTES

Notes to Chapter 7 — The Signal

[1] Choi, W.-G., Hilleary, R., Swanson, S. J., Kim, S.-H., & Gilroy, S. (2016). “Rapid, Long-Distance Electrical and Calcium Signaling in Plants.” *Annual Review of Plant Biology*, 67, 287–307. DOI: 10.1146/annurev-arplant-043015-112130.

[2] Wang, S., Payne, G. F., & Bentley, W. E. (2020). “Quorum Sensing Communication: Molecularly Connecting Cells, Their Neighbors, and Even Devices.” *Annual Review of Chemical and Biomolecular Engineering*, 11, 447–468. DOI: 10.1146/annurev-chembioeng-101519-124728. See also Aframian, N., & Eldar, A. (2020). “A Bacterial Tower of Babel: Quorum-Sensing Signaling Diversity and Its Evolution.” *Annual Review of Microbiology*, 74, 587–606. DOI: 10.1146/annurev-micro-012220-063740.

[3] Simard, S. W., Perry, D. A., Jones, M. D., Myrold, D. D., Durall, D. M., & Molina, R. (1997). “Net transfer of carbon between ectomycorrhizal tree species in the field.” *Nature*, 388, 579–582. DOI: 10.1038/41557. The study demonstrated bidirectional carbon transfer between *Betula papyrifera* and *Pseudotsuga menziesii* under the reported field conditions and argued that transfer was primarily through a direct hyphal pathway; it does not by itself establish generalized forest-level communication, preferential altruism, agency, or individuality.

[4] Karst, J., Jones, M. D., & Hoeksema, J. D. (2023). “Positive citation bias and overinterpreted results lead to misinformation on common mycorrhizal networks in forests.” *Nature Ecology & Evolution*, 7, 501–511. DOI: 10.1038/s41559-023-01986-1. An author correction

removed four extraneous references from the original version; the corrected article remains the cited source. DOI for correction: 10.1038/s41559-023-02035-7.

[5] Gordon, D. M. (2019). “The Ecology of Collective Behavior in Ants.” *Annual Review of Entomology*, 64, 35–50. DOI: 10.1146/annurev-ento-011118-111923. Used here only to support distributed/local-interaction mechanisms in ant collective behavior, not a universal claim that all ant colonies satisfy one higher-level individuality criterion.

Notes to Chapter 8 — The Memory

[1] Jiang, Y., & Hao, N. (2021). “Memorizing environmental signals through feedback and feedforward loops.” *Current Opinion in Cell Biology*, 69, 96–102. DOI: 10.1016/j.ceb.2020.11.008.

[2] Lam, N., Lee, Y., & Farber, D. L. (2024). “A guide to adaptive immune memory.” *Nature Reviews Immunology*, 24(11), 810–829. DOI: 10.1038/s41577-024-01040-6.

[3] Fineran, P. C., & Charpentier, E. (2012). “Memory of viral infections by CRISPR-Cas adaptive immune systems: Acquisition of new information.” *Virology*, 434(2), 202–209. DOI: 10.1016/j.virol.2012.10.003.

[4] Harris, C. J., Amtmann, A., & Ton, J. (2023). “Epigenetic processes in plant stress priming: Open questions and new approaches.” *Current Opinion in Plant Biology*, 75, 102432. DOI: 10.1016/j.pbi.2023.102432.

[5] Boisseau, R. P., Vogel, D., & Dussutour, A. (2016). “Habituation in non-neural organisms: evidence from slime moulds.” *Proceedings of the Royal Society B: Biological Sciences*, 283(1829), 20160446. DOI: 10.1098/rspb.2016.0446.

[6] Johnstone, J. F., Allen, C. D., Franklin, J. F., Frelich, L. E., Harvey, B. J., Higuera, P. E., Mack, M. C., Meentemeyer, R. K., Metz, M. R., Perry, G. L. W., Schoennagel, T., & Turner, M. G. (2016). “Changing disturbance regimes, ecological memory, and forest

resilience.” *Frontiers in Ecology and the Environment*, 14(7), 369–378. DOI: 10.1002/fee.1311.

Notes to Chapter 9 — The Anticipation

[1] Jabbur, M. L., & Johnson, C. H. (2022). “Spectres of Clock Evolution: Past, Present, and Yet to Come.” *Frontiers in Physiology*, 12, 815847. DOI: 10.3389/fphys.2021.815847. Supports circadian clocks as widespread biological oscillators that entrain to environmental cycles and use phase relationships to anticipate predictable daily phenomena.

[2] Sterling, P. (2012). “Allostasis: a model of predictive regulation.” *Physiology & Behavior*, 106(1), 5–15. DOI: 10.1016/j.physbeh.2011.06.004. Supports the use of predictive/anticipatory regulation language for preparation before physiological needs arise. The broader Macroscopic Life distinctions are project synthesis, not Sterling's taxonomy.

[3] Hill, K. A. P., Pfennig, K. S., & Pfennig, D. W. (2024). “Assessment and the regulation of adaptive phenotypic plasticity.” *Development*, 151(20), dev203101. DOI: 10.1242/dev.203101. Supports the role of environmental assessment, cue reliability, unreliable cues, and adaptive versus maladaptive plastic responses.

[4] Tagkopoulos, I., Liu, Y.-C., & Tavazoie, S. (2008). “Predictive behavior within microbial genetic networks.” *Science*, 320(5881), 1313–1317. DOI: 10.1126/science.1154456. Supports predictive-behavior terminology, correlations in *E. coli* transcriptional responses associated with environmental transitions, and rapid decoupling of internal correlations in novel environments. It is not used here as evidence for consciousness or an explicit internal forecast model.

[5] Bialek, W., Nemenman, I., & Tishby, N. (2001). “Predictability, complexity, and learning.” *Neural Computation*, 13(11), 2409–2463. DOI: 10.1162/089976601753195969. Supports predictive-information formalism. Macroscopic Life's observer-versus-system-use distinction is project synthesis.

[6] Palmer, S. E., Marre, O., Berry, M. J. II, & Bialek, W. (2015). “Predictive information in a sensory population.” *Proceedings of the National Academy of Sciences*, 112(22), 6908–6913. DOI: 10.1073/pnas.1506855112. Supports empirical measurement of predictive information in a biological sensory population.

[7] Hewing, L., Wabersich, K. P., Menner, M., & Zeilinger, M. N. (2020). “Learning-Based Model Predictive Control: Toward Safe Learning in Control.” *Annual Review of Control, Robotics, and Autonomous Systems*, 3, 269–296. DOI: 10.1146/annurev-control-090419-075625. Supports MPC as model-based constrained control using predictive system models; the biological comparison is illustrative rather than an implementation claim.

[8] Summerfield, C., & de Lange, F. P. (2014). “Expectation in perceptual decision making: neural and computational mechanisms.” *Nature Reviews Neuroscience*, 15, 745–756. DOI: 10.1038/nrn3838. Supports expectation/predictive-coding context and the distinction between expected and unexpected sensory events.

[9] Corlett, P. R., Mollick, J. A., & Kober, H. (2022). “Meta-analysis of human prediction error for incentives, perception, cognition, and action.” *Neuropsychopharmacology*, 47, 1339–1349. DOI: 10.1038/s41386-021-01264-3. Supports the breadth of prediction-error research across reward, perception, cognition, and action; it is not treated as a universal definition for prediction error.

Notes to Chapter 10 — The Goal

1. Richard I. Vane-Wright and Peter A. Corning, “Teleonomy in Living Systems: An Overview,” *Biological Journal of the Linnean Society* 139, no. 4 (2023): 341–356, DOI: 10.1093/biolinnean/blad037. This is evidence for an active literature on teleonomy and goal-directedness, not evidence that one definition of goal is field-wide consensus.

2. Walter B. Cannon, “Organization for Physiological Homeostasis,” *Physiological Reviews* 9, no. 3 (1929): 399–431, DOI: 10.1152/physrev.1929.9.3.399; Peter Sterling, “Allostasis: A Model of Predictive Regulation,” *Physiology & Behavior* 106, no. 1 (2012): 5–15, DOI: 10.1016/j.physbeh.2011.06.004; Jay Schulkin and Peter Sterling, “Allostasis: A Brain-Centered, Predictive Mode of Physiological Regulation,” *Trends in Neurosciences* 42, no. 10 (2019): 740–752, DOI: 10.1016/j.tins.2019.07.010. The chapter treats allostasis as an influential framework, not a universally accepted replacement for homeostasis.

3. Michael Levin, “The Biophysics of Regenerative Repair Suggests New Perspectives on Biological Causation,” *BioEssays* 42, no. 2 (2020): e1900146, DOI: 10.1002/bies.201900146; Michael Levin, “Bioelectric Signaling: Reprogrammable Circuits Underlying Embryogenesis, Regeneration, and Cancer,” *Cell* 184, no. 8 (2021): 1971–1989, DOI: 10.1016/j.cell.2021.02.034. These sources support the specific regenerative/bioelectric research program discussed here; they do not establish target morphology as settled field-wide ontology.

4. Hiroaki Kitano, “Biological Robustness,” *Nature Reviews Genetics* 5, no. 11 (2004): 826–837, DOI: 10.1038/nrg1471; James M. Whitacre, “Degeneracy: A Link Between Evolvability, Robustness and Complexity in Biological Systems,” *Theoretical Biology and Medical Modelling* 7 (2010): 6, DOI: 10.1186/1742-4682-7-6. These sources ground robustness and degeneracy as serious alternatives to a stronger goal-ownership inference.

5. Vicente Raja, Dinesh Valluri, Edward Baggs, Anthony Chemero, and Michael L. Anderson, “The Markov Blanket Trick: On the Scope of the Free Energy Principle and Active Inference,” *Physics of Life Reviews* 39 (2021): 49–72, DOI: 10.1016/j.plrev.2021.09.001; Michael David Kirchhoff, Julian Kiverstein, and Ian Robertson, “The Literalist Fallacy and the Free Energy Principle: Model Building, Scientific Realism, and Instrumentalism,” *The British Journal for the Philosophy of Science* 76, no. 3 (2025): 639–662, DOI: 10.1086/720861. These papers represent different sides of debates about scope and model-to-world interpretation; neither is used here as a consensus refutation of active inference.

6. Diane Marie Keeling, Patricia Garza, Charisse Michelle Nartey, and Anne-Ruxandra Carvunis, “The Meanings of ‘Function’ in Biology and the Problematic Case of De Novo Gene Emergence,” *eLife* 8 (2019): e47014, DOI: 10.7554/eLife.47014; Matteo Mossio, Cristian Saborido, and Alvaro Moreno, “An Organizational Account of Biological Functions,” *The British Journal for the Philosophy of Science* 60, no. 4 (2009): 813–841, DOI: 10.1093/bjps/axp036. These support pluralism about biological function and an organizational alternative; the chapter does not treat any one account as settled consensus.

7. Maël Montévil and Matteo Mossio, “Biological Organisation as Closure of Constraints,” *Journal of Theoretical Biology* 372 (2015): 179–191, DOI: 10.1016/j.jtbi.2015.02.029; Matteo Mossio and Leonardo Bich, “What Makes Biological Organisation Teleological?” *Synthese* 194, no. 4 (2017): 1089–1114, DOI: 10.1007/s11229-014-0594-z. These provide a serious organizational account of biological self-maintenance and teleology. Macroscopic Life’s question is whether comparable organization can be justified at a proposed larger level.

Notes to Chapter 11 — The Choice

1. Xabier E. Barandiaran, Ezequiel Di Paolo, and Marieke Rohde, “Defining Agency: Individuality, Normativity, Asymmetry, and Spatio-temporality in Action,” *Adaptive Behavior* 17, no. 5 (2009): 367–386, <https://doi.org/10.1177/1059712309343819>; Álvaro Moreno and Arantza Etxeberria, “Agency in Natural and Artificial Systems,” *Artificial Life* 11, nos. 1–2 (2005): 161–175, <https://doi.org/10.1162/1064546053278919>.

2. P. Kirk Visscher, “Group Decision Making in Nest-Site Selection Among Social Insects,” *Annual Review of Entomology* 52 (2007): 255–275, <https://doi.org/10.1146/annurev.ento.51.110104.151025>; Thomas D. Seeley, Kevin M. Passino, and P. Kirk Visscher, “Group Decision Making in Honey Bee Swarms,” *American Scientist* 94, no. 3 (2006): 220–229, <https://doi.org/10.1511/2006.59.220>.

3. Thomas D. Seeley and P. Kirk Visscher, “Quorum Sensing During Nest-Site Selection by Honeybee Swarms,” *Behavioral Ecology and Sociobiology* 56 (2004): 594–601, <https://doi.org/10.1007/s00265-004-0814-5>.

4. Iain D. Couzin, Jens Krause, Nigel R. Franks, and Simon A. Levin, “Effective Leadership and Decision-Making in Animal Groups on the Move,” *Nature* 433 (2005): 513–516, <https://doi.org/10.1038/nature03236>.

5. Wolfram Schultz, “Dopamine Reward Prediction Error Coding,” *Dialogues in Clinical Neuroscience* 18, no. 1 (2016): 23–32, <https://doi.org/10.31887/DCNS.2016.18.1/wschultz>; Wolfram Schultz, “Dopamine Reward Prediction-Error Signalling: A Two-Component Response,” *Nature Reviews Neuroscience* 17 (2016): 183–195, <https://doi.org/10.1038/nrn.2015.26>.

PART IV — THE INDIVIDUAL

Notes to Chapter 12 — The Individual

1. J. F. Harper, *Population Biology of Plants* (London: Academic Press, 1977), foundational treatment of modular/clonal plant organization and genet–ramet distinctions; see also modern clonal-plant literature on physiological integration and ramet autonomy. The chapter uses the distinction only to show that clonal lineage, physical connectedness, and physiological autonomy need not coincide perfectly.

2. David C. Queller and Joan E. Strassmann, “Beyond Society: The Evolution of Organismality,” *Philosophical Transactions of the Royal Society B: Biological Sciences* 364 (2009): 3143–3155, <https://doi.org/10.1098/rstb.2009.0095>; Joan E. Strassmann and David C. Queller, “The Social Organism: Congresses, Parties, and Committees,” *Evolution* 64, no. 3 (2010): 605–616, <https://doi.org/10.1111/j.1558-5646.2009.00929.x>.

3. Ellen Clarke, “The Problem of Biological Individuality,” *Biological Theory* 5 (2010): 312–325. This chapter treats individuality

pluralistically and framework-explicitly rather than claiming that all individuality concepts reduce to one criterion.

4. Stuart A. West, Roberta M. Fisher, Andy Gardner, and E. Toby Kiers, “Major Evolutionary Transitions in Individuality,” *Proceedings of the National Academy of Sciences* 112 (2015): 10112–10119; Richard E. Michod, “Evolution of the Individual,” *The American Naturalist* 150, suppl. 1 (1997): S5–S21, <https://doi.org/10.1086/286047>; Richard E. Michod and Denis Roze, “Transitions in Individuality,” *Proceedings of the Royal Society B: Biological Sciences* 264 (1997): 853–857, <https://doi.org/10.1098/rspb.1997.0119>. These sources represent major-transition/evolutionary-individuality work; they do not establish one universal definition for every form of biological individuality.

5. Andrew F. G. Bourke, “Conflict and Conflict Resolution in the Major Transitions,” *Proceedings of the Royal Society B: Biological Sciences* 290 (2023): 20231420, <https://doi.org/10.1098/rspb.2023.1420>. Used here to keep the conflict discussion conditional rather than treating one conflict-suppression mechanism as a universal prerequisite.

6. Joan E. Strassmann and David C. Queller, “Evolution of Cooperation and Control of Cheating in a Social Microbe,” *Proceedings of the National Academy of Sciences* 108 (2011): 10855–10862, PMID PMC3131822, PMID 21690338; Pauline Schaap, “Evolutionary Crossroads in Developmental Biology: Dictyostelium discoideum,” *Development* 138 (2011): 387–396, <https://doi.org/10.1242/dev.048934>.

7. Angela E. Douglas and John H. Werren, “Holes in the Hologenome: Why Host-Microbe Symbioses Are Not Holobionts,” *mBio* 7, no. 2 (2016): e02099-15, <https://doi.org/10.1128/mBio.02099-15>.

8. Kevin R. Theis, Nolwenn M. Dheilly, Jonathan L. Klassen, Robert M. Brucker, John F. Baines, Thomas C. G. Bosch, John F. Cryan, Scott F. Gilbert, Charles J. Goodnight, Elisabeth A. Lloyd, Jan Sapp, Philippe Vandenkoornhuys, Ilana Zilber-Rosenberg, Eugene Rosenberg, and Seth R. Bordenstein, “Getting the Hologenome Concept Right: an Eco-Evolutionary Framework for Hosts and Their Microbiomes,” *mSystems* 1, no. 2 (2016): e00028-16,

<https://doi.org/10.1128/mSystems.00028-16>. Theis et al. defend holobiont/hologenome terminology as a multilevel eco-evolutionary framework; this is paired deliberately with the Douglas–Werren critique rather than presented as settled consensus.

Notes to Chapter 13 — The Reproduction

1. Peter Godfrey-Smith, “Reproduction, Symbiosis, and the Eukaryotic Cell,” *Proceedings of the National Academy of Sciences of the United States of America* 112, no. 33 (2015): 10120–10125. DOI: 10.1073/pnas.1421378112. The article develops a framework distinguishing simple, collective, and scaffolded reproduction and analyzes parent–offspring lineage, causal responsibility, and reconstructed recurring structures.

2. Christian Peeters and Fuminori Ito, “Colony Dispersal and the Evolution of Queen Morphology in Social Hymenoptera,” *Annual Review of Entomology* 46 (2001): 601–630. DOI: 10.1146/annurev.ento.46.1.601.

3. Adam L. Cronin, Mathieu Molet, Claudie Doums, Thibaud Monnin, and Christian Peeters, “Recurrent Evolution of Dependent Colony Foundation Across Eusocial Insects,” *Annual Review of Entomology* 58 (2013): 37–55. DOI: 10.1146/annurev-ento-120811-153643.

4. John Odling-Smee, Douglas H. Erwin, Eric P. Palkovacs, Marcus W. Feldman, and Kevin N. Laland, “Niche Construction Theory: A Practical Guide for Ecologists,” *The Quarterly Review of Biology* 88, no. 1 (2013): 3–28. DOI: 10.1086/669266.

Notes to Chapter 14 — The Inheritance

1. Étienne Danchin, Anne Charmantier, Frances A. Champagne, Alex Mesoudi, Benoit Pujol, and Simon Blanchet, “Beyond DNA: Integrating Inclusive Inheritance into an Extended Theory of Evolution,” *Nature Reviews Genetics* 12 (2011): 475–486, DOI 10.1038/nrg3028. The review treats non-genetic inheritance as a heterogeneous set of processes contributing to parent–offspring resemblance and evolutionary dynamics rather than as one mechanism.

2. Maximilian H. Fitz-James and Giacomo Cavalli, “Molecular Mechanisms of Transgenerational Epigenetic Inheritance,” *Nature Reviews Genetics* 23 (2022): 325–341, DOI 10.1038/s41576-021-00438-5. The review covers DNA methylation, histone modifications, non-coding RNAs, taxonomic variation, and mechanisms by which epigenetic states can be maintained or reconstructed across generations.

3. Sanne D. van Otterdijk and Karin B. Michels, “Transgenerational Epigenetic Inheritance in Mammals: How Good Is the Evidence?” *The FASEB Journal* 30 (2016): 2457–2465, DOI 10.1096/fj.201500083. The review distinguishes true transgenerational inheritance from intrauterine/intergenerational exposure effects and concludes that comparable mechanisms in humans remain unclear apart from established phenomena such as imprinting.

4. Peter M. Visscher, William G. Hill, and Naomi R. Wray, “Heritability in the Genomics Era—Concepts and Misconceptions,” *Nature Reviews Genetics* 9 (2008): 255–266, DOI 10.1038/nrg2322. Heritability is a population parameter whose estimate depends on population and circumstances; it is not a percentage of an individual's phenotype caused by genes.

5. F. John Odling-Smee, Douglas H. Erwin, Eric P. Palkovacs, Marcus W. Feldman, and Kevin N. Laland, “Niche Construction Theory: A Practical Guide for Ecologists,” *The Quarterly Review of Biology* 88 (2013): 3–28, DOI 10.1086/669266. The paper develops ecological-inheritance and niche-construction concepts relevant to persistent organism-modified environmental legacies.

Notes to Chapter 15 — The Selection

1. George R. Price, “Selection and Covariance,” *Nature* 227 (1970): 520–521, DOI 10.1038/227520a0; George R. Price, “Extension of Covariance Selection Mathematics,” *Annals of Human Genetics* 35 (1972): 485–490, DOI 10.1111/j.1469-1809.1957.tb01874.x. The chapter uses Price-style partitioning as a descriptive mathematical framework and does not treat a covariance term as a standalone causal detector.

2. Steven A. Frank, “George Price’s Contributions to Evolutionary Genetics,” *Journal of Theoretical Biology* 175 (1995): 373–388, DOI 10.1006/jtbi.1995.0148; David C. Queller, “Quantitative Genetics, Inclusive Fitness, and Group Selection,” *The American Naturalist* 139 (1992): 540–558, DOI 10.1086/285343; contextual-analysis and Price approaches answer related but nonidentical questions, and causal interpretation depends on model and biological structure.

3. Richard E. Michod and Denis Roze, “Transitions in Individuality,” *Proceedings of the Royal Society B: Biological Sciences* 264 (1997): 853–857, DOI 10.1098/rspb.1997.0119; Richard E. Michod and Denis Roze, “Cooperation and Conflict in the Evolution of Multicellularity,” *Heredity* 86 (2001): 1–7, DOI 10.1046/j.1365-2540.2001.00808.x. These sources support the use of conflict, cooperation, and altered lower-/higher-level fitness relationships as evidence-rich features of evolutionary transitions without making any single feature universal.

4. Stuart A. West, Roberta M. Fisher, Andy Gardner, and E. Toby Kiers, “Major Evolutionary Transitions in Individuality,” *Proceedings of the National Academy of Sciences* 112 (2015): 10112–10119, DOI 10.1073/pnas.1421402112; Samir Okasha, *Evolution and the Levels of Selection* (Oxford: Oxford University Press, 2006). The chapter treats evolutionary transitions and fitness reorganization as framework-dependent rather than as a single binary fitness-decoupling test.

5. Samir Okasha, *Evolution and the Levels of Selection* (Oxford: Oxford University Press, 2006), especially the discussion of MLS-1 and

MLS-2; Elliott Sober and David Sloan Wilson, *Unto Others: The Evolution and Psychology of Unselfish Behavior* (Cambridge, MA: Harvard University Press, 1998). Terminology varies across authors; the chapter uses MLS-1/MLS-2 only as an orienting distinction between effects tracked through particle fitness and stronger collective-level fitness/reproduction claims.

6. George C. Williams, *Adaptation and Natural Selection: A Critique of Some Current Evolutionary Thought* (Princeton, NJ: Princeton University Press, 1966); Stephen Jay Gould and Richard C. Lewontin, “The Spandrels of San Marco and the Panglossian Paradigm: A Critique of the Adaptationist Programme,” *Proceedings of the Royal Society of London. Series B, Biological Sciences* 205 (1979): 581–598, DOI 10.1098/rspb.1979.0086. These sources ground the chapter's caution that current utility or current selection does not by itself establish historical adaptation.

Notes to Chapter 16 — The Transition

1. John Maynard Smith and Eörs Szathmáry, *The Major Transitions in Evolution* (Oxford University Press, 1995); Richard E. Michod, *Darwinian Dynamics: Evolutionary Transitions in Fitness and Individuality* (Princeton University Press, 1999); Stuart A. West, Andy Gardner, Angus S. Griffin, and colleagues' social-evolution work provides broader context for cooperation and conflict across levels. These traditions differ in emphasis; the chapter does not treat any single mechanism as universally necessary.

2. The cancer passage is used only to illustrate contemporary within-organism selection and conflict. It is not presented as a historical reconstruction of the origin of multicellularity.

3. The collective-lifecycle emphasis is a Macroscopic Life operational choice for the stronger lineage-based evolutionary-transition claim and should be read alongside the reproduction, inheritance, and selection gates established in Chapters 13–15.

4. William C. Ratcliff, R. Ford Denison, Mark Borrello, and Michael Travisano, “Experimental evolution of multicellularity,” *Proceedings of the National Academy of Sciences* 109 (2012): 1595–1600, DOI: 10.1073/pnas.1115323109. The experiment selected for rapid settling and produced attached multicellular clusters with multicellular propagules and a simple multicellular life cycle.

5. Later snowflake-yeast work has extended the system well beyond the original 2012 experiment. Publication lock should cite only the specific later primary studies needed for retained claims about collective size or other emergent multicellular traits; no early cell-death result is treated here as automatic proof of germ–soma specialization.

6. Matthew D. Herron and Richard E. Michod, “Evolution of complexity in the volvocine algae: transitions in individuality through Darwin's eye,” *Evolution* 62 (2008): 436–451, DOI: 10.1111/j.1558-5646.2007.00304.x. Their phylogenetic reconstruction reports multiple origins and reversals among traits associated with volvocine multicellular complexity. See also Armin Hallmann, “Evolution of reproductive development in the volvocine algae,” *Sexual Plant Reproduction* 24 (2011): 97–112, for reproductive-development context.

7. David C. Queller and Joan E. Strassmann, “Beyond society: the evolution of organismality,” *Philosophical Transactions of the Royal Society B* 364 (2009): 3143–3155, DOI: 10.1098/rstb.2009.0095; Joan E. Strassmann and David C. Queller, “The social organism: congresses, parties, and committees,” *Evolution* 64 (2010): 605–616, DOI: 10.1111/j.1558-5646.2009.00929.x. Their cooperation/conflict framework is one influential account, not a universal definition of individuality.

8. Pierrick Bourrat, Guilhem Doulcier, Caroline J. Rose, Paul B. Rainey, and Katrin Hammerschmidt, “Tradeoff breaking as model of evolutionary transitions in individuality and the limits of the fitness-decoupling metaphor,” *eLife* 11 (2022): e73715, DOI: 10.7554/eLife.73715. This supports caution about treating particle- and collective-level fitness as a single quantity literally transferred upward.

9. Joseph Felsenstein, “Phylogenies and the comparative method,” *The American Naturalist* 125 (1985): 1–15, DOI: 10.1086/284325. Species connected by a branching phylogeny are statistically non-independent; comparative inference must account for shared ancestry.

10. The planetary paragraph deliberately separates empirically testable life–environment feedbacks from the stronger requirements for evolutionary individuality. No claim is made here that Gaia-style regulation establishes planetary reproduction, heredity, or selection.